



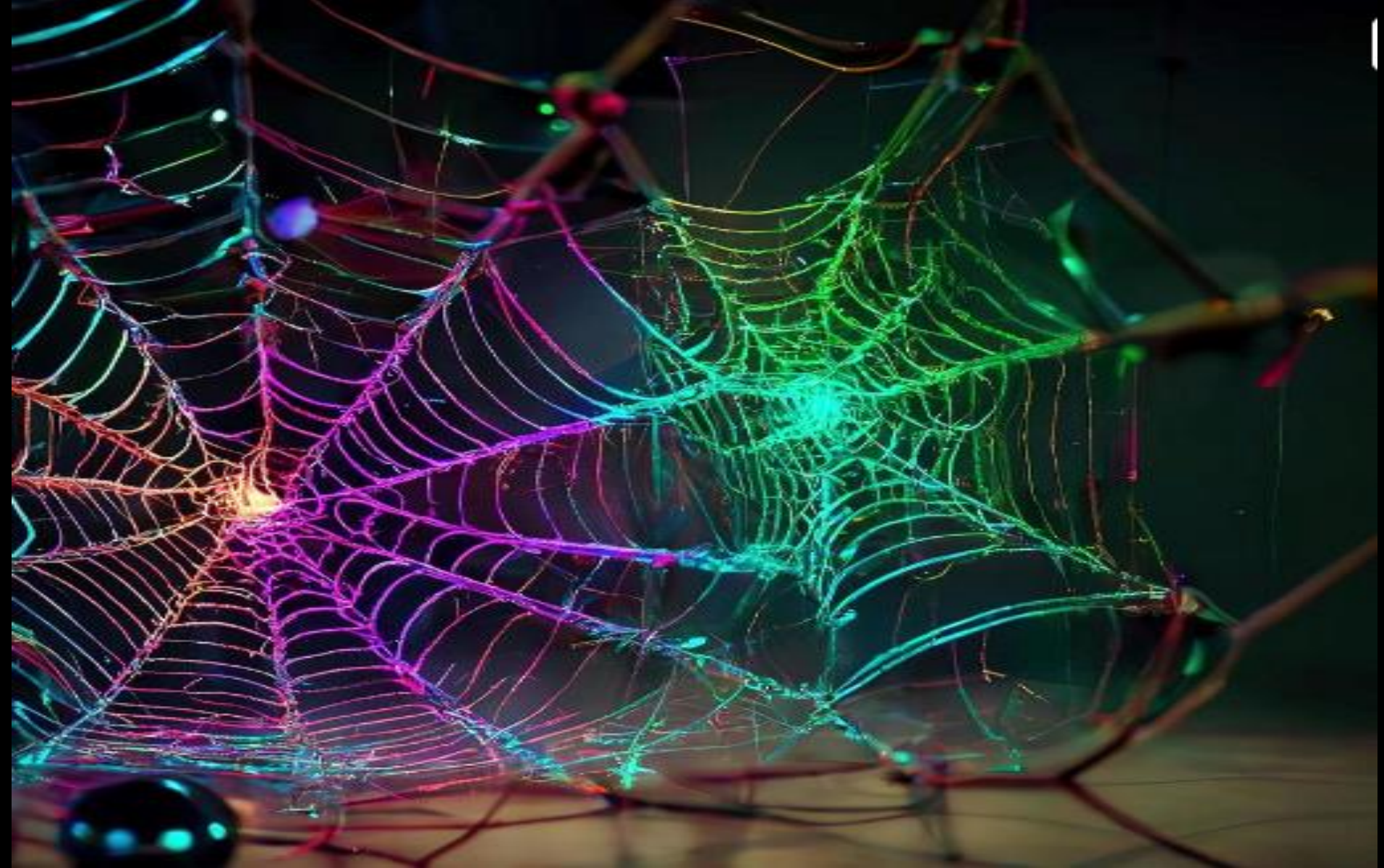
DATA COMMUNICATION AND NETWORKING

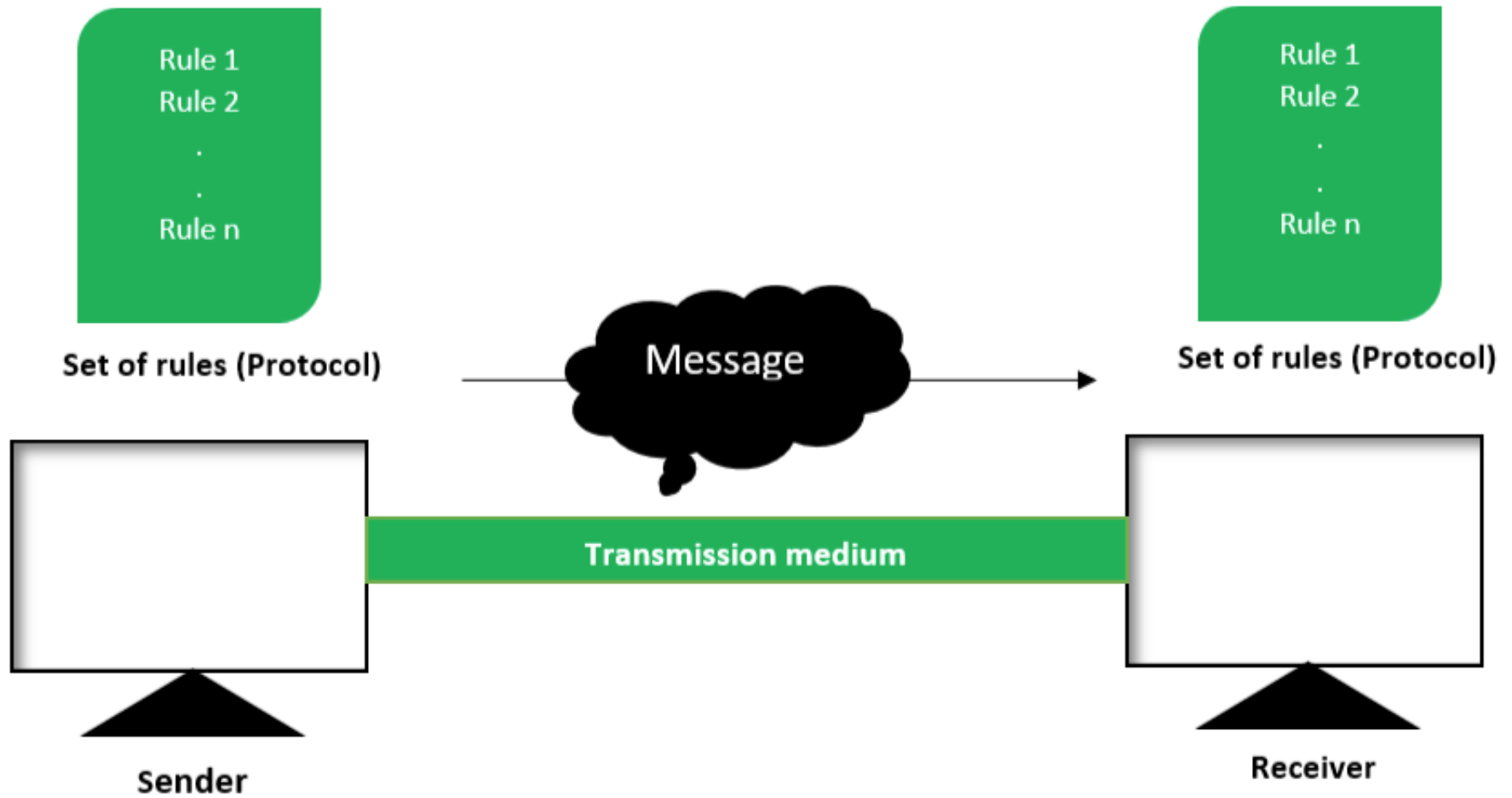
BY
ING.PRINCE KENNETH
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The Computer Network

- A **computer network** is a group of computers/devices(**Nodes**) that use a set of common communication **protocols** over digital **interconnections** for the purpose of sharing resources located on or provided by the network nodes.
- The **nodes** of a computer network may include personal computers, servers, networking hardware, or other specialised or general-purpose hosts.
- The **interconnections** between **nodes** are formed from a broad spectrum of telecommunication network technologies, based on physically wired, optical, and wireless technologies.
- A **communication protocol** is a set of rules for exchanging information over a network.

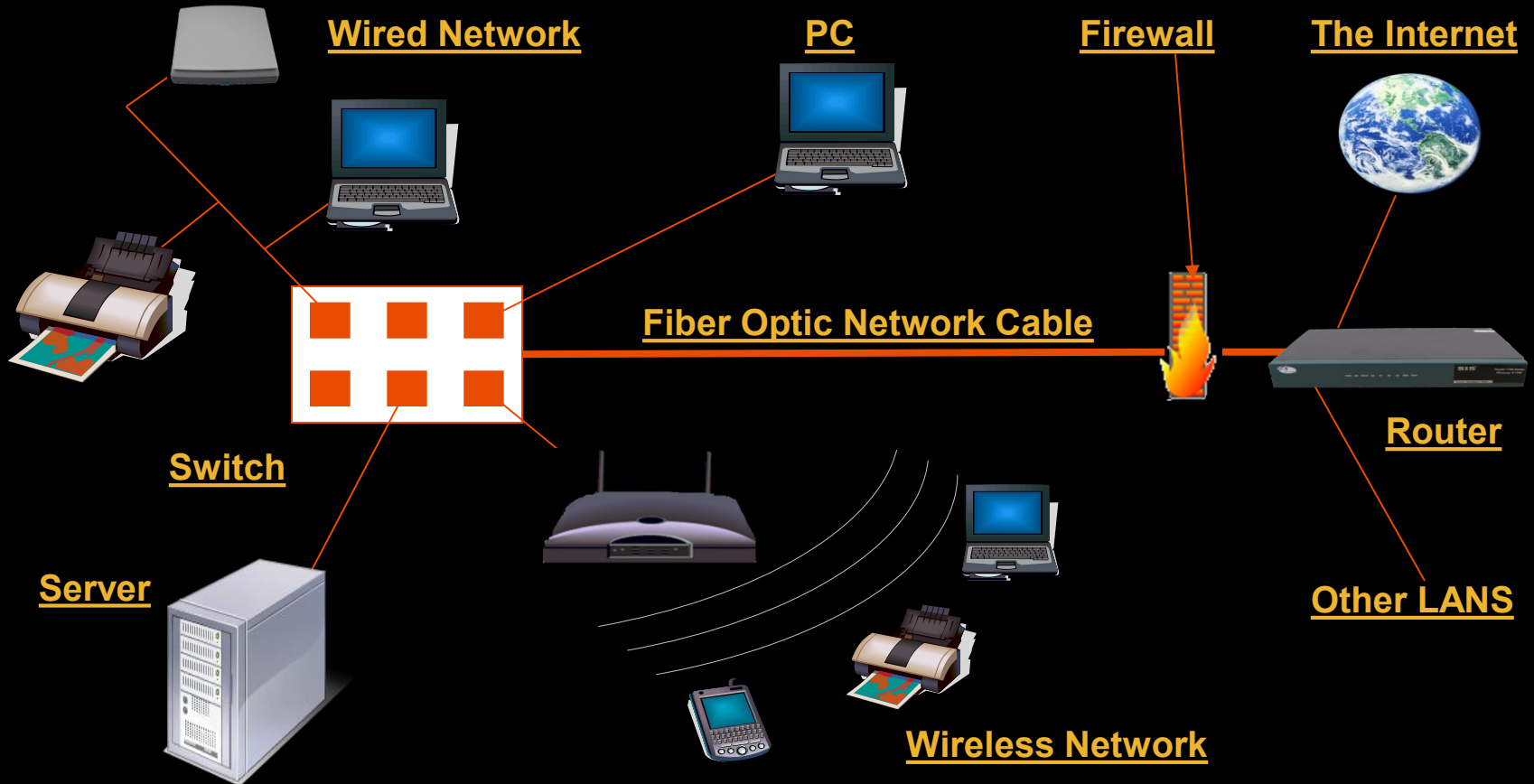
Computer Networking is the practice of connecting computer systems and networking devices together to enable communication and data exchange between them. In general, Computer Network is a collection of two or more computers. It helps users to communicate more easily.





The Network Diagram

(Click on the Words Below and Learn More About Each Component)



The Advantages/Uses of Network

Simultaneous Access

- There are moments in any business when several workers may need to use the same data at the same time.

Shared Peripheral Devices

Personal Communications

- Videoconferencing
- Voice over Internet Protocol (VoIP):-VoIP transmits the sound of voice over a computer network using the Internet Protocol (IP) rather than sending the signal over traditional phone wires

Easier Data Backup

The Networking Devices(Nodes)

1. NIC Card
2. Repeater
3. Hub
4. Switch
5. Bridge
6. Router
7. Gateway
8. Firewall

1. Network Interface Card

- NIC is used to physically connect host devices to the network media.
- A NIC is a printed circuit board that fits into the expansion slot of a bus on a computer motherboard.
- It can also be a peripheral device. NICs are sometimes called network adapters.
- Each NIC is identified by a unique code called a Media Access Control (MAC) address.
- This address is used to control data communication for the host on the network.



2. Repeaters

- A repeater is a network device used to regenerate a signal.
- Repeaters regenerate analog or digital signals that are distorted by transmission loss due to attenuation.
- A repeater does not make an intelligent decision concerning forwarding packets



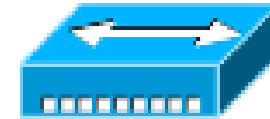
3. Hubs

- Hubs concentrate on connections.
- In other words, they take a group of hosts and allow the network to see them as a single unit. This is done passively, without any other effect on the data transmission.
- Active hubs concentrate hosts and also regenerate signals.

100BaseT Hub

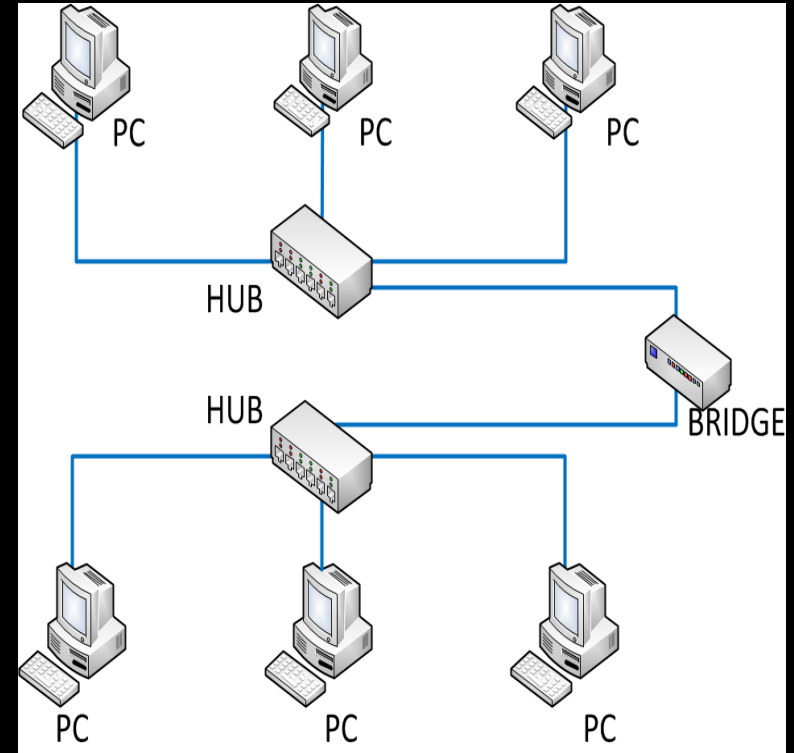


10BaseT Hub



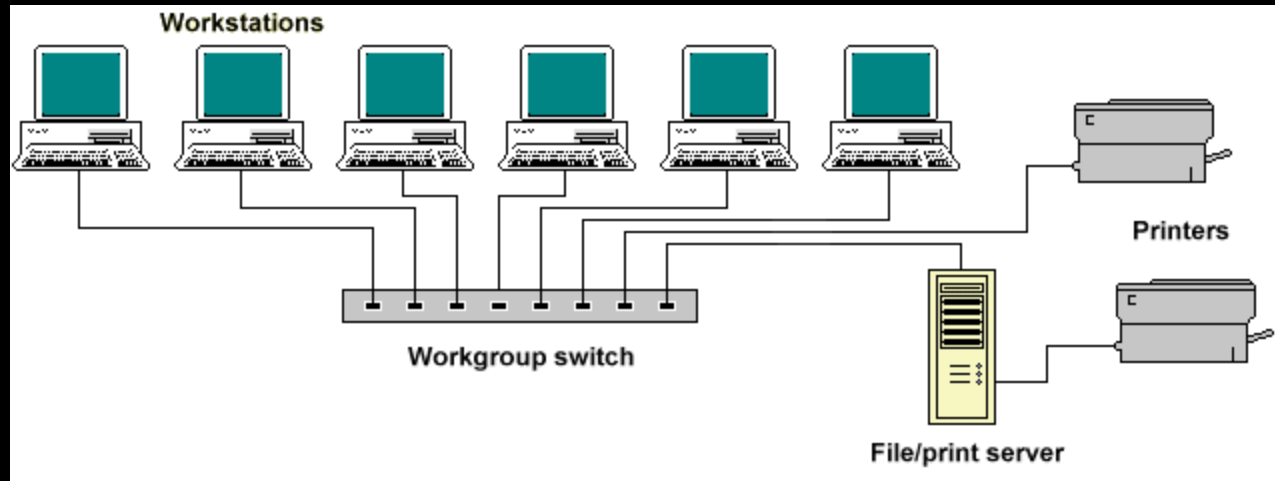
4. Bridges

- Bridges convert network data formats and perform basic data transmission management.
- Bridges provide connections between LANs.
- They also check data to determine if it should cross the bridge. This makes each part of the network more efficient



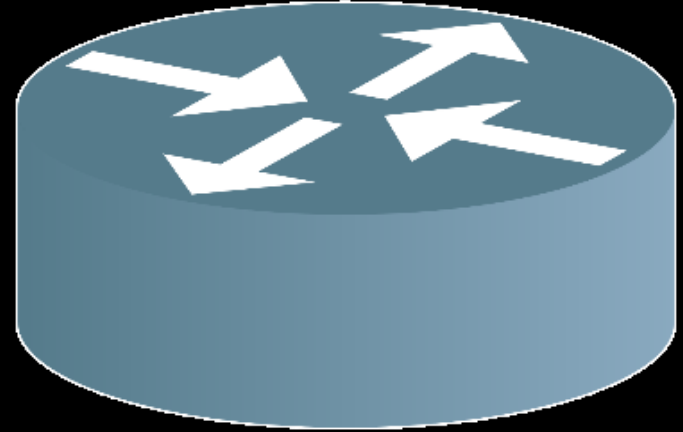
5. Switches

- Switches add more intelligence to data transfer management.
- They can determine if data should remain on a LAN and transfer data only to the connection that needs it.
- Another difference between a bridge and switch is that a switch does not convert data transmission formats



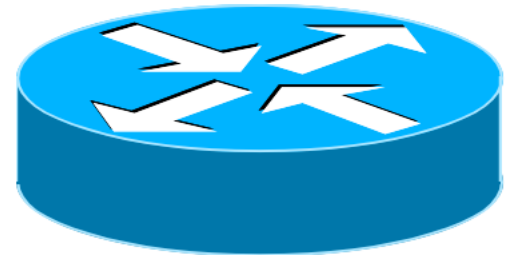
6. Routers

- Routers have all the capabilities listed above.
- Routers can regenerate signals, concentrate multiple connections, convert data transmission formats, and manage data transfers.
- They can also connect to a WAN, which allows them to connect LANs that are separated by great distances.



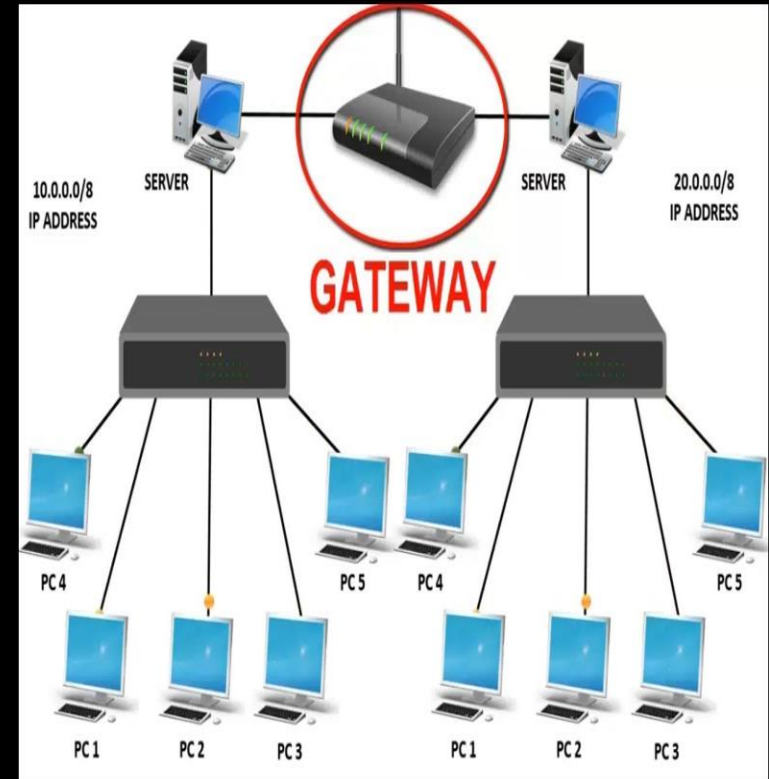
A day in a life of a router

- find path
- forward packet, forward packet, forward packet, forward packet...
- find alternate path
- forward packet, forward packet, forward packet, forward packet...
- repeat until powered off



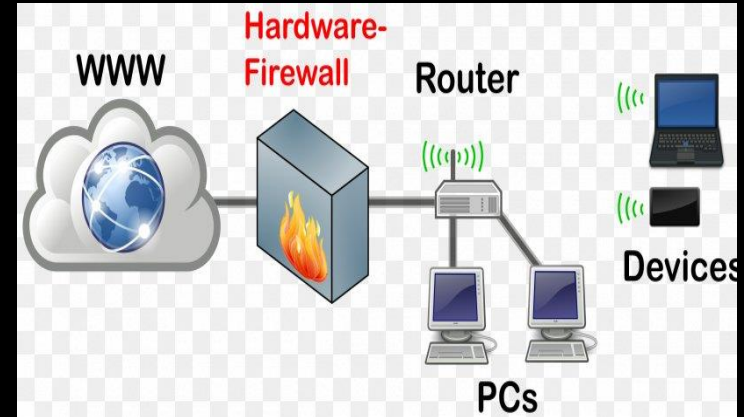
7. Gateway

- A **gateway** is a piece of networking hardware used in telecommunications for telecommunications networks that allows data to flow from one discrete network to another.
- Gateways are distinct from routers or switches in that they communicate using more than one protocol to connect a bunch of networks



8. Firewall

- A firewall is a network device or software for controlling network security and access rules.
- Firewalls are inserted in connections between secure internal networks and potentially insecure external networks such as the Internet.
- Firewalls are typically configured to reject access requests from unrecognized sources while allowing actions from recognized ones.
- The vital role firewalls play in network security grows in parallel with the constant increase in cyber attacks.



Network Media

The function of the media is to carry a flow of information through a LAN.

A. Wired Media:- widely adopted *family* that uses copper and fiber media in local area network (LAN) technology are collectively known as Ethernet

1. Copper Cable

- a. Coaxial Cables
- b. Shielded Twisted Pair(STP)
- c. Unshielded Twisted Pair

2. Fibre Optic Cable

B. Wireless Media:- use the atmosphere, or space, as the medium.

Fiber Optic Cable

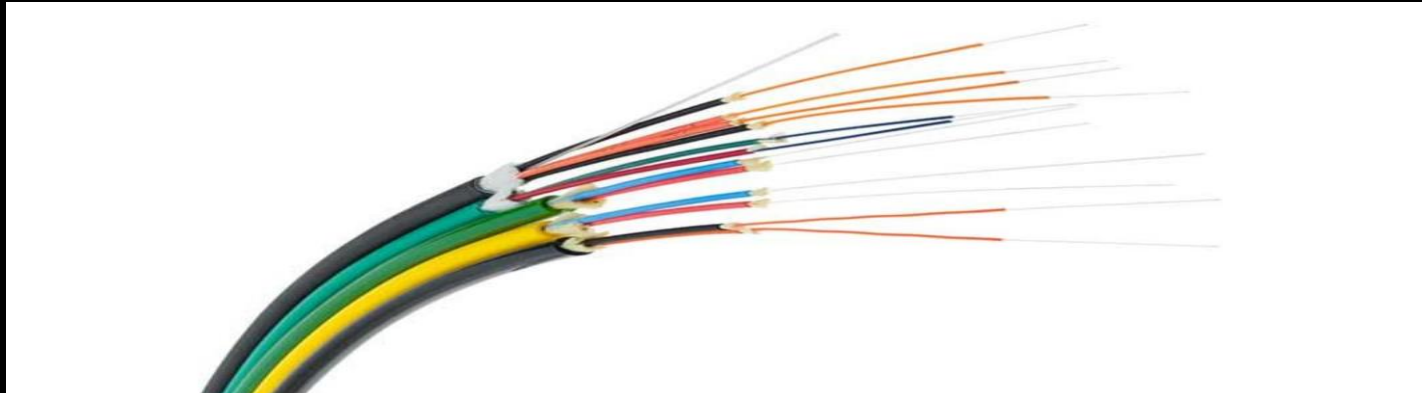
- Glass fiber carrying light pulses, each pulse a bit.
- Based on the Total Internal Reflection of Light.
- High-speed point-to-point transmission 10-100's Gbps
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



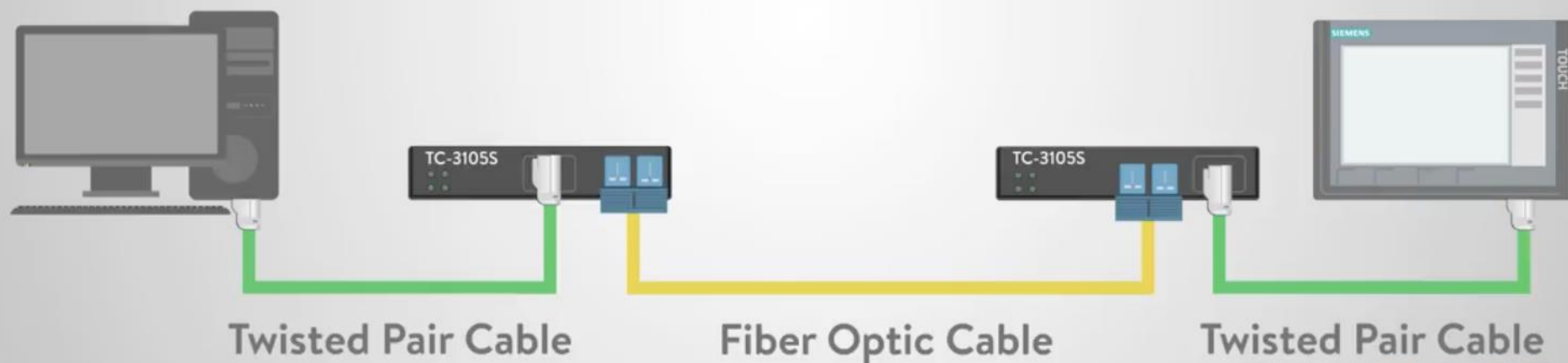
Fiber Optic works on the properties of light. When light ray hits at critical angle, it tends to refract at 90 degree. This property has been used in fiber optic. The core of fiber optic cable is made of high quality glass or plastic. From one end of it light is emitted, it travels through it and at the other end light detector detects light stream and converts it to electric data.

Fiber Optic provides the highest mode of speed. It comes in two modes, one is single mode fiber and second is multimode fiber. Single mode fiber can carry a single ray of light whereas multimode is capable of carrying multiple beams of light.

Fiber Optic also comes in unidirectional and bidirectional capabilities. To connect and access fiber optic special type of connectors are used. These can be Subscriber Channel (SC), Straight Tip (ST), or MT-RJ.



Physical Layer Cabling



1. Copper Cable

- The most common, easiest, quickest, and cheapest form of network media to install.
- The disadvantage of sending data over copper wire is that the further the signal travels, the weaker it becomes.

10BASE5 - "Thicknet"



10BASE2 - "Thinnet"

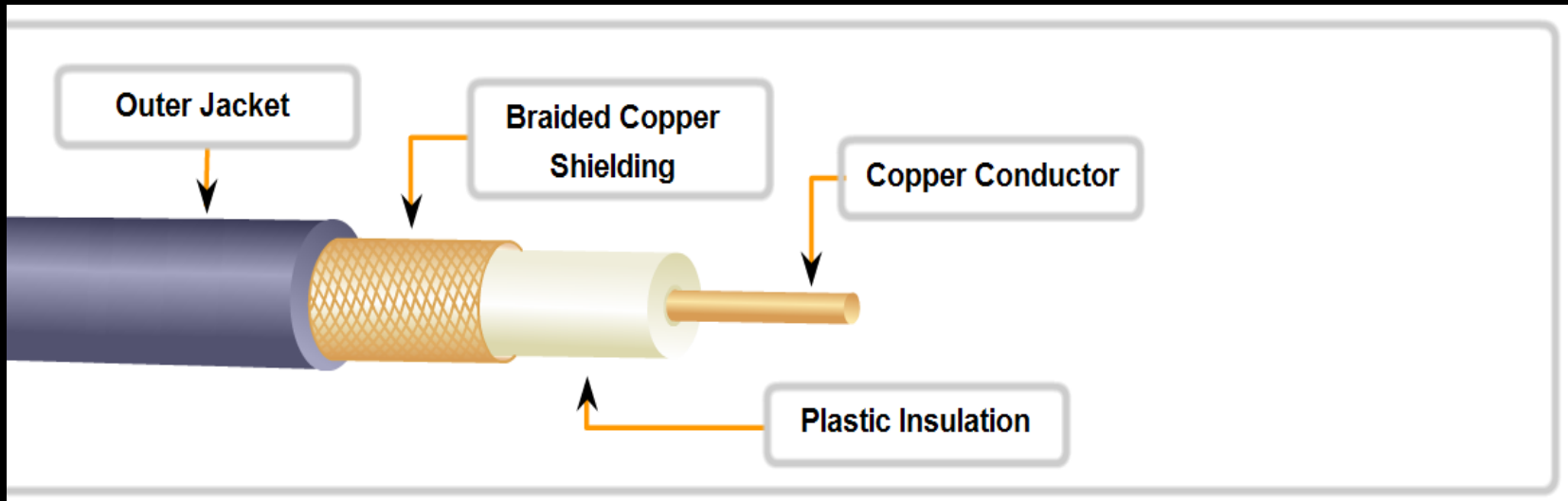


10BASE-T



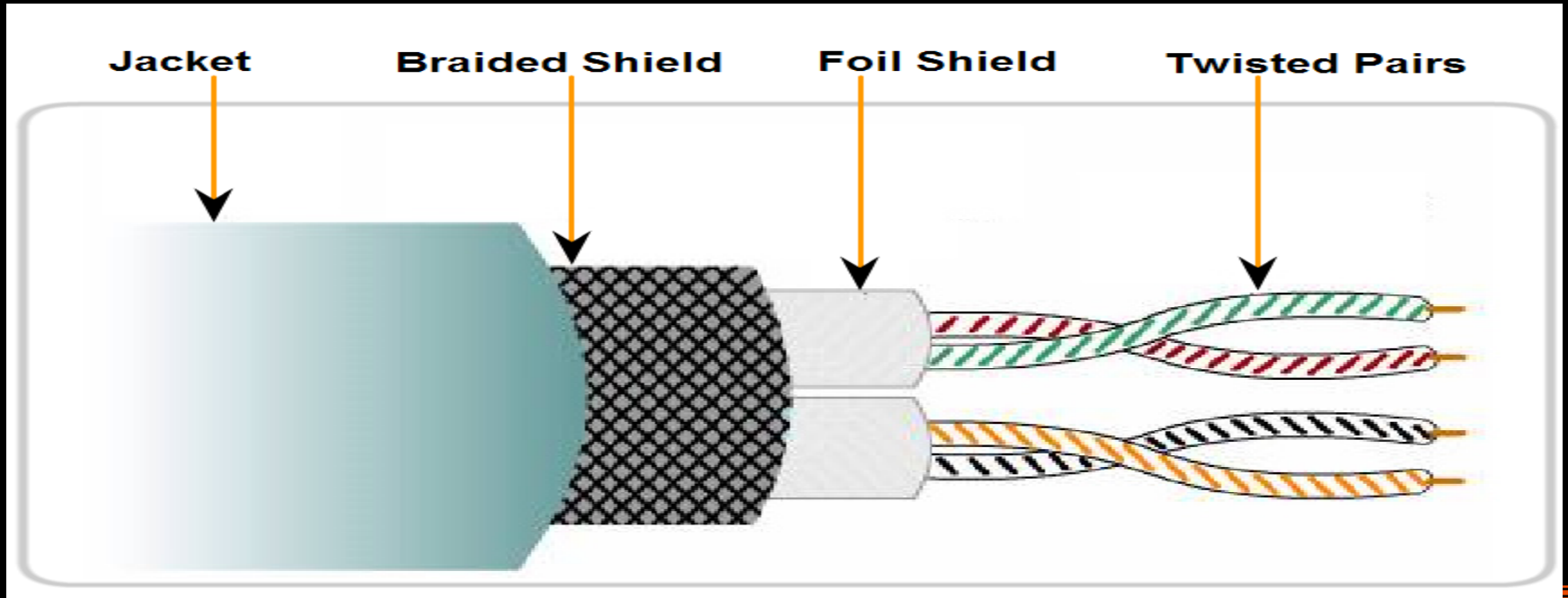
a. Coaxial Cable

- It can be run longer distances than Twisted pair Cables.
- Speed: 10-100Mbps
- Cost: Inexpensive
- Media and connector size: Medium
- Maximum cable length: 500m



b. Shielded Twisted Pair(STP)

- Speed: 0-100Mbps
- Cost: Moderate
- Media and connector size: Medium to large
- Maximum cable length: 100m



c. Unshielded Twisted Pair

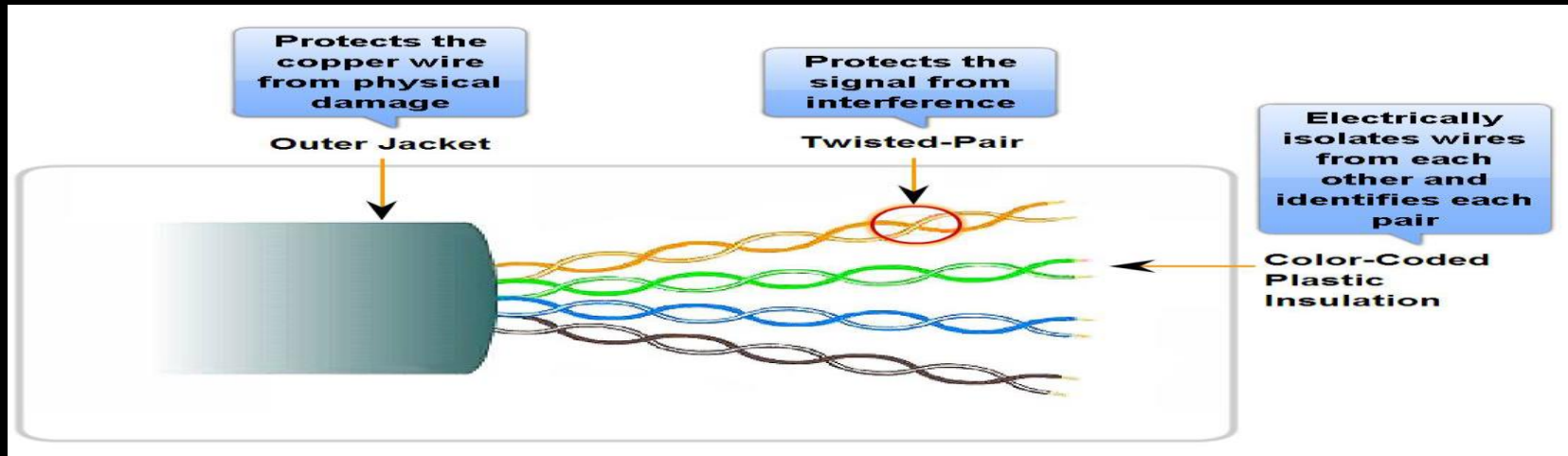
- UTP is a four-pair wire medium used in a variety of networks.
- Each of the eight copper wires in the UTP cable is covered by insulating material

Speed: 10-100-1000 Mbps*

Cost: Least Expensive

Media and connector size: Small

Maximum cable length: 100m * (Depending on the quality/category of cable)



UTP Implementation

- EIA/TIA specifies an RJ-45 connector for UTP cable.
- The letters RJ stand for registered jack.

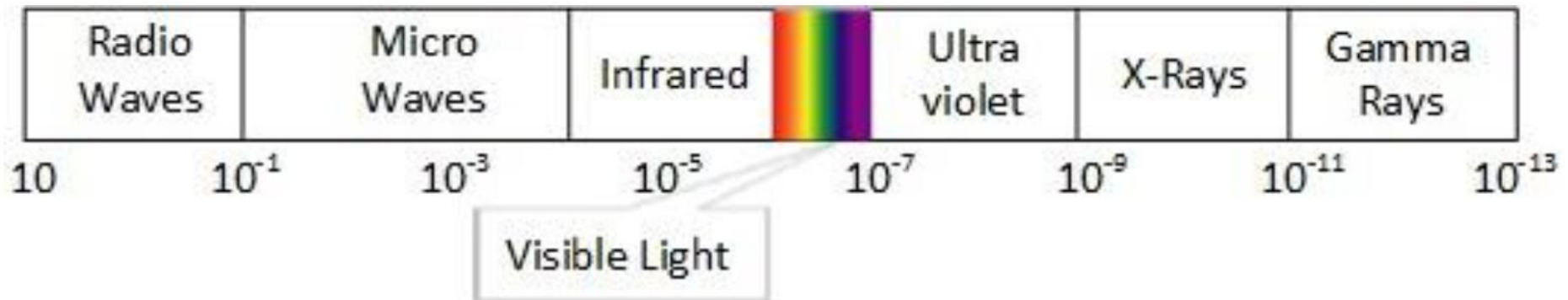


Wireless transmission is a form of unguided media. Wireless communication involves no physical link established between two or more devices, communicating wirelessly.

Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.

When an antenna is attached to electrical circuit of a computer or wireless device, it converts the digital data into wireless signals and spread all over within its frequency range. The receptor on the other end receives these signals and converts them back to digital data.

A little part of electromagnetic spectrum can be used for wireless transmission

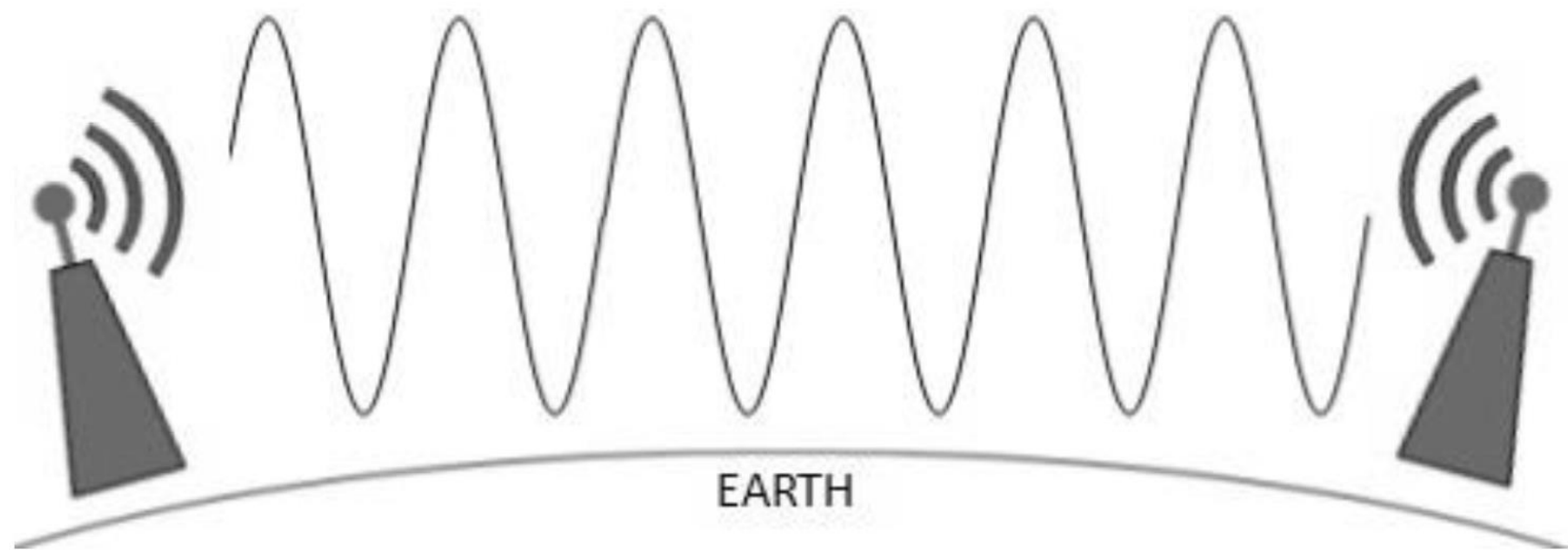


Radio Transmission

Radio frequency is easier to generate and because of its large wavelength it can penetrate through walls and structures alike. Radio waves can have wavelength from 1mm – 100,000km and have frequency ranging from 3Hz (Extremely Low Frequency) to 300 GHz (Extremely High Frequency). Radio frequencies are subdivided into six bands.

Radio waves at lower frequencies can travel through walls whereas higher RF can travel in straight line and bounce back. The power of low frequency waves decreases sharply as they cover long distance. High frequency radio waves have more power.

Lower frequencies such as VLF, LF, MF bands can travel on the ground up to 1000 kilometers, over the earth's surface

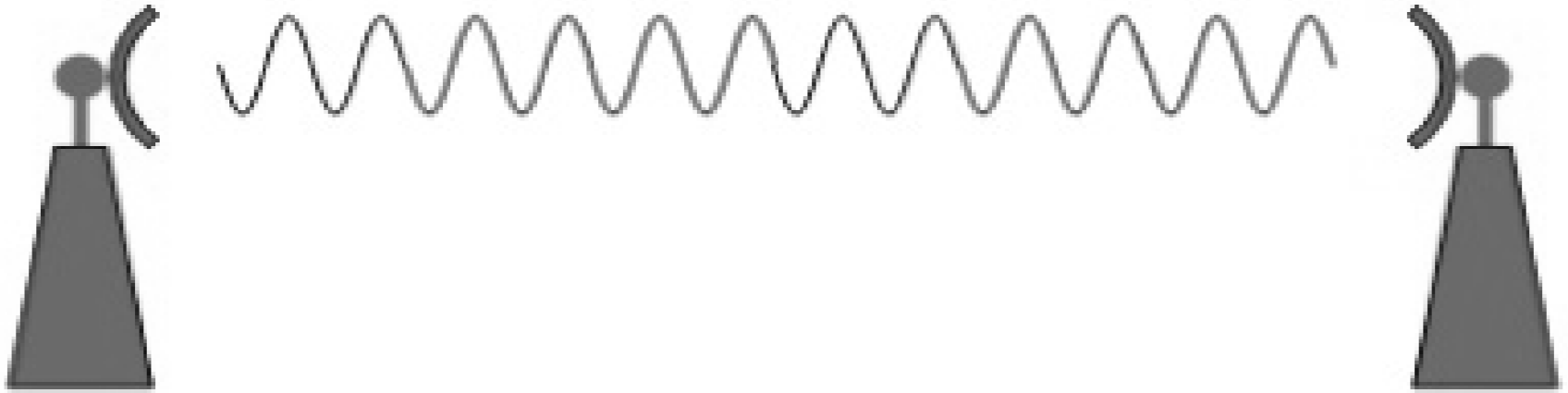


Microwave Transmission

Electromagnetic waves above 100MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station.

Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.

Microwaves can have wavelength ranging from 1mm – 1meter and frequency ranging from 300MHz to 300GHz



Communication Protocols

Internet Protocol Suite

- Also called TCP/IP, is the foundation of all modern networking.
- It defines the addressing, identification, and routing specifications for IPv4 and for IPv6.
- It is the defining set of protocols for the Internet.

IEEE 802

- It is a family of IEEE standards dealing with local area networks and metropolitan area networks.
- They operate mostly at levels 1 and 2 of the OSI model.

Ethernet

- It is a family of protocols used in wired LANs, described by a set of standards together called IEEE 802.3

Types of Networks

1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Campus Area Network (CAN)
4. Metropolitan Area Network (MAN)
5. Wide Area Network (WAN)
6. Storage-Area Network (SAN)
7. Virtual Private Network (VPN)
8. Client Server Network
9. Peer to Peer Network (P2P)

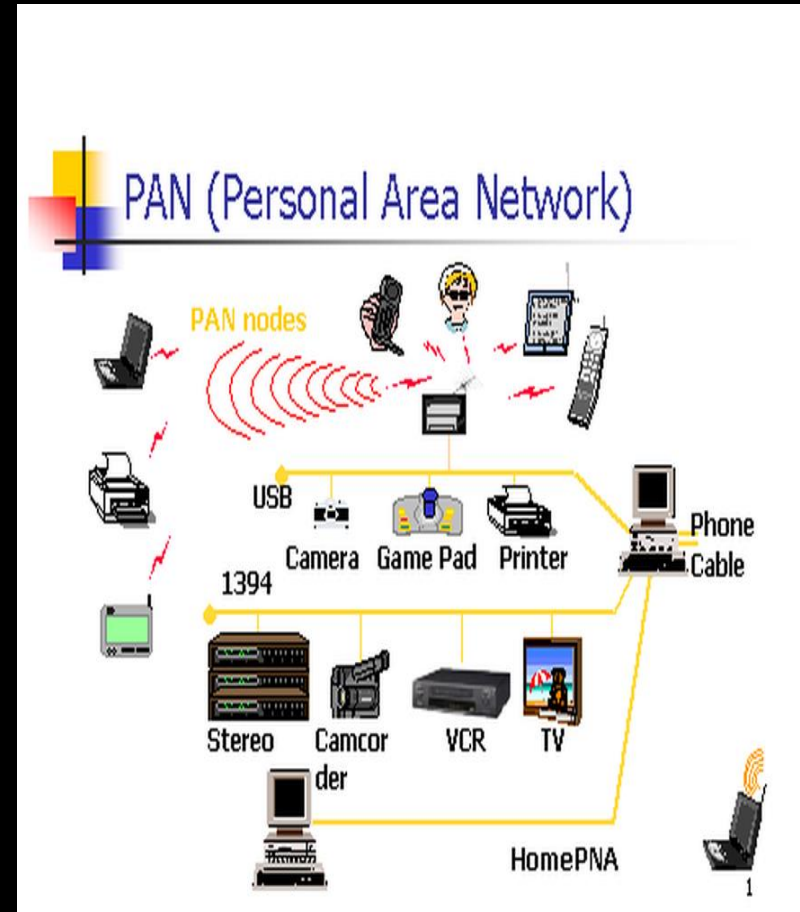
5. Wide Area Network



- A Wide Area Network exist over a large area
- Data travels through telephone or cable lines
- Usually requires a Modem
- The world's largest Wide Area Network in the Internet

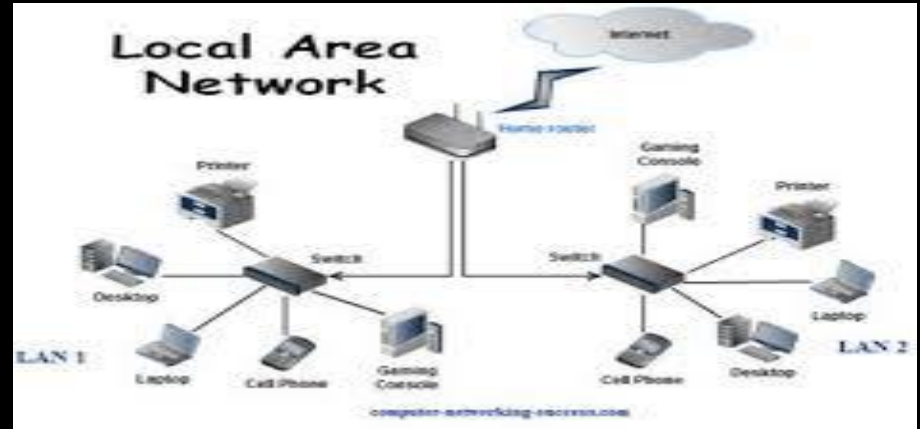
1. Personal Area Network

1. Personal Area Network (PAN) is a computer network used for data transmission amongst devices such as computers, telephones, tablets and personal digital assistants.
2. Also Known as HAN (Home Area Network)
3. PANs can be used for communication amongst the personal devices themselves (interpersonal communication), or for connecting to a higher level network and the Internet (an uplink) where one "master" device takes up the role as internet router.



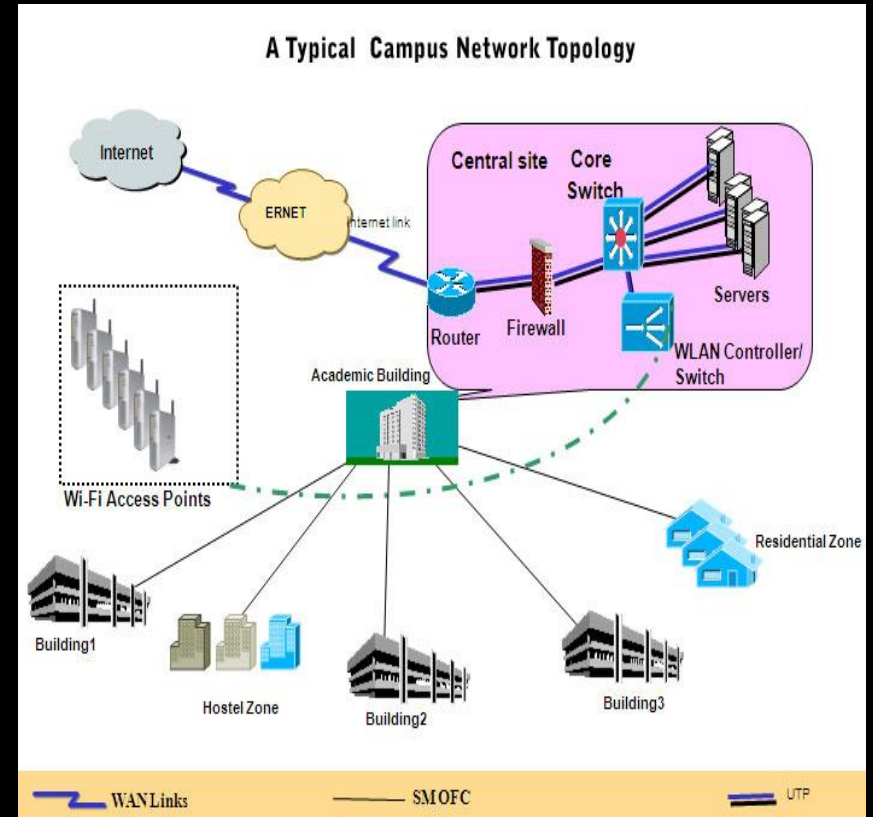
2. Local Area Network

- Xerox Corporation worked in collaboration with DEC and Intel to create Ethernet, which is the most pervasive LAN architecture used today.
- Ethernet has evolved and has seen significant improvements in regard to speed and efficiency.
- An upside of a LAN is fast data transfer with data speed that can reach up to 10Gbps.
- Other significant LAN technologies are Fiber Distributed Data Interface (FDDI) and token ring.

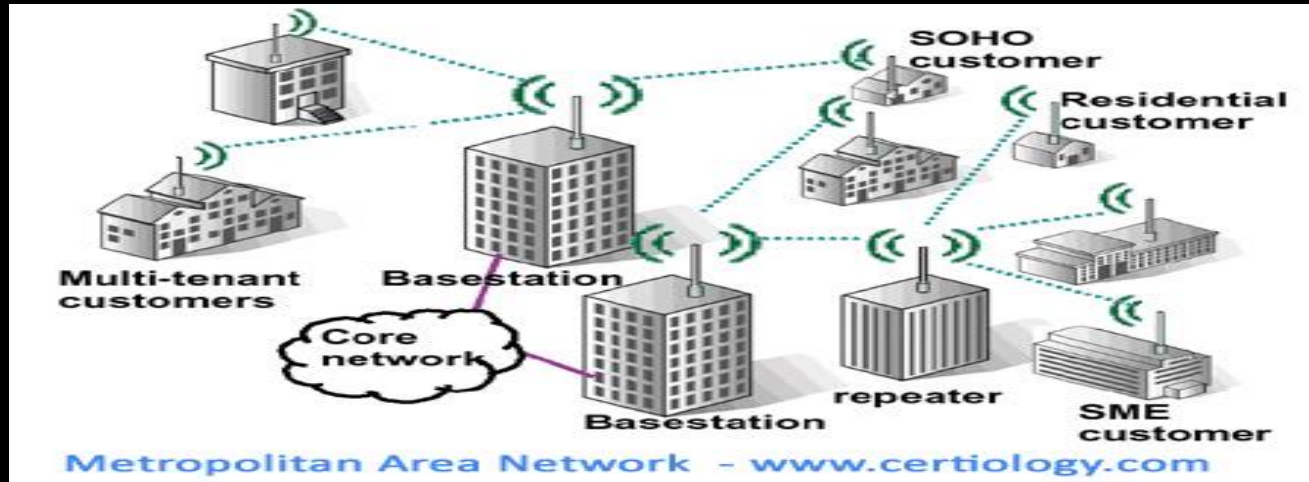


3. Campus Area Network

- Larger than LANs, but smaller than metropolitan area networks these types of networks are typically seen in universities, large K-12 school districts or small businesses.
- They can be spread across several buildings that are fairly close to each other so users can share resources

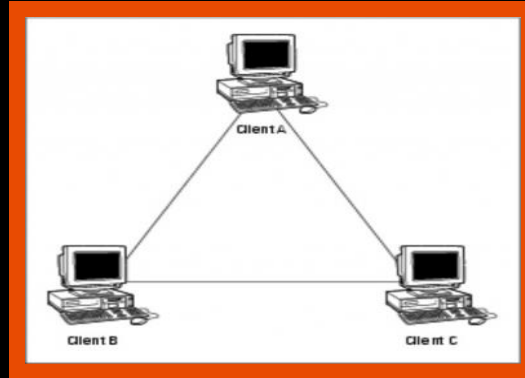


4. Metropolitan Area Network



1. A MAN is larger than a LAN but smaller than or equal in size to a WAN.
2. The size range anywhere from 5 to 50km in diameter.
3. MANs are typically owned and managed by a single entity.
4. This could be an ISP or telecommunications company that sells its services to end-users in that metropolitan area.
5. For all intents and purposes, a MAN has the same characteristics as a WAN with distance constraints.

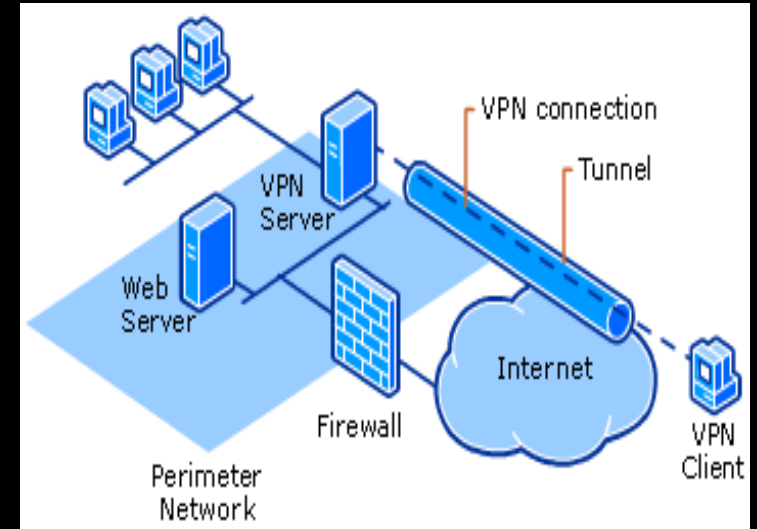
9. Peer to Peer Network



- Usually very small networks
- Each workstation has equivalent capabilities and responsibilities
- Does not require a switch or a hub.
- These types of networks do not perform well under heavy data loads.

7. Virtual Private Network

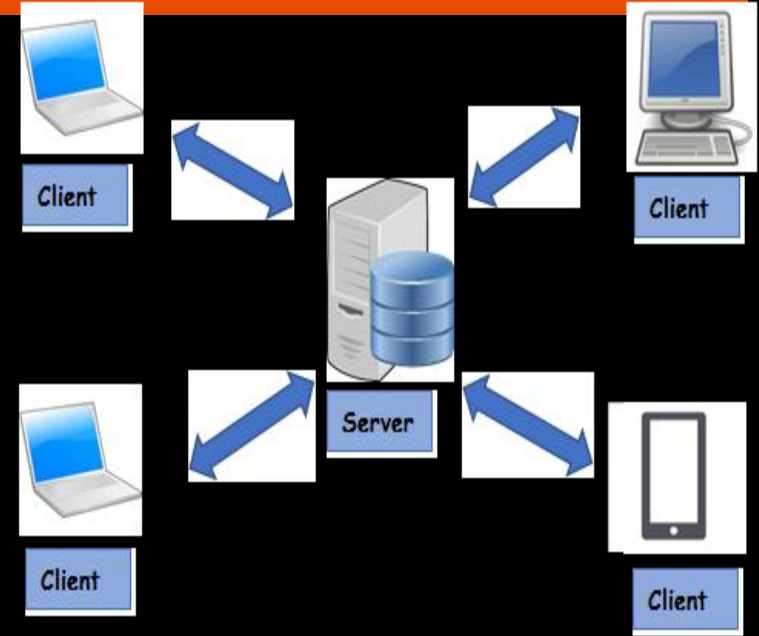
- VPN is a private network that can access public networks remotely. VPN uses encryption and security protocols to retain privacy while it accesses outside resources.
- When employed on a network, VPN enables an end user to create a virtual tunnel to a remote location. Typically, telecommuters use VPN to log in to their company networks from home.



- **Authentication** is provided to validate the identities of the two peers.
- **Confidentiality** provides encryption of the data to keep it private from prying eyes.
- **Integrity** is used to ensure that the data sent between the two devices or sites has not been tampered with.

8. Client/Server Network

- In a client/server arrangement, network services are located on a dedicated computer called a server.
- The server responds to the requests of clients.
- The server is a central computer that is continuously available to respond to requests from clients for file, print, application, and other services.
- Most network operating systems adopt the form of a client/server relationship.
- Typically, desktop computers function as clients, and one or more computers with additional processing power, memory, and specialized software function as servers.



Communication Protocols

Wireless LAN

- It is standardized by IEEE 802.11 and shares many properties with wired Ethernet.

SONET/SDH

- Synchronous optical networking (SONET) and Synchronous Digital Hierarchy (SDH) are standardized multiplexing protocols that transfer multiple digital bit streams over optical Fibre using lasers.

Asynchronous Transfer Mode(ATM)

- It uses asynchronous time-division multiplexing and encodes data into small, fixed-sized cells.
- Good choice for a network that handle both traditional high-throughput data traffic, and real-time, low-latency content such as voice and video.

Network Topologies

Network topology defines the structure of the network.

A. Physical topology:- It define the actual layout of the wire or media.

1. Bus
2. Ring
3. Star
4. Tree(Hierarchical)
5. Mesh

B. Logical topology:- It defines how the hosts access the media to send data.

1. Broadcast
2. Token passing

C. Hybrid Topology

A broadcast in a computer network is a message that is **transmitted to all participants in a network** and does not require a response. A computer in a network sends a data packet simultaneously to all other participants in the network. The sender does not have to specify any recipient addresses — this distinguishes the broadcast method from unicast, in which only a single, known recipient is addressed.

A broadcast is a multipoint connection in a computer network. In this case, a data packet is transmitted from one point to all participants in a message network. This is done using the broadcast address.

The general advantage of a broadcast is that **information can be widely distributed** and does not have to be transmitted multiple times. To implement the procedure, a special address is required that replaces the respective recipient addresses. This broadcast IP is mainly used when the addresses of the individual network participants are unknown

Basics of broadcast technology

Addressing is used to transfer data for communication in a network via the Internet protocol from one system to another. The **IP address** enables the unique forwarding and delivery of data packets from the source to the destination. Like a telephone number, this address can be subdivided into a prefix and a call number - except that IP addresses are named network and host parts.

This **subdivision is done with the help of the netmask**, which is placed over the IP address like a template. A netmask is just as long as an IPv4 address. It describes which bit positions within the IP address function as the network or host part.

MAC address (Media Access Control)

Each device that is integrated into a computer network requires a network adaptor. This adapter receives a worldwide unique identification number from the manufacturer: the MAC address. This enables devices like desktop computers, tablets, or mobile phones to be identified in the network and addressed as required.

The MAC address is the address of a network device. It is usually written as a hexadecimal number, such as *08-00-20-ae-fd-7e* or *08:00:20:ae:fd:7e* (or also: *080020aefdf7e*).

Broadcast address

The broadcast address sends the data and information to all devices in a network. The network components are responsible for receiving and processing the data. The task of the broadcast IP address is to **connect all network devices with each other**.

We provide more information on this in our detailed article on the [broadcast address](#)

How does a broadcast work?

A **connection setup via broadcast** is initiated by the sender. The sender sends their address, which the recipients can use to contact them. A broadcast therefore functions similarly to a mailing list, in which the **recipients are not visible** — and the sender does not need to know the addresses of the network participants. Only when the participants contact the sender do they reveal their own addresses. The network participants can open the data packets and then interpret the information, execute instructions, or reject them.

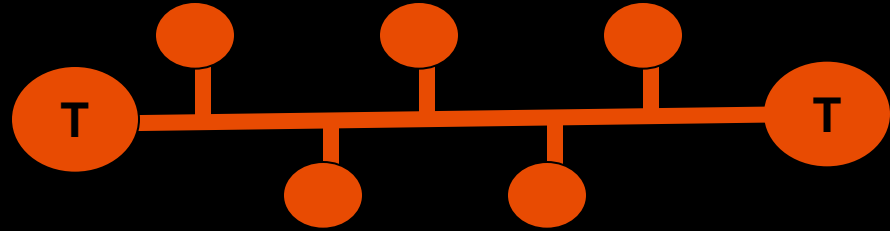
IP address

An IP address is a numerical designation. Each device connected to a **computer network based on Internet technology** is assigned an IP address. This IP address uses the Internet protocol for communication. An IP address performs two main tasks: Identifying host or network interfaces and addressing location.

1. Bus Topology

All devices are connected to a central cable, called bus or backbone.

There are terminators at each end of the bus that stops the signal and keeps it from traveling backwards.



Advantages:

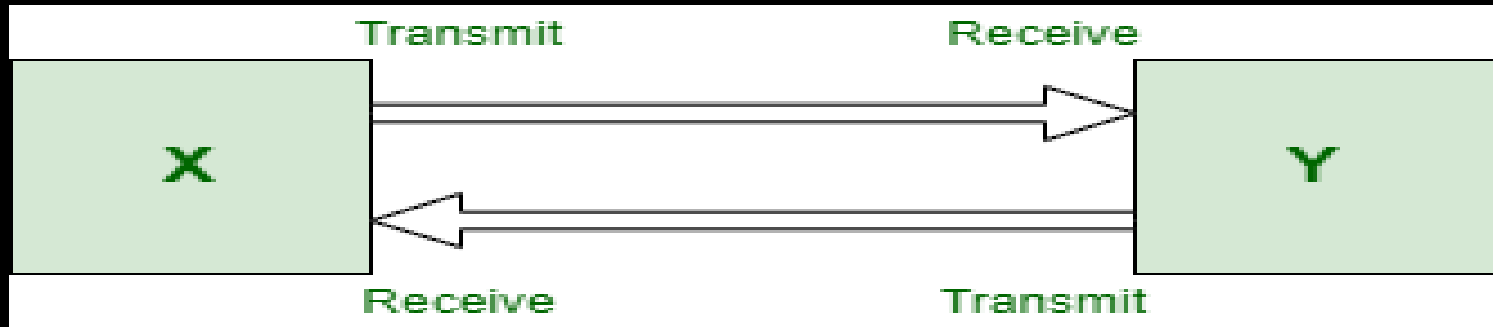
1. There is no central controller.
2. Control resides in each station
3. The less interconnecting wire is required.
4. Ease of installation.
5. Backbone cable can be laid along the most efficient path, and then connected to the nodes by drop lines of various lengths

Disadvantages:

1. It is possible that more than one station may attempt transmission simultaneously (collision or contention).
2. Difficult reconfiguration and fault isolation.
3. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem.
4. The damaged area reflects signals in the direction of origin, creating noise in both directions

Point to Point Topology

Point-to-Point Topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



2. Ring Topology

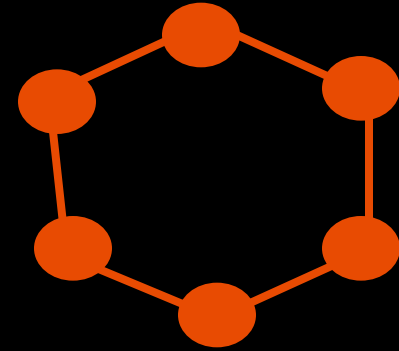
- All devices are connected to one another in the shape of a closed loop.
- Each device is connected directly to two other devices, one on either side of it.

Advantages:

1. Avoids the collisions that are possible in the bus topology.
2. Each pair of stations has a point-to-point connection.
3. A signal is passed along the ring in one direction, from device to another, until it reaches its destination.
4. Each device incorporates a repeater.
5. Relatively easy to install and reconfigure.
6. Fault isolation is simplified.

Disadvantages:

1. A break in the ring (such as station disabled) can disable the entire network.
2. Unidirectional traffic.



3. Star Topology

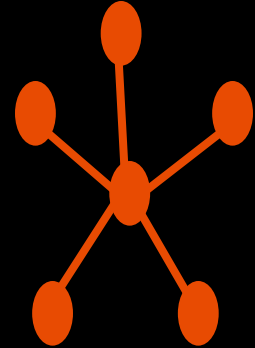
- All devices are connected to a central hub.
- Nodes communicate across the network by passing data through the hub or switch.

Advantages:

1. Easy to install and reconfigure.
2. Robustness, if one link fails; only that link is affected. All other links remain active.
3. Easy fault identification and isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

Disadvantages:

1. The devices are not linked to each other.
2. If one device wants to send data to another, it sends it to the controller, which then relays the data to the other connected device.



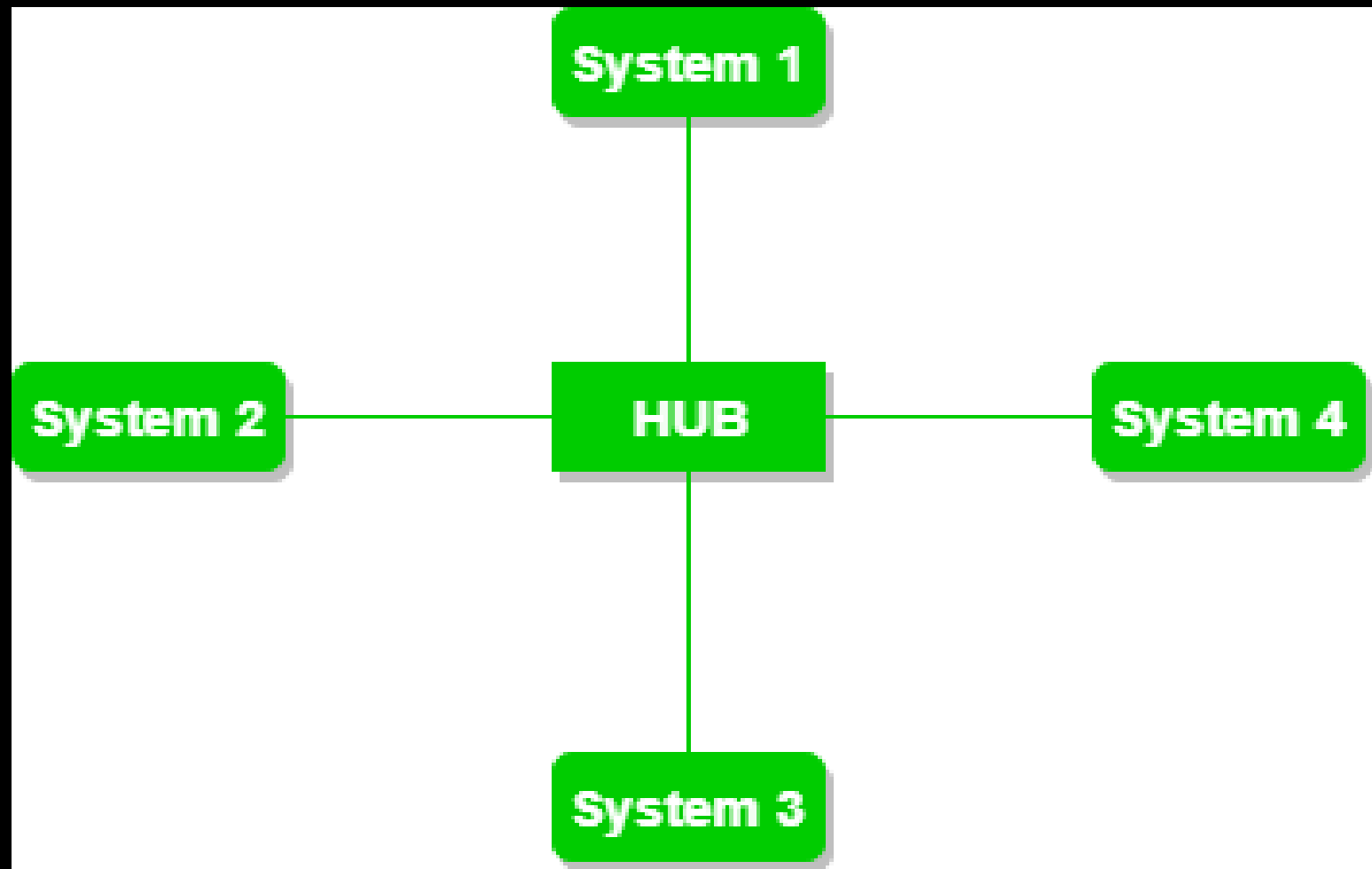


Figure 2: A star topology having four systems connected to a single point of connection i.e. hub.

Advantages of Star Topology

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N . So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Drawbacks of Star Topology

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

A common example of star topology is a local area network (LAN) in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.

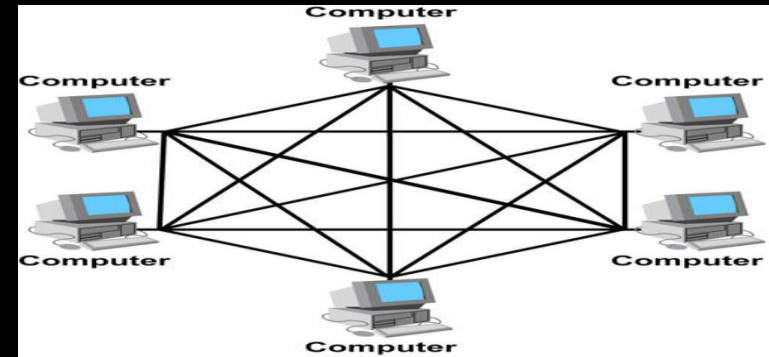
In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them.

Coaxial cables or RJ-45 cables are used to connect the computers. In Star Topology, many popular Ethernet LAN protocols are used as CD(Collision Detection), CSMA (Carrier Sense Multiple Access),

6. Mesh Topology

Each host has its connections to all other hosts. Mesh topology is implemented to provide as much protection as possible from interruption of service.

1. A nuclear power plant might use a mesh topology in the networked control systems.
2. Although the Internet has multiple paths to any one location, it does not adopt the full mesh topology.



Advantages:

1. The use of dedicated links guarantees that each connection can carry its data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. It is robust, if one link becomes unusable, it does not incapacitate (affect) the entire system.
3. Privacy and Security (every message sent travels along a dedicated line; only the intended recipient sees it).
4. Point-to-point links make fault identification and fault isolation easy.

Disadvantages:

1. A large amount of cabling required.
2. A large amount of I/O ports required.
3. Installation and reconfiguration are difficult.
4. The sheer bulk of the wiring can be greater than the available space (in the walls, ceiling, or floors) can accommodate.
5. The hardware required to connect each link (I/O ports and cables) can be prohibitively expensive.

Note:

1. Hub, Repeater, Modem, and Cables are Physical Layer devices.
2. Network Layer, Data Link Layer, and Physical Layer are also known as **Lower Layers** or **Hardware Layers**.

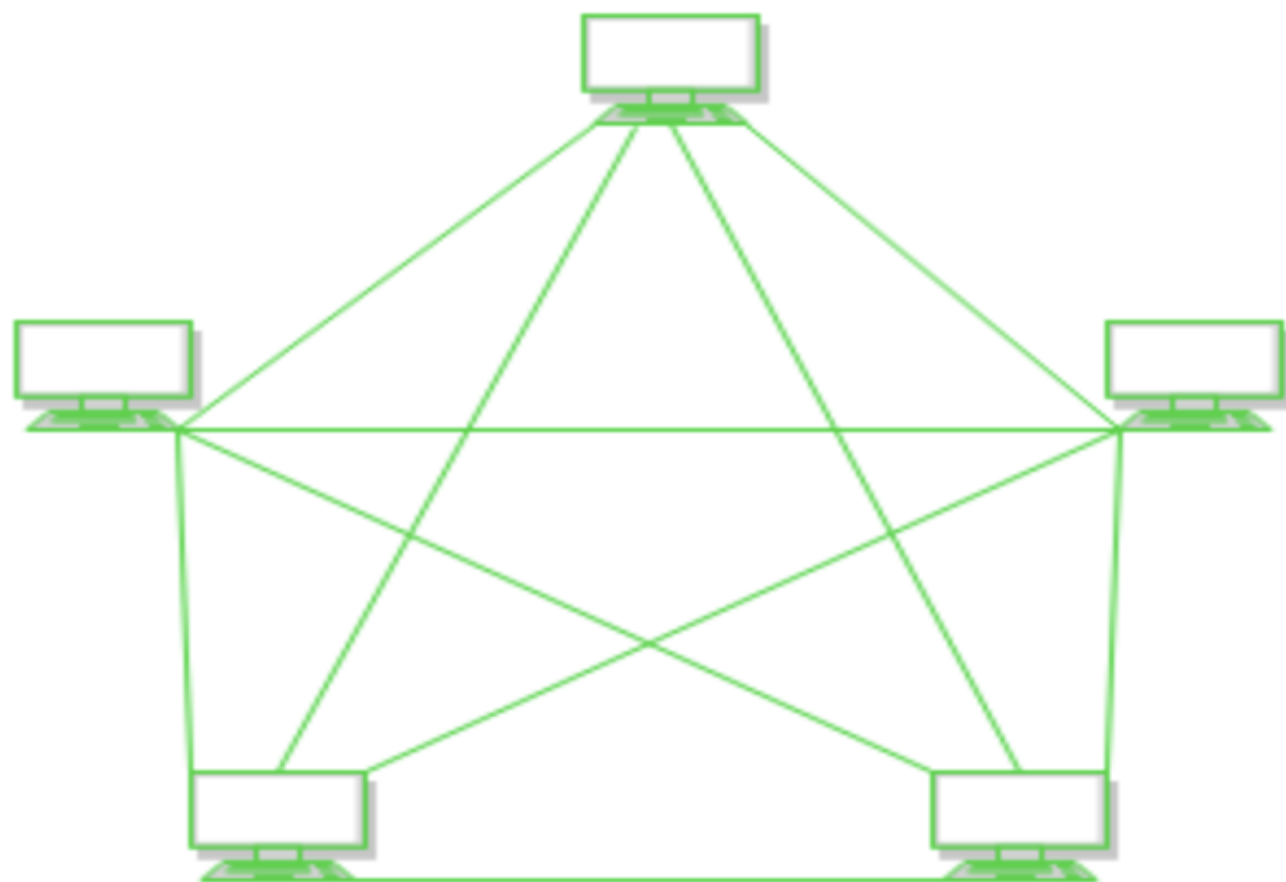
Advantages of Mesh Topology

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Drawbacks of Mesh Topology

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

A common example of mesh topology is the internet backbone, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.



Mesh Topology

Figure 1: Every device is connected to another via dedicated channels. These channels are known as links.

- Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is $N-1$. In Figure 1, there are 5 devices connected to each other, hence the total number of ports required by each device is 4. The total number of ports required $= N * (N-1)$.

- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is ${}^N C_2$ i.e. $N(N-1)/2$. In Figure 1, there are 5 devices connected to each other, hence the total number of links required is $5*4/2 = 10$.

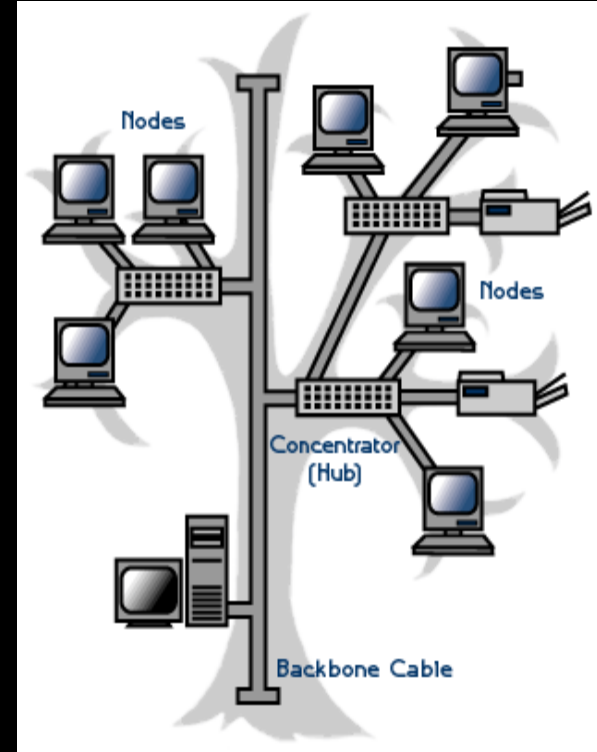
4. Tree/Hierarchical Topology

Advantages:

1. It allows more devices to be attached to a single central hub and can therefore increase the distance a signal can travel between devices.
2. It allows the network to isolate and prioritize communications from different computers.

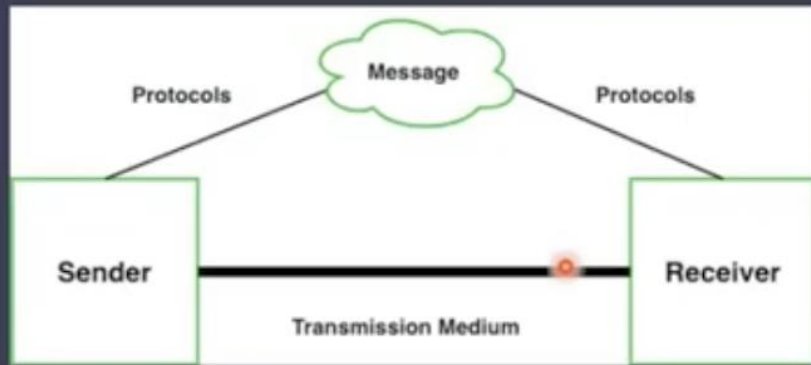
Disadvantages:

1. The devices are not linked to each other.
2. If one device wants to send data to another, it sends it to the controller, which then relays the data to the other connected device.
3. The addition of secondary hubs brings two further advantages.



What is Protocol

A protocol is a set of rules and guidelines for communicating data. Rules are defined for each step and process during communication between two or more computers. Networks have to follow these rules to successfully transmit data



Key Element of Protocol

- **Syntax** : syntax refers to the structure or the format of the data that gets exchanged between the devices. Syntax of message includes the type of data, composition of message and sequencing of message. The starting 8 bits of data is considered as the address of the sender. The next 8 bits is considered to be the address of the receiver. The remaining bits are considered as the message itself.
- **Semantics** : Semantics defines data transmitted between devices. It provides rules and norms for understanding message or data element values and actions.
- **Timing** : Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing data loss, collisions and other timing related issues.

- **Sequence control** : Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it gets received, and the retransmission of lost data. Through this mechanism the data is delivered in correct order.
- **Flow Control** : Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.
- **Error Control** : Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.
- **Security** : Network security safeguards data confidentiality, integrity, and authenticity, which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

Protocols

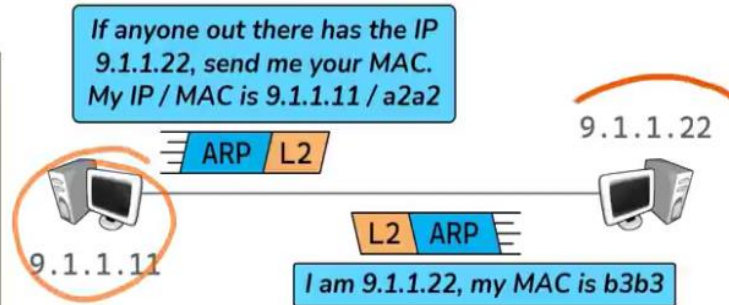
- Set of **rules** and **messages** that form an **Internet standard**
 - **ARP** – Address Resolution Protocol – RFC 826
 - Resolves IP to MAC mappings
 - ARP Requests / ARP Responses

RFC 826:

An Ethernet Address Resolution Protocol

Abstract

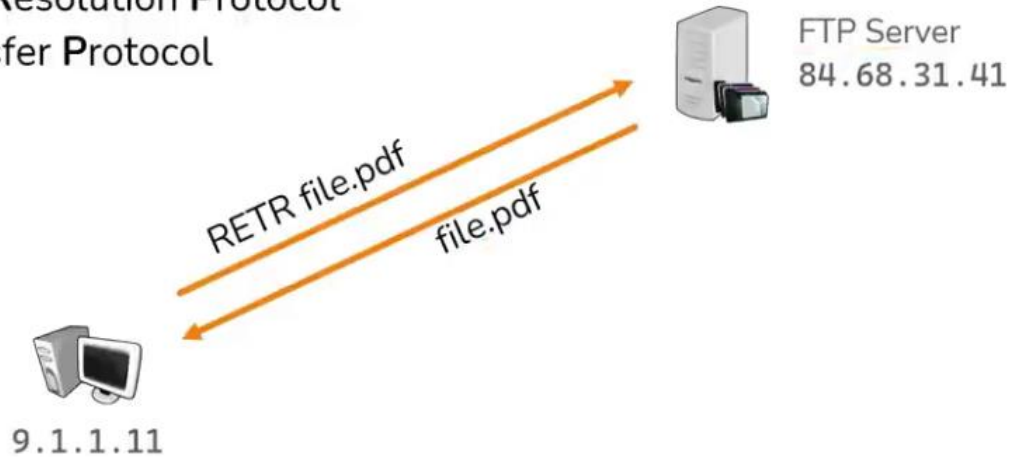
Presented here is a protocol that allows dynamic distribution of the information needed to build tables to translate an address in protocol P's address space into a 48.bit Ethernet address.



Protocols

- Set of **rules** and **messages** that form an **Internet standard**

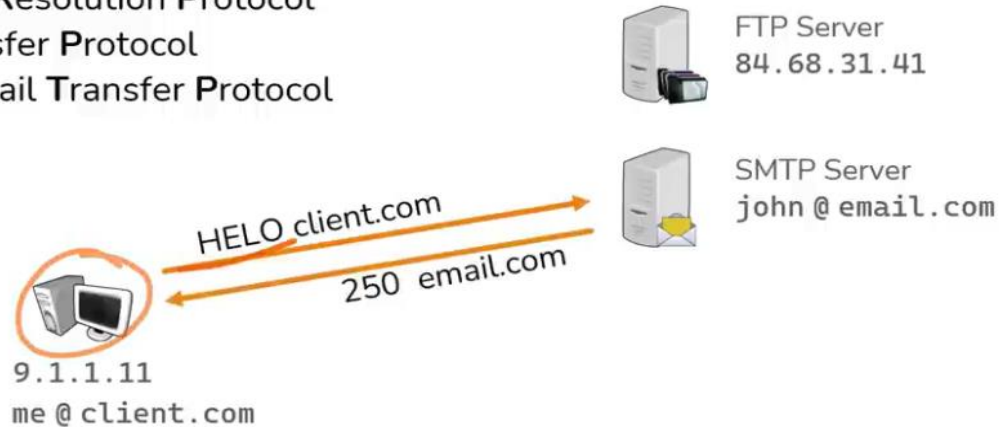
- ARP – Address Resolution Protocol
- FTP – File Transfer Protocol



Protocols

- Set of **rules** and **messages** that form an **Internet standard**

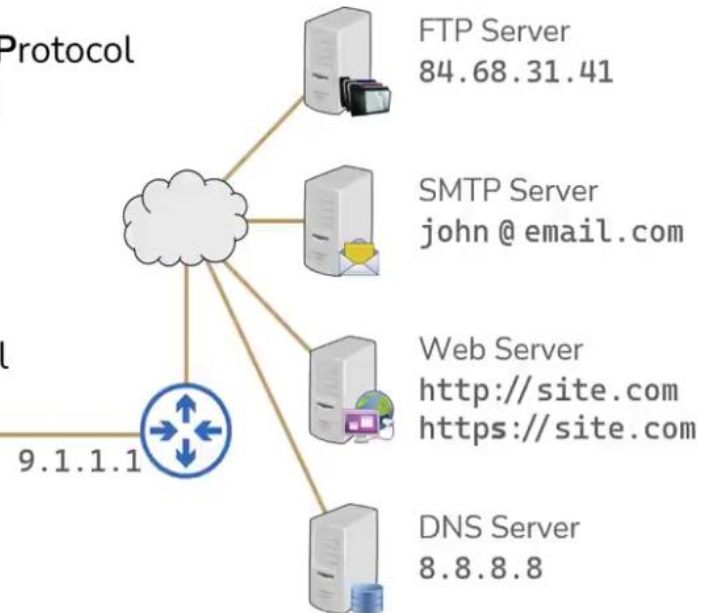
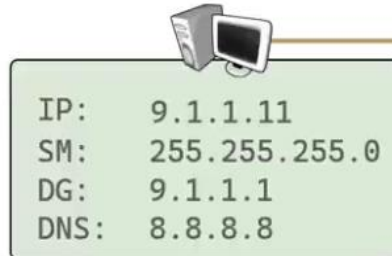
- **ARP** – Address Resolution Protocol
- **FTP** – File Transfer Protocol
- **SMTP** – Simple Mail Transfer Protocol



Protocols

- Set of **rules** and **messages** that form an **Internet standard**

- **DNS** – **D**omain **N**ame **S**ystem
- **DHCP** – **D**ynamic **H**ost **C**onfiguration **P**rotocol
- **HTTP** – **H**yper **T**ext **T**ransfer **P**rotocol
- **SSL** – **S**ecure **S**ockets **L**ayer
- **TLS** – **T**ransport **L**ayer **S**ecurity
- **HTTPS** – **HTTP** secured with **SSL/TLS**
- **FTP** – **F**ile **T**ransfer **P**rotocol
- **SMTP** – **S**imple **M**ail **T**ransfer **P**rotocol



Network Devices

Host

IP Address

Network

Repeater

Hub

Bridge

Switch

Router

Networking Fundamentals
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- **Hosts** are any device which **sends** or **receive** traffic
 - **Clients** and **Servers**



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- Client initiate requests, Servers respond
 - Relative to specific communication



Update Server

Server



File Server

Client



Difference between Server and Mainframe

1. Server :

Server, as name suggests, is basically computer or PC by hardware with higher reliability or quality parts that is connected to local area network, wide area network and internet. It simply accesses, hold and transfer all data and files from other computer machine over computer network.

Examples : Linux, NetWare , Web server, etc.

2. Mainframe :

Mainframe, as name suggests, is a class of computer that handle and control large number of connected users, high volume of transactions, and provide more reliable performance. It works well with running scalable software and dealing with massively parallel operations. It is considered the best platform to execute millions of transactions in every single second.

Examples : IBM zSeries , System z9, System z10, NonStop , etc.

<i>Server</i>	<i>Mainframe</i>
It controls and handle access to hardware, software, and other resources on network.	It controls and handle hundred and thousands of connected users simultaneously.
It provides centralized storage for programs, data, and information.	It provides storage for instructions, information, and huge amount of data.
These are small in size as compared to mainframe therefore require less floor space.	These are huge in size as compared to server therefore requires more floor space.
It is more cost effective as compared to mainframes.	It is more costly as compared to servers as its initial cost, cost of running, and maintenance cost is very high.
It provides various functionalities such as sharing data or resources among users, performing computation for client.	It provides e-commerce and mobile world more reliable, available and scalable way to perform their basic business functions.
They are designed to not only help business with data storage but also to improve efficiency and productivity.	These computers are designed to handle large-scale processing, data storage and other tasks.
It cannot handle huge volume of transactions, high reliability and different types of workloads.	It supports and handles huge volume of transactions, high reliability and support different types of workloads.
It is used in digital applications such as Big data, Edge, Cloud applications and <u>IoT</u> applications, etc.	It is used in Banking applications, high volume OLTP (Online Transaction Processing) applications, back-end systems, etc.
Cloud options are limitless and available in huge quantity.	Cloud options are limited in mainframes.

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www.site.com

- Client initiate requests, Servers respond
 - Relative to specific communication
- Servers are simply computers with software installed which responds to specific requests

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Networking Fundamentals
(C) Practical Networking .net

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 - Clients and Servers



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Client

- Client initiate requests Servers respond
 - Relative to specific communication



File Server

Server

Network Devices

Host

IP Address

Network

Repeater

Hub

Bridge

Switch

Router

- **Hosts** are any device which **sends** or **receive** traffic



Computer



Laptop



Phones



Printers



Servers



Cloud Server

- Also any **Internet of Things (IoT)** devices

- TV
- Speakers
- Smart Watches
- Thermometers
- Lights
- Refrigerator

- An **IP Address** is the identity of each host

- IP addresses are 32 bits
 - Bit = 1 or 0
 - Represented as four Octets

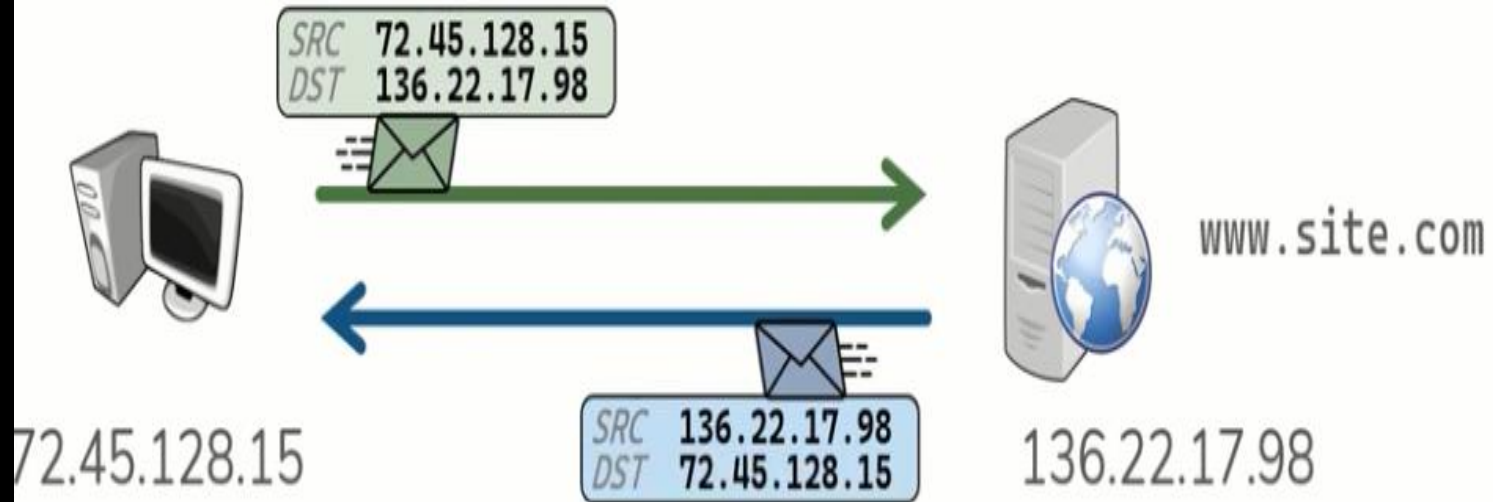


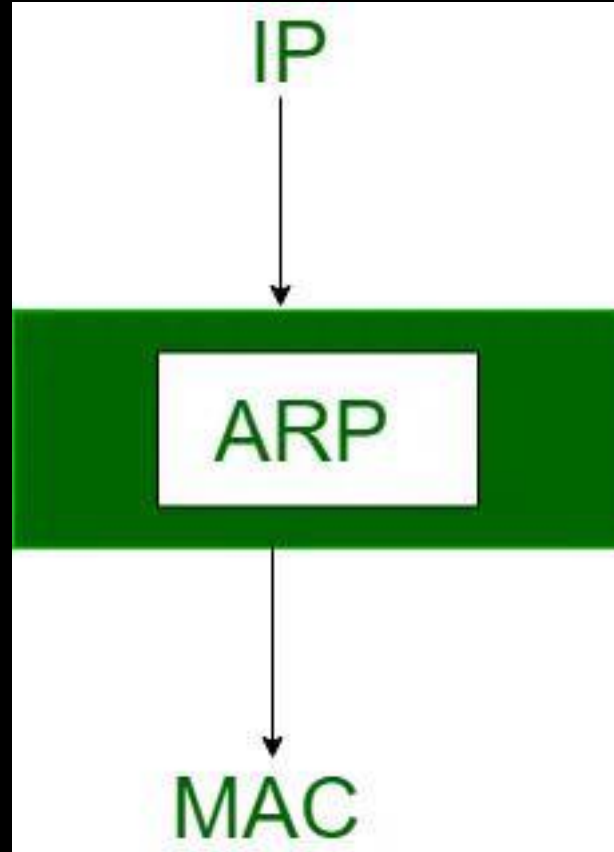
136.22.17.98

1 0 0 0 1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 1 0 1 1 0 0 0 1 0

1000 1000 . 0001 0110 . 0001 0001 . 0110 0010

- An **IP Address** is the identity of each host





DNS Server: DNS stands for **Domain Name System**. DNS is basically a server that translates web addresses or URLs (ex: `www.google.com`) into their corresponding IP addresses. We don't have to remember all the IP addresses of each and every website. The command '`nslookup`' gives you the IP address of the domain you are looking for. This also provides information on our DNS Server.

S.NO	MAC Address	IP Address
1.	MAC Address stands for Media Access Control Address.	IP Address stands for Internet Protocol Address.
2.	MAC Address is a six byte hexadecimal address.	IP Address is either a four-byte (IPv4) or a sixteen-byte (IPv6) address.
3.	A device attached with MAC Address can retrieve by ARP protocol.	A device attached with IP Address can retrieve by RARP protocol.
4.	<u>NIC</u> Card's Manufacturer provides the MAC Address.	Internet Service Provider provides IP Address.
5.	MAC Address is used to ensure the physical address of a computer.	IP Address is the logical address of the computer.
6.	MAC Address operates in the data link layer.	IP Address operates in the network layer.
7.	MAC Address helps in simply identifying the device.	IP Address identifies the connection of the device on the network.
8.	MAC Address of computer cannot be changed with time and environment.	IP Address modifies with the time and environment.
9.	MAC Addresses can't be found easily by a third party.	IP Addresses can be found by a third party.

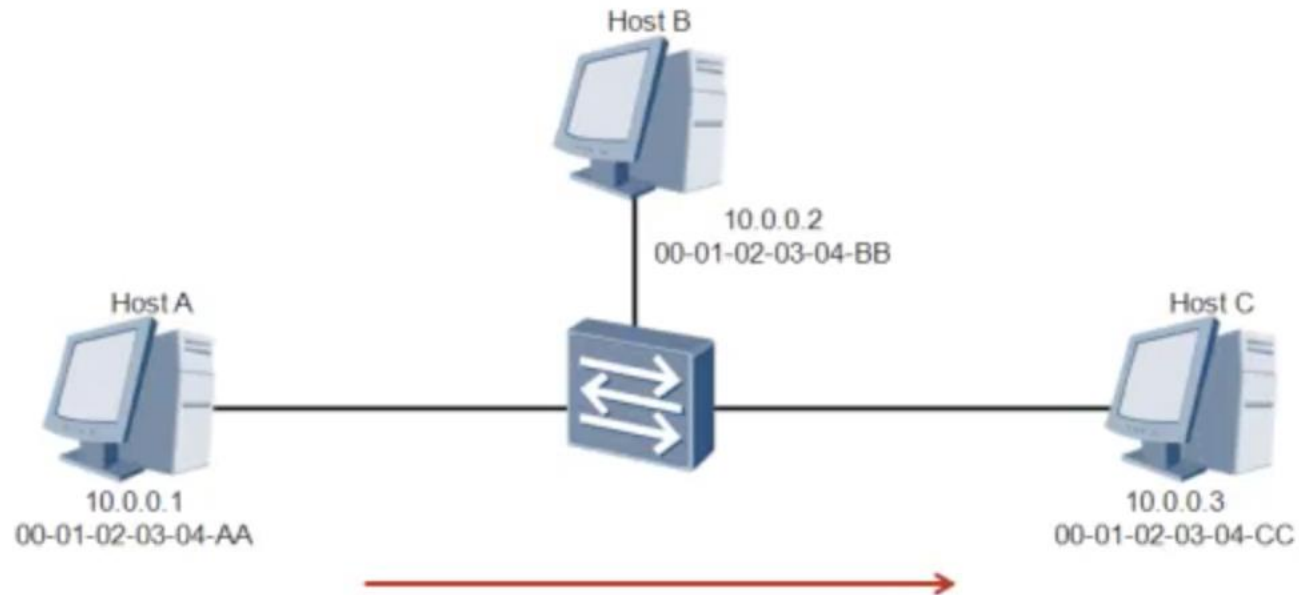
S.NO	MAC Address	IP Address
10.	It is a 48-bit address that contains 6 groups of 2 hexadecimal digits, separated by either hyphens (-) or colons(.). Example: <u>00:FF:FF:AB:BB:AA</u> or 00-FF-FF-AB-BB-AA	IPv4 uses 32-bit addresses in dotted notations, whereas IPv6 uses 128-bit addresses in hexadecimal notations. Example: IPv4 192.168.1.1 IPv6 FFFF:F200:3204:0B00
11.	No classes are used for MAC addressing.	IPv4 uses A, B, C, D, and E classes for IP addressing.
12.	MAC Address sharing is not allowed.	In IP address multiple client devices can share the IP address.
13.	MAC address help to solve IP address issue.	IP addresses never able to solve MAC address issues.
14.	MAC addresses can be used for broadcasting.	The IP address can be used for broadcasting or multicasting.
15.	MAC address is hardware oriented.	IP address is software oriented.
16.	While communication, Switch needs MAC address to forward data.	While communication, Router need IP address to forward data.

Most computer programs/applications use logical addresses (IP Addresses) to send/receive messages. However, the actual communication happens over the **Physical Address (MAC Address)** that is from layer 2 of the OSI model. So our mission is to get the destination MAC Address which helps in communicating with other devices. This is where ARP comes into the picture, its functionality is to translate IP addresses to Physical Addresses.

Address Resolution Protocol (ARP) which is one of the most important protocols of the Data link layer in the OSI model. It is responsible to find the hardware address of a host from a known IP address. There are three basic ARP terms.

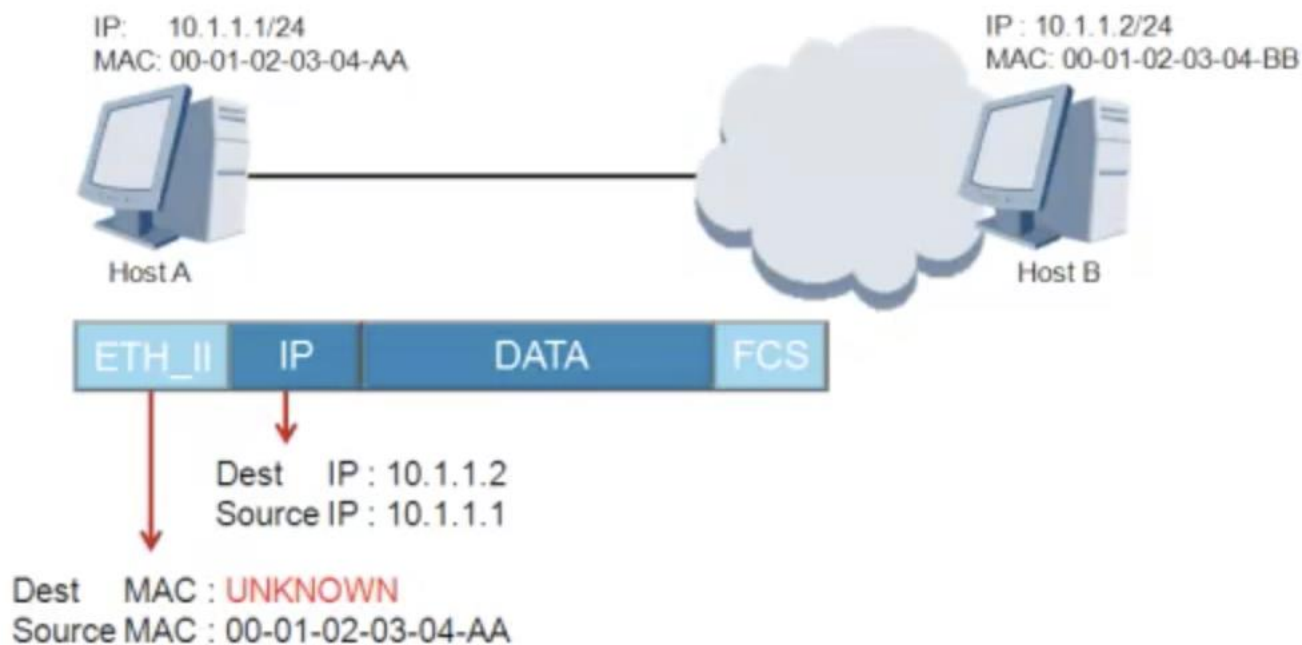
Note: ARP finds the hardware address, also known as the Media Access Control (MAC) address, of a host from its known IP address.

ARP Process



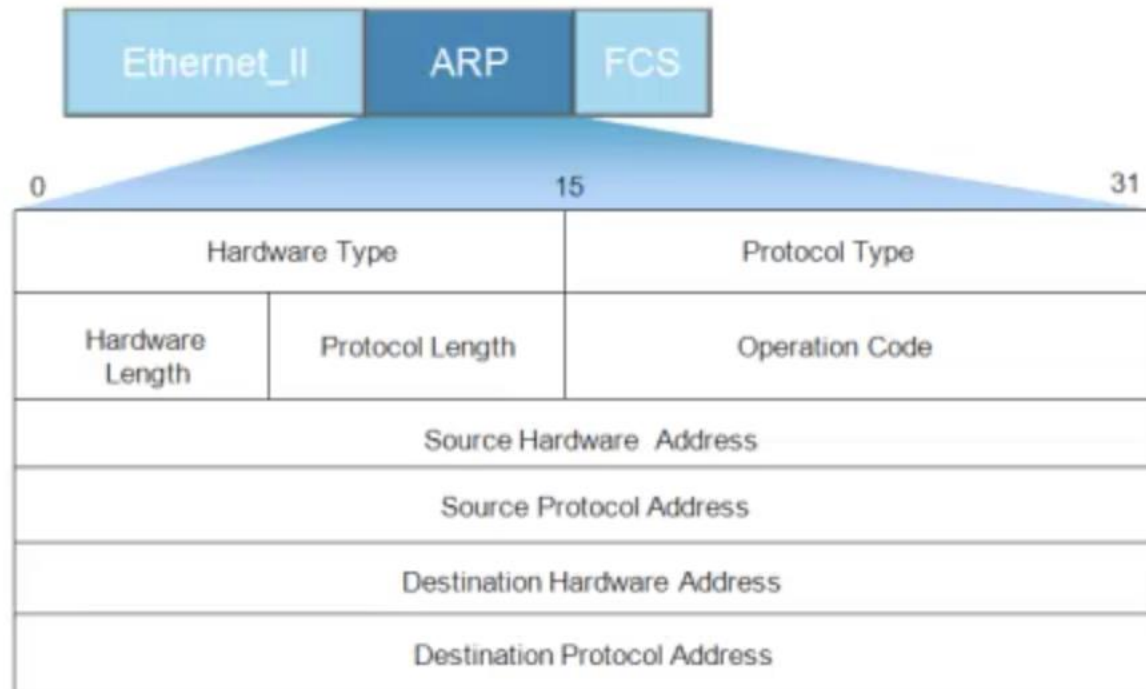
- Host A wishes to forward data to Host C, but must identify whether it is able to reach the destination at the data link layer.

ARP



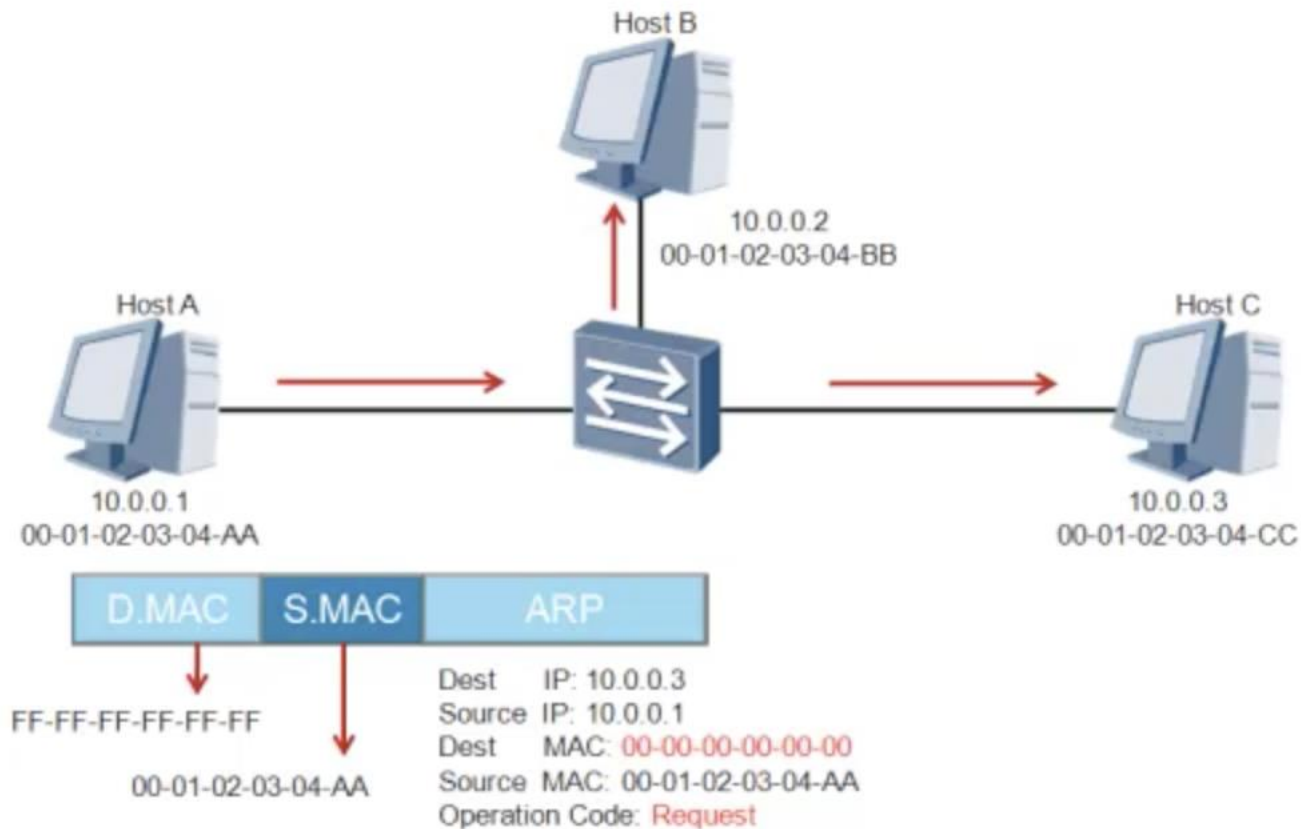
- Data link forwarding relies on knowledge of the MAC address of the data link layer destination.

ARP Format



- The ARP packet operates within the boundaries of the data link layer, as can be understood by the absence of an IP header.

ARP Request Process



Bandwidth, or precisely **network bandwidth**, is the maximum rate at which data transfer occurs across any particular path of the network. Bandwidth is basically a measure of the amount of data that can be sent and received at any instance of time. That simply means that the higher the bandwidth of a network, the larger the amount of data the network can be sending to and from across its path. Do not to confuse bandwidth with closely related terms such as the [data](#)

bps (Bits per second)

Mbps (Megabits per second)

Gbps (Gigabits per second)

Example: Here, a bandwidth of 10 bps for a channel, is just another way of saying that a maximum of 10 bits can be transferred using that link for any given time. It has no relation with the transfer speed of the channel.

FTP (File Transfer Protocol)

It is a protocol that is used to transfer or copy the file from one host to another host. But there may be some problems like different file names and different file directories while sending and receiving a files in different hosts or systems. And in FTP, a secure channel is not provided to transfer the files between the hosts or systems. It is used in port no-21.

FTP is used for transferring of web pages and used for downloading the files from other different servers. Basically, it is used for transferring the files from one system to another system more reliably and efficiently.

PROS-

- Speedy data transfer
- Transfer multiple files and directories
- Multitasking (transferring and downloading can go in parallel)
- Continuous Transfer i.e. transfers can be resumed from a given point
- Many FTP clients have scripting features.

•CONS

- Few FTP providers do not offer encryption
- Possibility of Eavesdropping because of the plain text used for sensitive information such as password, etc.
- When sending files via FTP, compliance can be a problem.

SFTP(Secure File Transfer Protocol)

It is a protocol that provides a secure channel, to transfer or copy the file from one host to another host or system. SFTP establishes the control connection under SSH protocol and It is used in port no-22.

Basically, it is a protocol for transferring large files over the web. It could handle sensitive data securely in all situations. So, a user who has a business can successfully transmit files containing secret information by using SFTP.

Advantages-

- Encryption
- Host Authentication
- Compliance
- Only one connection is used as DATA connection is not required in SFTP.

Disadvantages-

- SSH keys are difficult to maintain and verify.
- Difficult to login because communication is binary.
- The standards categorize some elements as optional or recommended, which causes some incompatibilities between software products from various vendors.
- Slow speed
- Communication is not human-readable as it is in BINARY FORM

FTP	SFTP
It stands for File Transfer Protocol.	It stands for Secure File Transfer Protocol.
It was developed by <u>Abhay Bhushan</u> and published as RFC in the year 1971.	<u>Sami Lehtinen</u> assisted the <u>Tatu Ylönen</u> in the development of SFTP in the year 1997.
In FTP, secure channel is not provided to transfer the files between the hosts.	In SFTP, a secure channel is provided to transfer the files between the hosts.
It is a part of the <u>TCP/IP</u> protocol.	It is a <u>SSH</u> protocol.
It usually runs on port no-21.	It runs on port no-22.
It establishes the connection under TCP protocol.	It establishes the control connection under SSH protocol.
It does not encrypt the data before sending.	It encrypted data before sending.
It works on direct method for file transfer.	It works on tunneling method for transferring of files.
It makes uploading and downloading of files without any security.	It maintains full security of the data by using SSH keys.
It uses two channels.	While; SFTP uses only one channel.
It doesn't require any authentication.	While in SFTP, the user needs to authenticate the FTP client with a username and password or SSH keys.
FTP is faster.	It is slower compared to FTP.
The direct transfer method is used to transfer the data in FTP.	The tunneling method is used to transfer the data in SFTP.

Difference between Bandwidth and Latency

Latency can be easily defined as how late the information is coming to the user and Bandwidth is simply the data coming to the user. This is the main characteristic difference between Bandwidth and Latency. An example of latency that we suffer almost every day is Buffering and Bandwidth can be referred to as simply the amount of data received.

Difference between Bandwidth and Throughput

Throughput is simply the data that is finally reaching the destination or in simple words, the product data and Bandwidth is the data coming to the user. This is the main difference between Bandwidth and Throughput. There are some factors that impact the Throughput is that Network Speed, Packet Loss, etc.

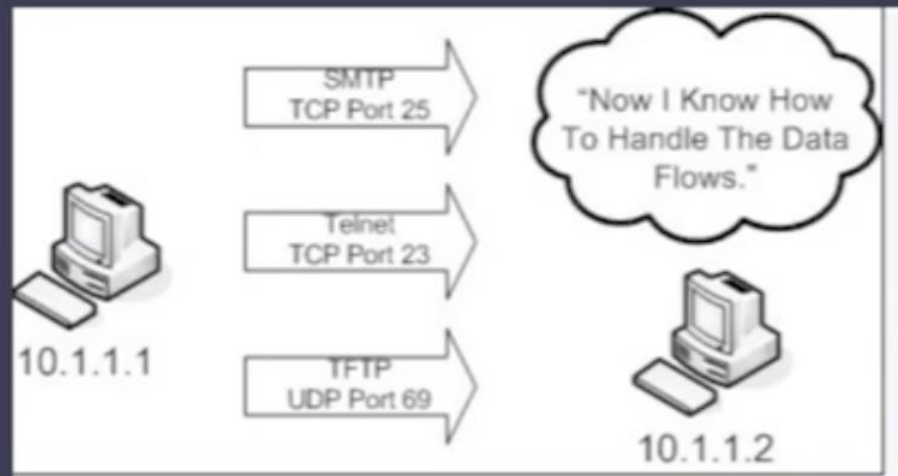
TCP/IP

- TCP/IP is a collection of the two main communication protocols (TCP and IP).
- TCP/IP is an internet communication protocol that defines the rules for the computers to communicate with each other online.
- Web browsers and internet servers use TCP/IP to connect to the internet.
- Email client programs such as Outlook Express etc. use TCP/IP to connect to the internet and send/receive emails.

- There are different types of protocols that reside inside the TCP/IP such as:

- ICMP
- DHCP
- DNS
- UDP

What is Port



- A Port is not a physical connection
- It's a logical connection between two programs
- Some of these port numbers are specifically defined and always associated with a specific type of service

Example, File Transfer Protocol (FTP) is always port number 21 and Hypertext Transfer Protocol web traffic is always port 80. These are called well-known ports and go from 0 to 1023

GROUP ASSIGNMENT (PRESENTATION WITH ANIMATION VIDEOS)

S No	Protocols (abbreviation)	Ports Numbers	Transport protocols	Meanings
1	File Transfer Protocol (FTP)	20 and 21	TCP	It is a protocol that carries data guarantees that data will be delivered properly.
2	Secure Shell (SSH)	22	TCP and UDP	It is a cryptographic network protocol used to secure data communication.
3	Telnet	23	TCP	It is the used for remote management protocol for managing network devices.
4	Simple Mail Transfer Protocol (SMTP)	25	TCP	It is a communication protocol which is used to transmit email messages over the internet to the destination server.
5	Domian Name System (DNS)	53	TCP and UDP	It is used in the performance of one simple task of converting IP address,To domain names that everyone can easily understand.
6	Trivial File Transfer Protocol (TFTP)	69	UDP	TFTP is typically used by devices to upgrade software and firmware and that include cisco.
7	Hyper Text Transfer Protocol (HTTP)	80	TCP	It is a kind of protocol used to define how data is transmitted and formatted and also used by www as a channel for communication.

S No	Protocols (abbreviation)	Ports Numbers	Transport protocols	Meanings
8	Dynamic Host Configuration Protocol (DHCP)	67 and 68	UDP	It is a kind of service used in the client and server model.
9	Network Time Protocol (NTP)	123	UDP	It is the synchronization of time between network devices in the network.
10	Simple Network Management Protocol (SNMP)	161 and 162	TCP and UDP	It has the ability to monitor, configure and control network devices.
11	Hyper Text Transfer Protocol Secure. (HTTPS)	443	TCP	It renders authentication and encryption that provides secure communication with the use of secure socket layer.
12	Address Resolution Protocol (ARP)	3389	TCP	It is used to resolve the network layer address into the link address.

Network Devices

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IP Address

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Networking Fundamentals
(C) Practical Networking .net

- **Hosts** are any device which **sends** or **receive** traffic



Computer



Laptop



Phones



Printers



Servers



Cloud Server

- Also any **Internet of Things (IoT)** devices
 - TV
 - Speakers
 - Smart Watches
 - Thermometers
 - Lights
 - Refrigerator
- *Anything that sends or receive traffic over a network*

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Standards

Standards are the set of rules for data communication that are needed for exchange of information among devices. It is important to follow Standards which are created by various Standard Organization like IEEE , ISO , ANSI etc.

Types of Standards

Standards are of two types :

- De Facto Standard.
- De Jure Standard.

De Facto Standard : The meaning of the word "*De Facto*" is "By Fact" or "By Convention". These are the standards that have not been approved by any Organization , but have been adopted as Standards because of its widespread use. Also , sometimes these standards are often established by Manufacturers.

For example : Apple and Google are two companies which established their own rules on their products which are different . Also they use some same standard rules for manufacturing for their products.

De Jure Standard : The meaning of the word “*De Jure*” is “By Law” or “By Regulations”.Thus , these are the standards that have been approved by officially recognized body like ANSI , ISO , IEEE etc. These are the standard which are important to follow if it is required or needed.

For example : All the data communication standard protocols like SMTP , TCP , IP , UDP etc. are important to follow the same when we needed

Types of Protocol

- **Network Layer Protocols** : Network layer protocols operate in the network layer which is also known as the Layer 3 of the network architecture. Network layer protocols are responsible for packet routing, forwarding and addressing of data packets throughout the network . IP and ICMP are the network layer protocols.
- **Transport layer Protocols** : Transport layer protocols works in transport layer which provides end-to-end service ensuring data transfer across apps on different devices. TCP and UDP are the most popular transport layer protocols.
- **Application Layer Protocol** : Application layer protocol working in the application layer of the network architecture provides communication between applications running on different devices. The application layer protocols enable cross-device communication. They format, exchange, and interpret

- **Wireless Protocols** : Wireless protocols basically used in wireless communication which enables data transfer through wireless networks. Bluetooth, Wi-Fi, and LTE protocols are examples.
- **Routing Protocols** : Routing protocol establishes the best/optimal network pathways throughout the network for fastest data transmission. Routers share information to develop and maintain routing tables. RIP, OSPF, and BGP are examples.
- **Security Protocols** : security protocol protects data confidentiality, integrity, and authenticity while transmission of data over the network. They include SSL and TLS, encryption methods, and authentication protocols for providing data security.
- **Internet Protocols** : IP identifies devices uniquely. Internet protocol provides data communication through routing and forwarding data packets from one device to another.

Protocol and Standard Compliance in Network Security

Protocol and standard compliance protects data, resources, and networks. Protocol and standard compliance are crucial to network security for these reasons:

- **Interoperability** : Protocols and standards allow devices and systems to communicate. These protocols ensure network components can function together, avoiding risks and security gaps produced by incompatible or unsupported systems.
- **Security Baseline** : Protocols and standards contain security principles and best practices that help secure network infrastructure. These protocols allow organizations to protect sensitive data via encryption, authentication, and access controls.
- **Vulnerability Management** : Network security protocols and standards help organizations find and fix vulnerabilities. Many standards requires regular security assessments, vulnerability scanning, and penetration testing to discover network infrastructure flaws. Organizations can prevent

OSI MODEL

The OSI model, created in 1984 by ISO, is a reference framework that explains the process of transmitting data between computers. It is divided into seven layers that work together to carry out specialised network functions, allowing for a more systematic approach to networking.

In starting days of computer networking, vendors created proprietary networking models to support their products. The biggest problem with these models was that they allowed communication only between the devices that were manufactured by the same vendor. For example, a computer manufactured by ***IBM was able to communicate only with the computer or networking device that was also manufactured by IBM.***

This situation was neither good nor profitable for both manufacturers and consumers. It not only forced manufacturers to make all essential networking devices but also forced consumers to buy all devices from the same manufacturer.

To break this barrier, leading hardware manufactures agreed to use or support a vendor-neutral networking model along with their proprietary networking model. To create a vendor-neutral networking model, two major efforts were made.

The first one was made by ISO, and the second one was made by DoD. ISO (The International Organization for Standardization) created the OSI (Open Systems Interconnection) model and DoD (U.S. Department of Defense) created the TCP/IP model.

Purposes of OSI model

OSI model was created for the following purposes:-

- To standardize data networking protocols to allow communication between all networking devices across the entire planet.
- To create a common platform for software developers and hardware manufactures that encourage the creation of networking products that can communicate with each other over the network.

Layer	Common Standards
1. Physical layer	RS-232c cable (LAN)
2. Data link layer	Ethernet (LAN)
3. Network layer	IP (Internet)
4. Transport layer	TCP (Internet)
5. Application layer	HTTP, HTML (Web) MPEG (audio/video)

TCP/IP Protocol Suite

TCP/IP Protocol Suite

OSI model layers

Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer

DARPA layers

Application Layer

Transport Layer

Internet Layer

Network Interface
Layer

HTTP

FTP

SMTP

DNS

RIP

SNMP

TCP

UDP

IGMP

ICMP

ND

MLD

ICMPv6

ARP

IP (IPv4)

IPv6

Ethernet

802.11
wireless
LAN

Frame Relay

ATM

TCP/IP Architecture Model

Layer	Network Infrastructure	Visibility
Application	host, server firewall proxy server application load balancer	HTTP, SSH, Telnet, DNS, BGP SNMP, FTP, DHCP, NTP, SMTP
Transport	firewall, load balancer	application ports (TCP/UDP)
Internet	router, multilayer switch, firewall	routing, IP addressing (IP)
Data Link	switch, access point, wireless controller	switching, wireless, MAC address (Ethernet)
Physical	cabling, transceiver, modem, antenna, repeater	transmit binary bits (1000Base-T)

PDU Information at Device: Router0

OSI Model

Inbound PDU Details

PDU Formats

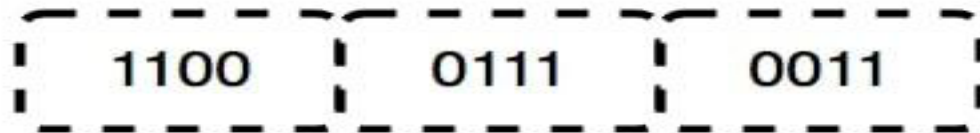
SRC IP: 192.168.1.1	
DST IP: 192.168.1.2	
OPT: 0x0	0x0
DATA (VARIABLE LENGTH)	

TCP

0	16	31	Bits
SRC PORT: 20		DEST PORT: 1028	
SEQUENCE NUM: 6212745			
ACK NUM: 1			
OFF.	RES.	ACK	WINDOW
CHECKSUM: 0x0		URGENT POINTER	
OPTION			PADDING
DATA (VARIABLE)			

Physical Layer – Layer 1

The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of **bits**. It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.



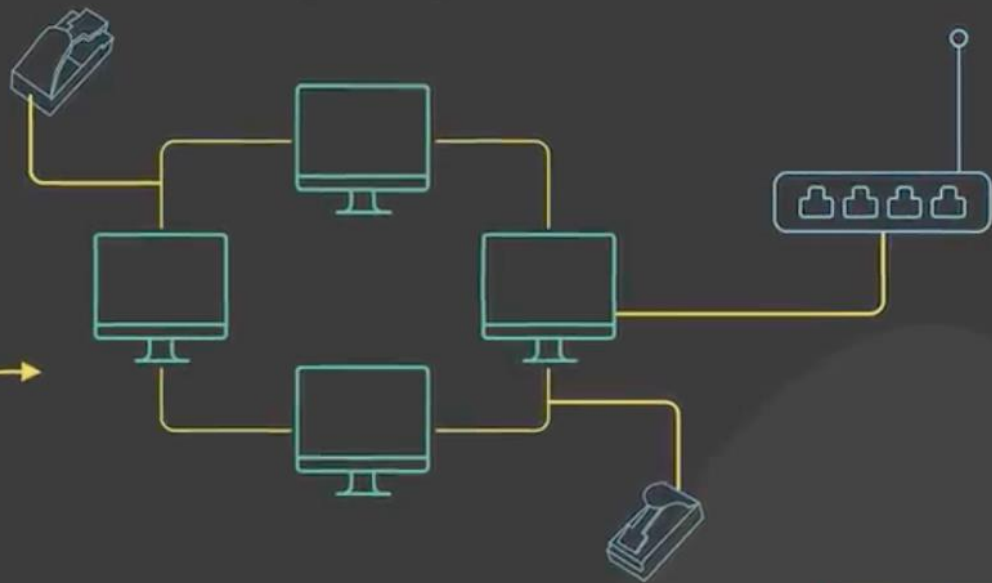
1100 0111 0011

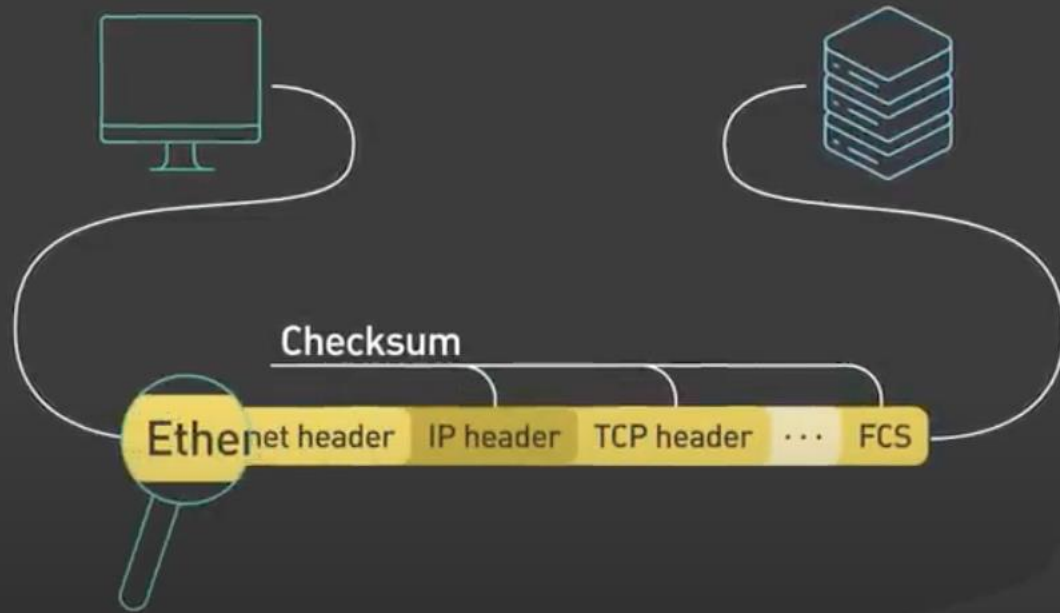
Functions of the Physical Layer

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.



Physical Layer





Data Link Layer (DLL) – Layer 2

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.

The Data Link Layer is divided into two sublayers:

1. Logical Link Control (LLC)
2. Media Access Control (MAC)

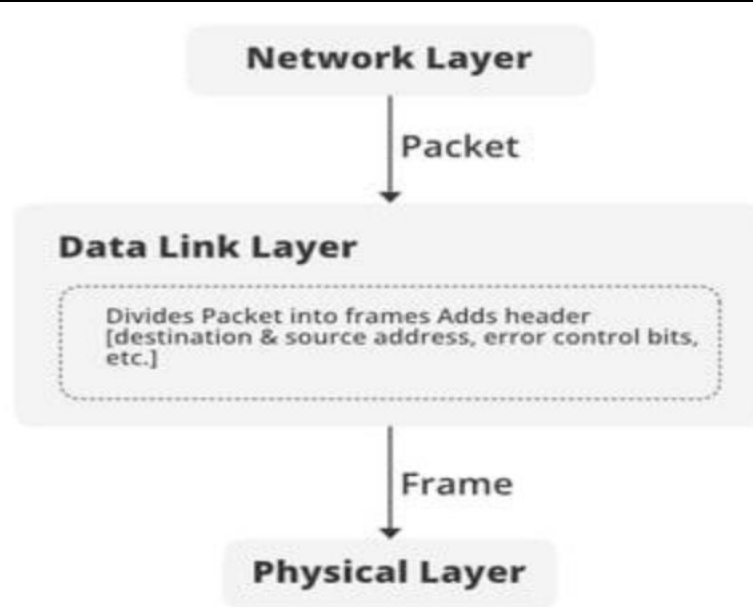
The packet received from the Network layer is further divided into frames depending on the frame size of the NIC(Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.

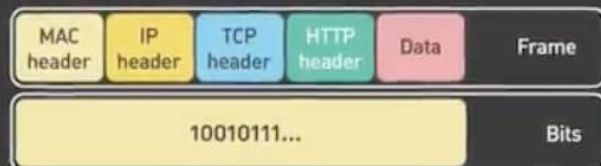
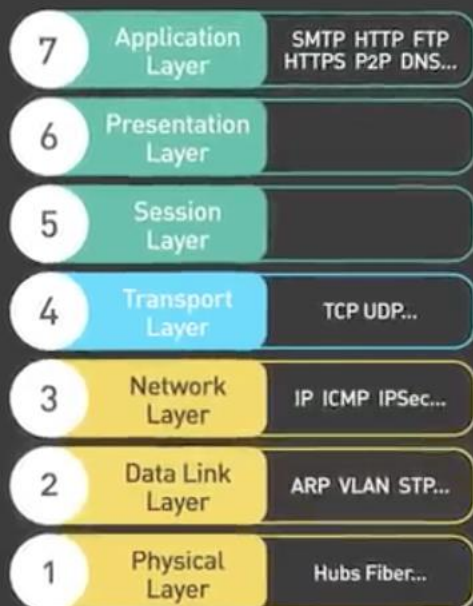
The Receiver's MAC address is obtained by placing an ARP(Address Resolution Protocol) request onto the wire asking "Who has that IP address?" and the destination host will reply with its MAC address.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.

- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer handles the channel.





Transport Layer – Layer 4

The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as *Segments*. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.

At the sender's side: The transport layer receives the formatted data from the upper layers, performs **Segmentation**, and also implements **Flow and error control** to ensure proper data transmission. It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.

Functions of the Transport Layer

- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

1.Connection-Oriented Service

2.Connectionless Service

1. Connection-Oriented Service: It is a three-phase process that includes

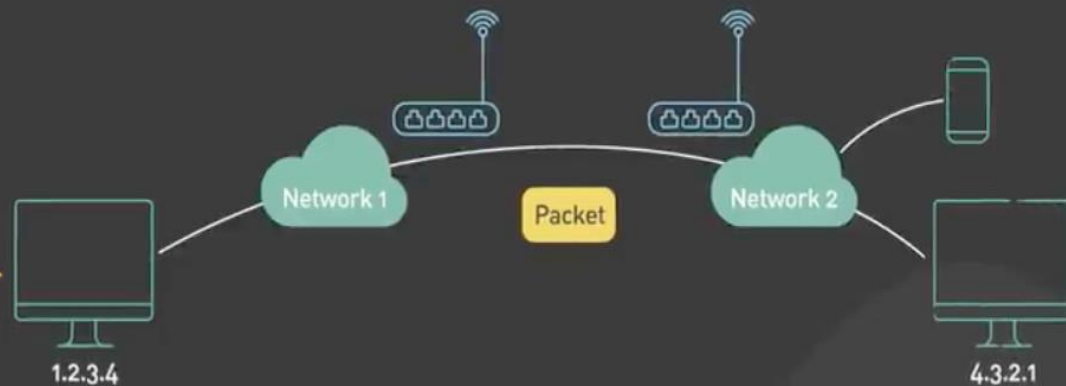
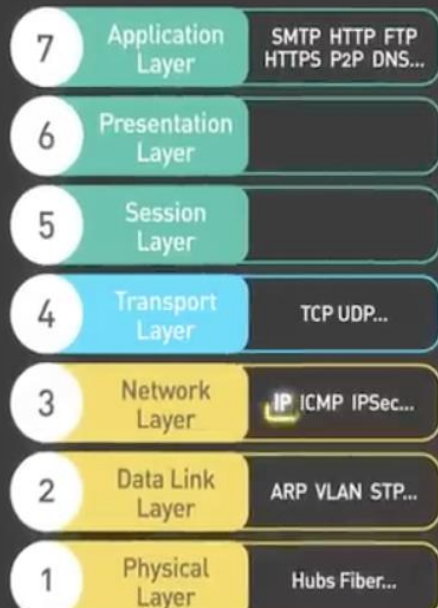
- Connection Establishment
- Data Transfer
- Termination/disconnection

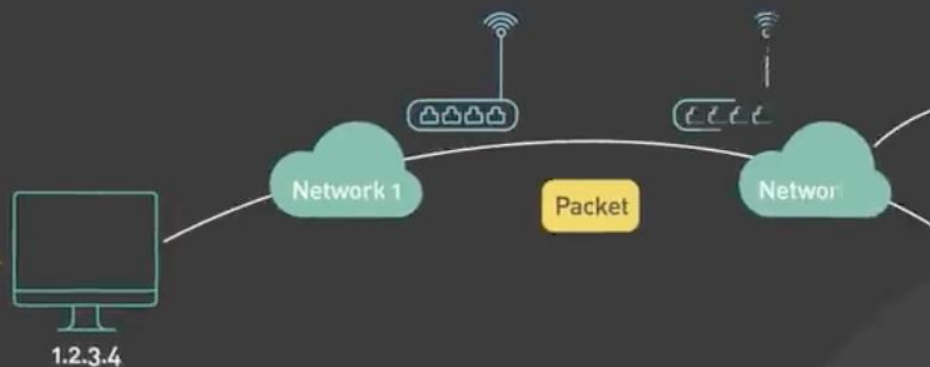
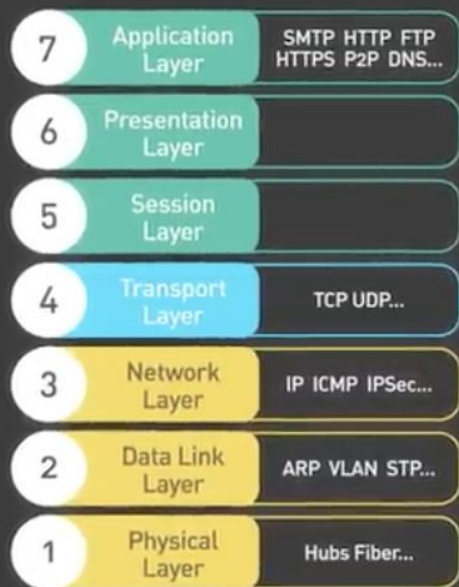
In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

Connectionless service: It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

Note:

- 1.Data in the Transport Layer is called **Segments**.
- 2.Transport layer is operated by the Operating System. It is a part of the OS and communicates with the Application Layer by making system calls.
- 3.The transport layer is called as **Heart of the OSI** model.
- 4.**Device or Protocol Use** : TCP, UDP NetBIOS, PPTP







Device A



3 2 1 4



Device B

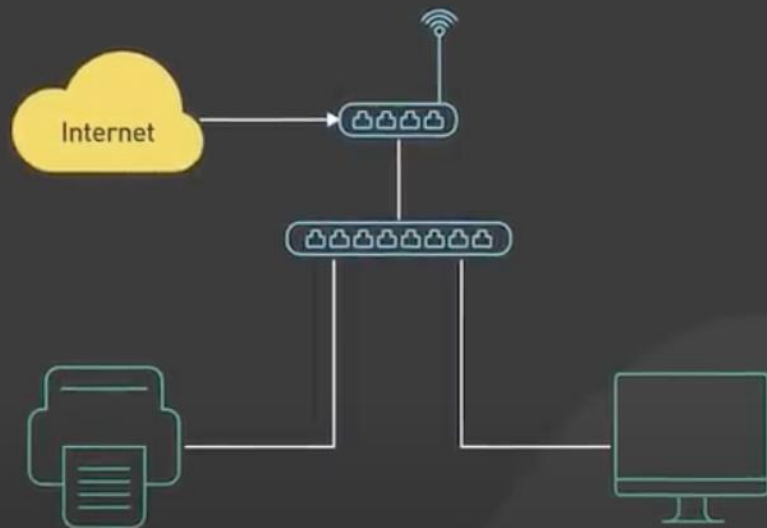
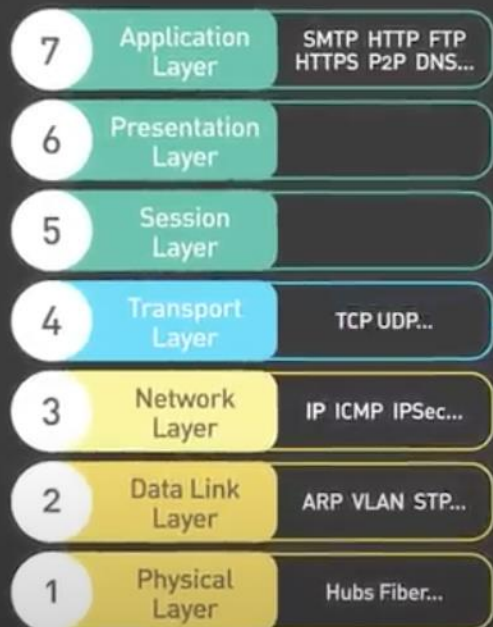
Segments

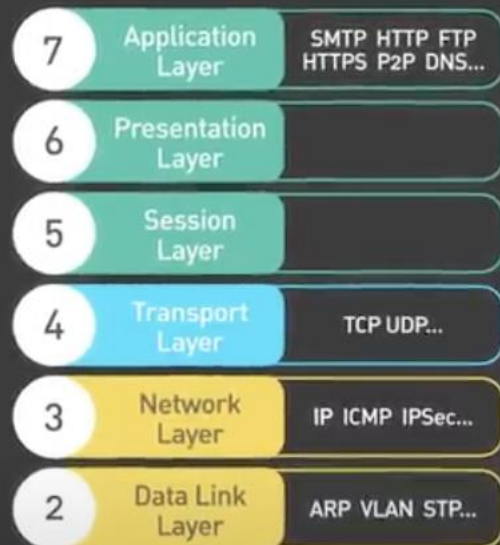
- Sequence num: 3
- Port num
- Data

- Sequence num: 2
- Port num
- Data

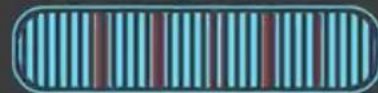
- Sequence num: 1
- Port num
- Data

- Sequence num: 4
- Port num
- Data





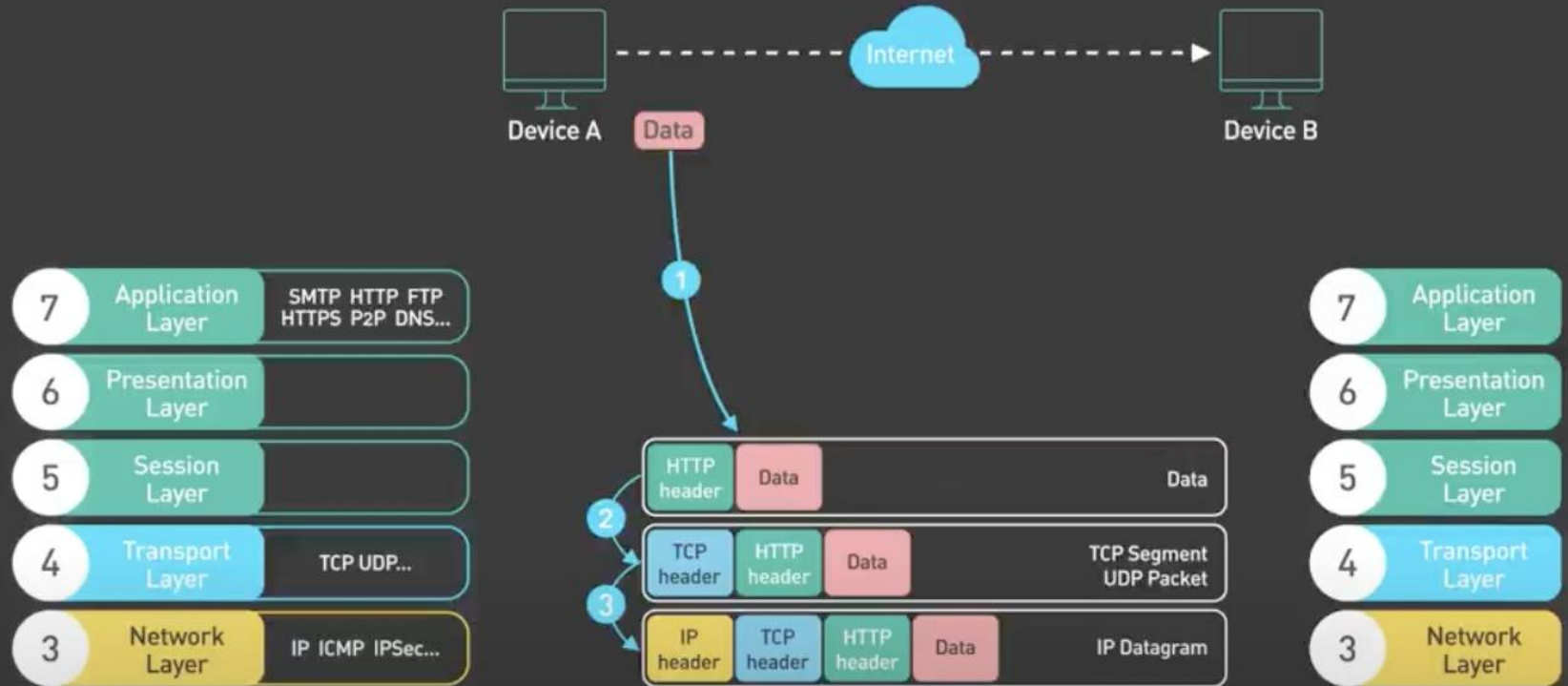
Downloading

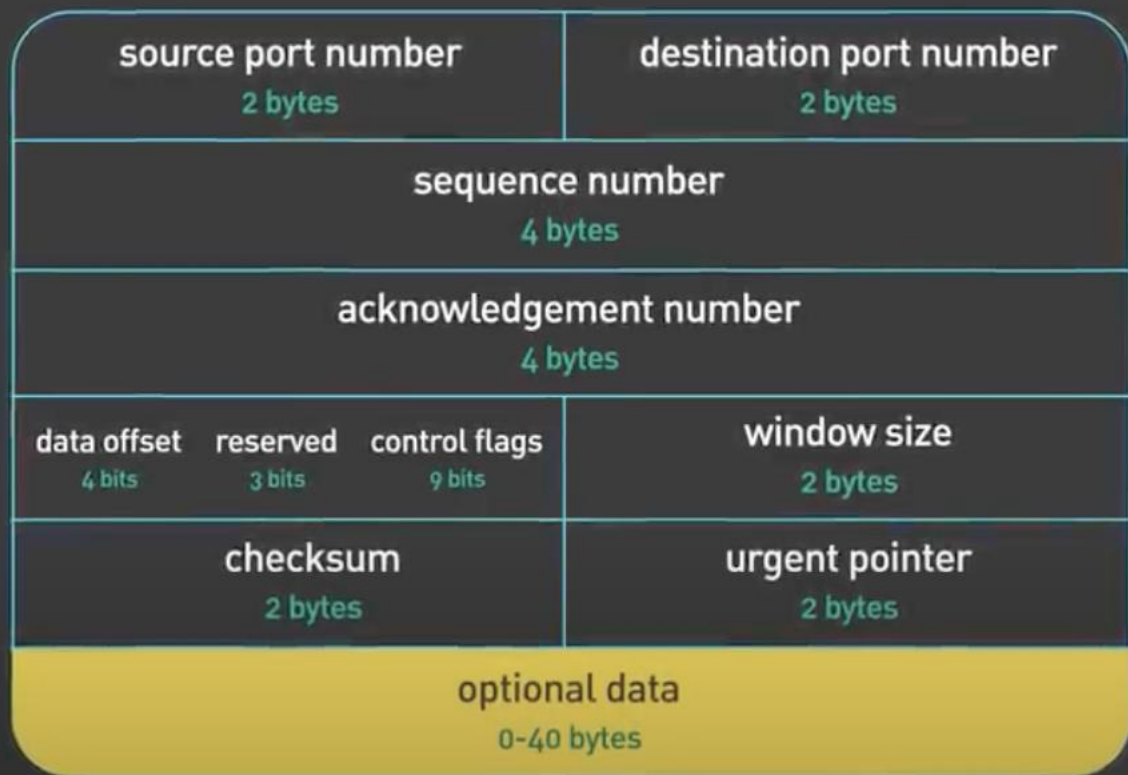


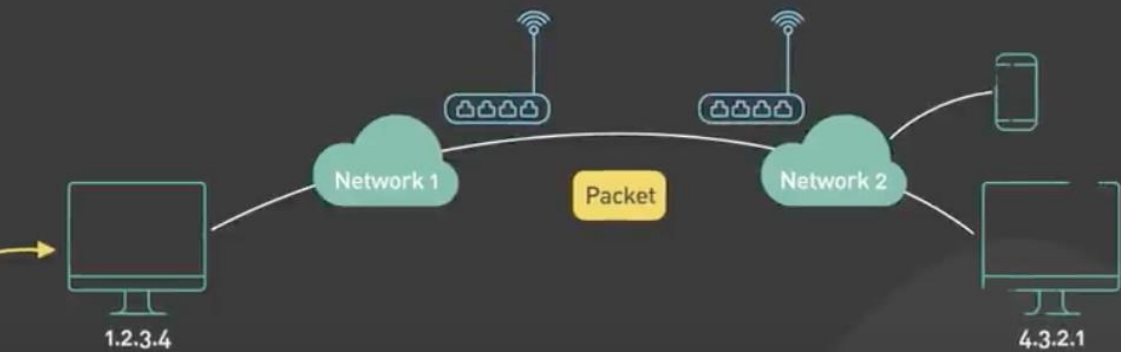
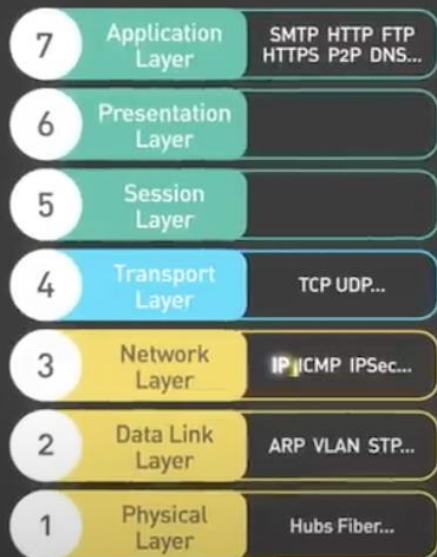


If an error is detected, the receiving end simply discards the packet.

OSI Model





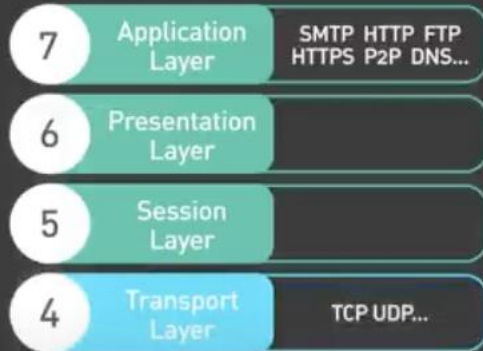


The IP layer (TCP/IP)



Data

1

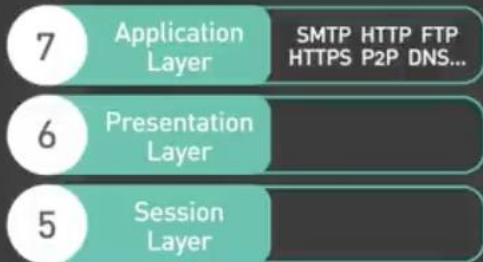


Session Layer – Layer 5

This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Functions of the Session Layer

- **Session establishment, maintenance, and termination:** The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.



Note:

- 1.All the below 3 layers(including Session Layer) are integrated as a single layer in the TCP/IP model as the "Application Layer".*
- 2.Implementation of these 3 layers is done by the network application itself. These are also known as Upper Layers or Software Layers.*
- 3.Device or Protocol Use : NetBIOS, PPTP.*

For example:-

Let us consider a scenario where a user wants to send a message through some Messenger application running in his browser. The "Messenger" here acts as the application layer which provides the user with an interface to create the data. This message or so-called Data is compressed, encrypted (if any secure data), and converted into bits (0's and 1's) so that it can be transmitted.

Application Layer – Layer 7

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data, which has to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

Example: Application – Browsers, Skype Messenger, etc.

Note: 1. The application Layer is also called Desktop Layer.

2. **Device or Protocol Use:** SMTP
The presentation layer is responsible for translation, encryption, and compression of data. Two communicating devices communicating may be using different encoding methods, so layer 6 is responsible for translating incoming data into a syntax that the application layer of the receiving device can understand.

The Application layer provides services for network applications with the help of protocols. Some of the most widely used application layer protocols are HTTP, HTTPS, FTP, NFS, DHCP, FMTP, SNMP, SMTP, Telnet, etc.

Functions of the Application Layer

The main functions of the application layer are given below.

- **Network Virtual Terminal:** It allows a user to log on to a remote host.
- **FTAM- File Transfer Access and Management:** This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- **Mail Services:** Provide email service.
- **Directory Services:** This application provides distributed database sources and access for global information about various objects and services.

Note: OSI model acts as a reference model and is not implemented on the Internet because of its late invention. The current model being used is the TCP/IP model

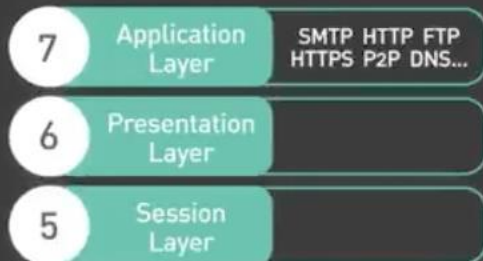
1.Following are the main functionalities of an application layer:

2.File Transfer: The Application layer mainly facilitates the file transfer between two network devices with the help of FTP(File Transfer Protocol).

3.Web Surfing: Web surfing is possible only in the application layer. Some protocols like HTTP(Hypertext Transfer Protocol), HTTPs(Hypertext Transfer Protocol Secure), etc. enables web surfing.

4.Emails: Electronic-mails can be sent from one device to another on the network only through the application layer. Some protocols like SMTP(Simple Mail Transfer Protocol), etc. are used for sending emails over the network.

5.Network Virtual Terminal: The Application layer facilitates the remote host login in the network with the help of protocols like Telnet, etc. It can also be referred to as the software version of the physical terminal in the network.



What is the Flow of Data in OSI Model?

When we transfer information from one device to another, it travels through 7 layers of OSI model.

First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.

Let's look at it with an Example:

Luffy sends an e-mail to his friend Zoro.

Step 1: Luffy interacts with e-mail application like Gmail, outlook, etc. Writes his email to send. (This happens in **Layer 7: Application layer**)

Step 2: Mail application prepares for data transmission like encrypting data and formatting it for transmission. (This happens in **Layer 6: Presentation Layer**)

Step 3: There is a connection established between the sender and receiver on the internet. (This happens in **Layer 5: Session Layer**)

Step 4: Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information. (This happens in **Layer 4: Transport Layer**)

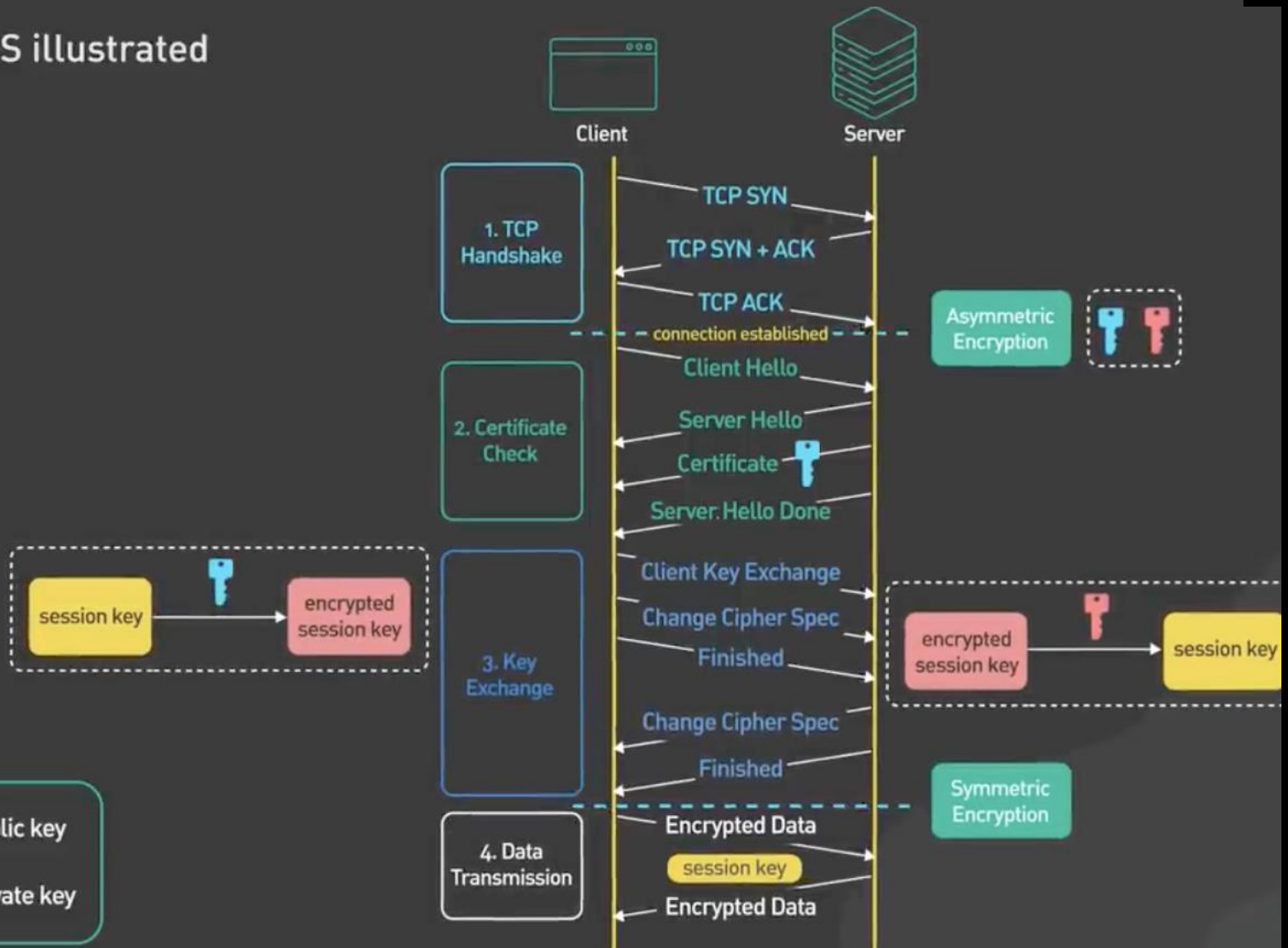
Step 5: Addressing of packets is done in order to find the best route for transfer. (This happens in **Layer 3: Network Layer**)

Step 6: Data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection. (This happens in **Layer 2: Data Link Layer**)

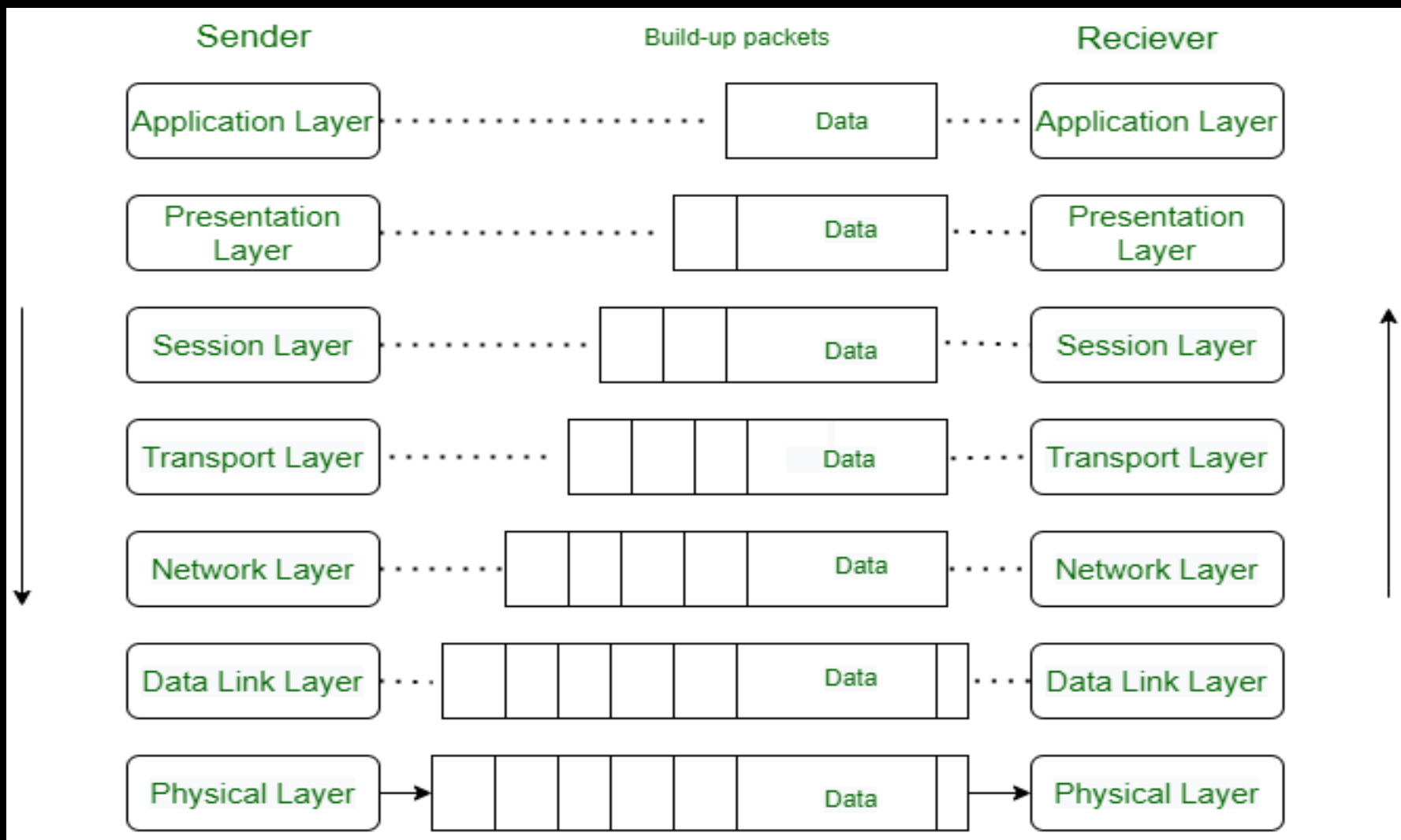
Step 7: Lastly Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

After the email reaches the receiver i.e. Zoro, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Zoro's email client.

HTTPS illustrated



PRESENTATION-GROUP



Layer No	Layer Name	Responsibility	Information Form(Data Unit)	Device or Protocol
7	Application Layer	Helps in identifying the client and synchronizing communication.	Message	SMTP
6	Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Message	JPEG, MPEG, GIF
5	Session Layer	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Message	Gateway
4	Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segment	Firewall
3	Network Layer	Transmission of data from one host to another, located in different networks.	Packet	Router

Layer No	Layer Name	Responsibility	Information Form(Data Unit)	Device or Protocol
2	Data Link Layer	Node to Node Delivery of Message.	Frame	Switch, Bridge
1	Physical Layer	Establishing Physical Connections between Devices.	Bits	Hub, Repeater, Modem, Cables

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

The OSI Model

Application

Access to the network

Presentation

Manipulate data (Translate, encrypt)

Session

Manage sessions (connections)

Transport

Provide reliable delivery

Network

Internetwork - move packets from source to destination

Data Link

Configure data for direct delivery by physical layer

Physical

Physical delivery - electrical specs etc

OSI and TCP/IP Model

Application

Presentation

Session

Transport

Network

Data Link

Physical

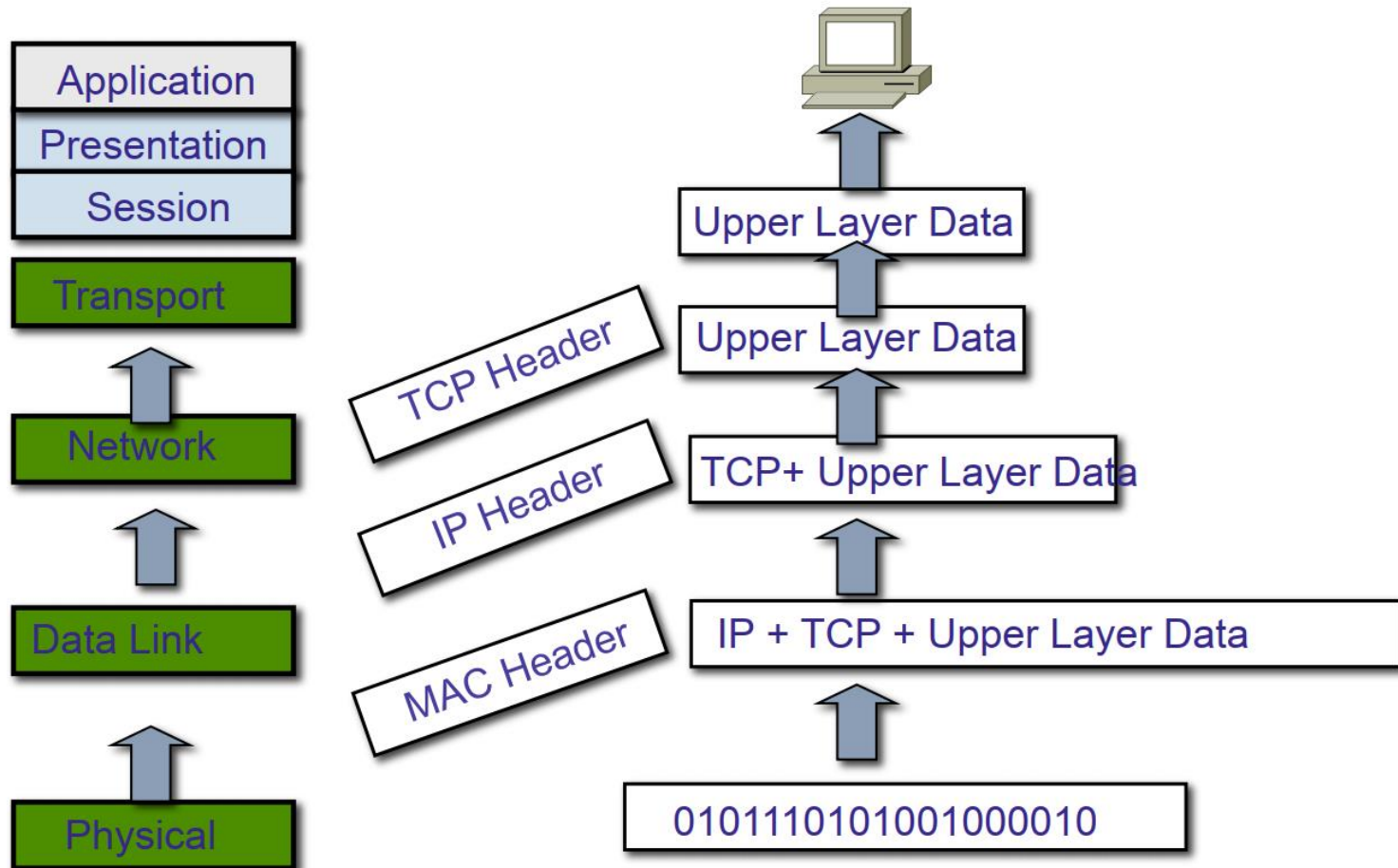
Application

Transport

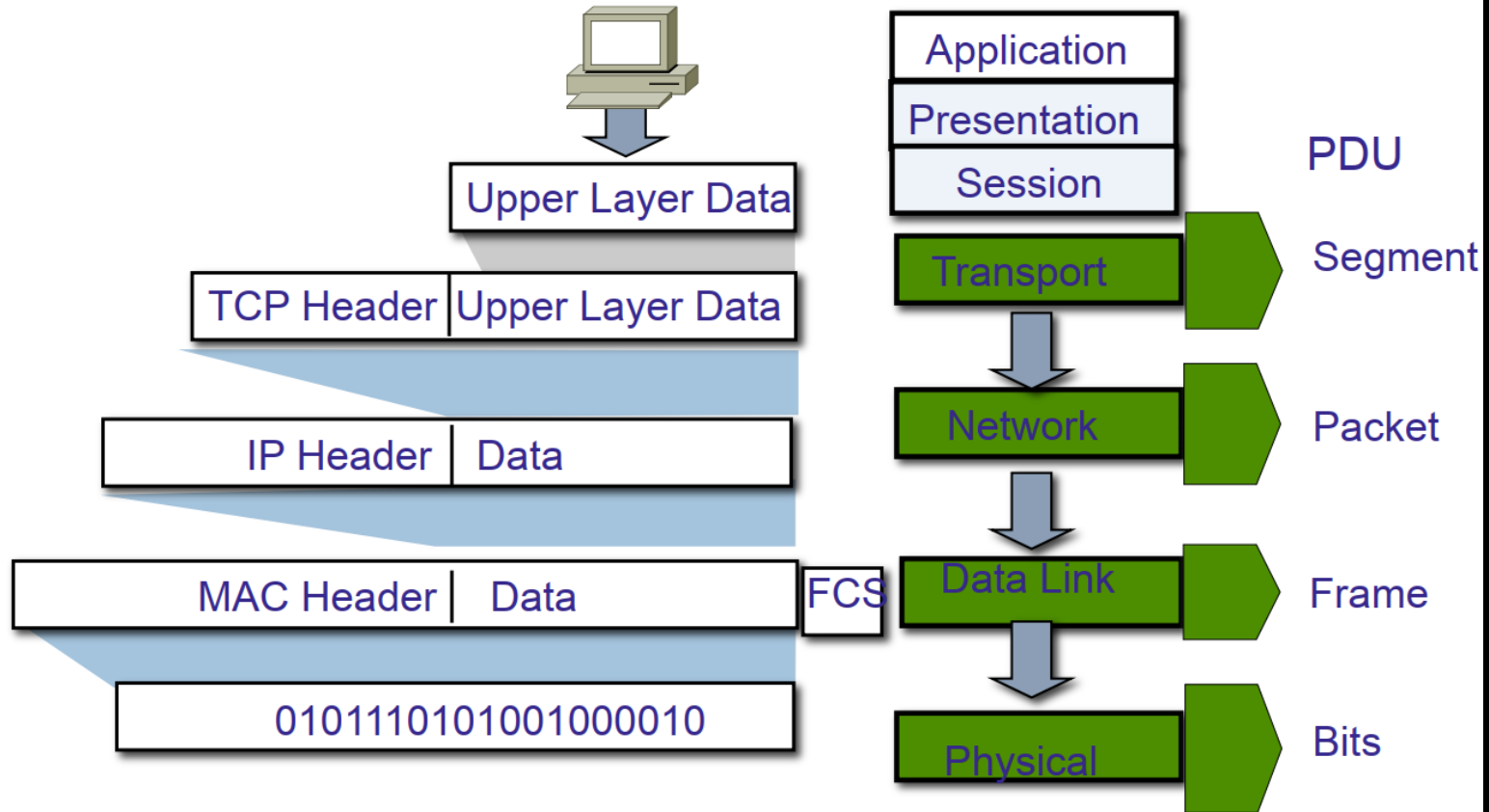
Internet

Network Access

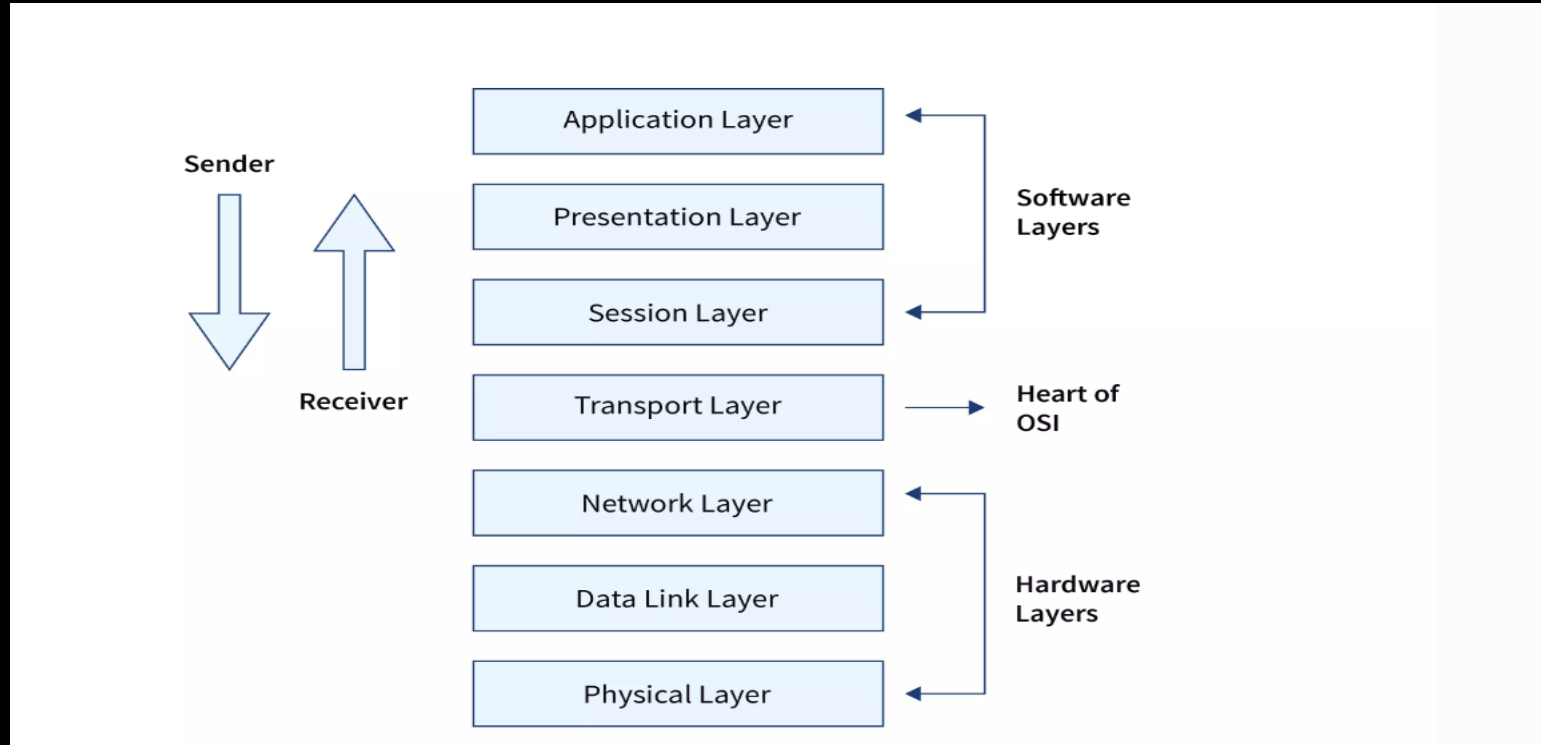
De-encapsulating Data



Encapsulating Data



OSI's HEART



The transport layer controls the reliability of a given link through flow control, segmentation and desegmentation, and error control. Some protocols are state- and connection-oriented. This means that the transport layer can keep track of the segments and retransmit those that fail.

OSI model

Layer	Name	Example protocols
7	Application Layer	HTTP, FTP, DNS, SNMP, Telnet
6	Presentation Layer	SSL, TLS
5	Session Layer	NetBIOS, PPTP
4	Transport Layer	TCP, UDP
3	Network Layer	IP, ARP, ICMP, IPSec
2	Data Link Layer	PPP, ATM, Ethernet
1	Physical Layer	Ethernet, USB, Bluetooth, IEEE802.11

Technology	Transmission Distance		Speed
Bluetooth	33ft	10 Meters	1 Mbps
Wireless b	375ft	115 Meters	11 Mbps
Wireless g	375ft	115 Meters	54 Mbps
Satellite	Worldwide		1 Mbps
GPRS	Cell tower		56 Kbps
3G	Cell tower		1 Mbps
4G	Cell tower		50 Mbps

TCP/IP (Transmission Control Protocol/Internet Protocol) is a suite of communication protocols that define the standards for transmitting data over computer networks, including the internet. The TCP/IP protocol is the foundation of the internet and enables devices to communicate with each other using a common language.

The TCP/IP protocol is divided into two layers: the Transport layer and the Internet layer. The Transport layer is responsible for ensuring that data is transmitted reliably from one device to another. This layer is comprised of two protocols: the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP). TCP is used for reliable data transmission, while UDP is used for fast transmission of data that can tolerate some packet loss.

The Internet layer is responsible for transmitting data packets between devices. This layer is comprised of two protocols: the Internet Protocol (IP) and the Address Resolution Protocol (ARP). IP is responsible for routing data packets between devices, while ARP is used to map IP addresses to physical addresses.

TCP/IP also includes a number of application layer protocols that are used to provide services to end-users. These include protocols such as HTTP (Hypertext Transfer Protocol) for web browsing, FTP (File Transfer Protocol) for file transfer, and SMTP (Simple Mail Transfer Protocol) for email.

TCP/IP stands for **Transmission Control Protocol/ Internet Protocol**. It is a set of conventions or rules and methods that are used to interconnect network devices on the Internet. The internet protocol suite is commonly known as TCP/IP, as the foundational protocols in the suite are Transmission Control Protocol and Internet Protocol. It chooses how the information will be traded over the web through end-to-end communications that incorporate how the information ought to be organized into bundles (bundles of data), addressed, sent, and received at the goal. This communication protocol can also be utilized to interconnect organize devices in a private network

History of TCP/IP:

The Defense Advanced Research Projects Office (DARPA), the investigation department of the U.S. Department of Defense, made the TCP/IP shown in the 1970s for utilization in ARPANET, a wide zone organize that gone before the web. TCP/IP was initially planned for the Unix working framework, and it has been built into all of the working frameworks that came after it

How TCP/ IP works?

- TCP/IP employs the client-server demonstration of communication in which a client or machine (a client) is given a benefit (like sending a webpage) by another computer (a server) within the network.
- Collectively, the TCP/IP suite of conventions is classified as stateless, which suggests each client request is considered new since it is irrelevant to past requests. Being stateless liberates up network paths so they can be utilized continuously.
- The transport layer itself, is stateful. It transmits a single message, and its connection remains open until all the packets in a message have been received and reassembled at the destination.
- The TCP/IP model differs from the seven-layer Open Systems Interconnection (OSI) model designed after it.

Application/Uses of TCP/IP

Some Real-Time Applications are:

- **Simple Mail Transfer Protocol(SMTP):** It helps to send email to another email address.
- **File Transfer Protocol(FTP):** It is used for sending large files.
- **Dynamic Host Configure Protocol(DHCP):** It assigns the IP address.
- **Telnet:** Bi-directional text communication via a terminal application.
- **HyperText Transfer Protocol(HTTTP):** Used to transfer the web pages.
- **Domain Name System(DNS):** It translates the website name to IP addresses.
- **Simple Network Time Protocol(SNTP):** It provides the time of a day to the network devices.

OSI vs TCP/IP Model

Some key differences between the OSI model and the TCP/IP Model are:

1. TCP/IP model consists of 4 layers but OSI model has 7 layers. Layers 5,6,7 of the OSI model are combined into the Application Layer of TCP/IP model and OSI layers 1 and 2 are combined into Network Access Layers of TCP/IP protocol.

2. The TCP/IP model is older than the OSI model, hence it is a foundational protocol that defines how should data be transferred online.

3. Compared to the OSI model, the TCP/IP model has less strict layer boundaries.

4. All layers of the TCP/IP model are needed for data transmission but in the OSI model, some applications can skip certain layers. Only layers 1,2 and 3 of the OSI model are necessary for data transmission.

UDP and TCP protocols come into play with VoIP because they structure the way web traffic travels through the Internet. TCP and UDP packets are sent from a source to your phone or computer, and if any of these packets are dropped, it will affect the quality of your call.

TCP/IP Layers

- **Application Layer** An application layer is the topmost layer within the TCP/IP model. When one application layer protocol needs to communicate with another application layer, it forwards its information to the transport layer.
- **Transport Layer** It is responsible for the reliability, flow control, and correction of data that is being sent over the network. There are two protocols used in this layer are User Datagram Protocol and Transmission control protocol.
- **Internet/Network Layer** It is the third layer of the TCP/IP Model and also known as the Network layer. The main responsibility of this layer is to send the packets from any network, and they arrive at the goal irrespective of the route they take.
- **Network Access Layer** It is the lowest layer of the TCP/IP Model. It is the combination of the Physical Layer and the Data link layer which present in the OSI Model. Its main responsibility is to the transmission of information over the same network between two devices.

Benefits of TCP/IP

- It is an **industry-standard demonstrate** that can be viably deployed in commonsense organizing problems.
- It is **interoperable**, i.e., it permits cross-platform communications among heterogeneous networks.
- It is an **open convention suite**. It isn't claimed by any specific established and so can be utilized by any individual or organization.
- It may be **versatile, client-server engineering**. This permits systems to be included without disturbing the current services.
- It allots an **IP address to each computer on the organize**, hence making each device to be identifiable over the arrange. It allots each location a space title. It gives the title and addresses determination administrations.

Challenges of TCP/IP:

- It is not generic in nature. So, it comes up short to represent any protocol stack other than the TCP/IP suite. For the case, it cannot depict the Bluetooth connection.
- It does not clearly isolate the concepts of services, interfacing, and protocols. So, it isn't appropriate to portray unused advances in modern networks.
- It does not recognize between the data link and the physical layers, which has exceptionally distinctive functionalities.
- The information interface layer ought to concern with the transmission of outlines. On the other hand, the physical layer ought to lay down the physical characteristics of the transmission.
- In this, model the transport layer does not guarantee

- Security: TCP/IP was originally designed for an open and trusting environment, and as a result, it is not inherently secure. This has led to a range of security challenges, including attacks such as DDoS, man-in-the-middle attacks, and other types of network-based attacks.
- Complexity: The TCP/IP protocol suite is highly complex, with many different protocols and layers that interact with each other. This complexity can make it difficult to troubleshoot network issues and can increase the likelihood of errors and misconfigurations.
- Scalability: While TCP/IP is highly scalable, there are limits to its scalability. As networks grow larger and more complex, it can become more difficult to manage and optimize TCP/IP-based networks.
- Congestion: TCP/IP was not designed with congestion management in mind, which can lead to issues such as network

- Legacy systems: TCP/IP is based on legacy technology that was designed in the 1970s and 1980s. While the protocol has been updated over the years, it can still struggle to support modern networking needs, such as real-time applications, mobile devices, and the Internet of Things.

- IPv4 address depletion: The IPv4 address space is limited and has been depleted in many regions, which has led to the widespread adoption of IPv6. However, the transition from IPv4 to IPv6 has been slow, and many networks still

Did you Know?

rely on IPv4.

TCP/IP protocol (Transfer Control Protocol/Internet Protocol) was created by U.S. Department of Defense's Advanced Research Projects Agency (ARPA) in 1970s.

Advantages of OSI Model

OSI Model defines the communication of a computing system into 7 different layers.

Advantages of OSI Model include:

- It divides network communication into 7 layers which makes it easier to understand and troubleshoot.
- It standardizes network communications, as each layer has fixed functions and protocols.
- Diagnosing network problems is easier with the OSI model.
- It is easier to improve with advancements as each layer can get updates separately

the required hardware and software to build their network.

- Encourage hardware manufacturers to create networking products that can communicate with each other over the network.
- Provide a teaching tool to understand the communication process used between networking components.
- Separate a complex function into simpler components.
- Make troubleshooting easier, as network administrators can troubleshoot issues more quickly and effectively by looking in a layer that is causing the issue rather than finding it in the entire network.

The OSI model helps users and operators of computer networks: Determine the required hardware and software to build their network. Understand and communicate the process followed by components communicating across a network.

Is OSI layer still used?

Yes, the OSI model is still used by networking professionals to understand data abstraction paths and processes better.

Multiplexing

Multiplexing is a technique to mix and send multiple data streams over a single medium. This technique requires system hardware called multiplexer (MUX) for multiplexing the streams and sending them on a medium, and de-multiplexer (DMUX) which takes information from the medium and distributes to different destinations.

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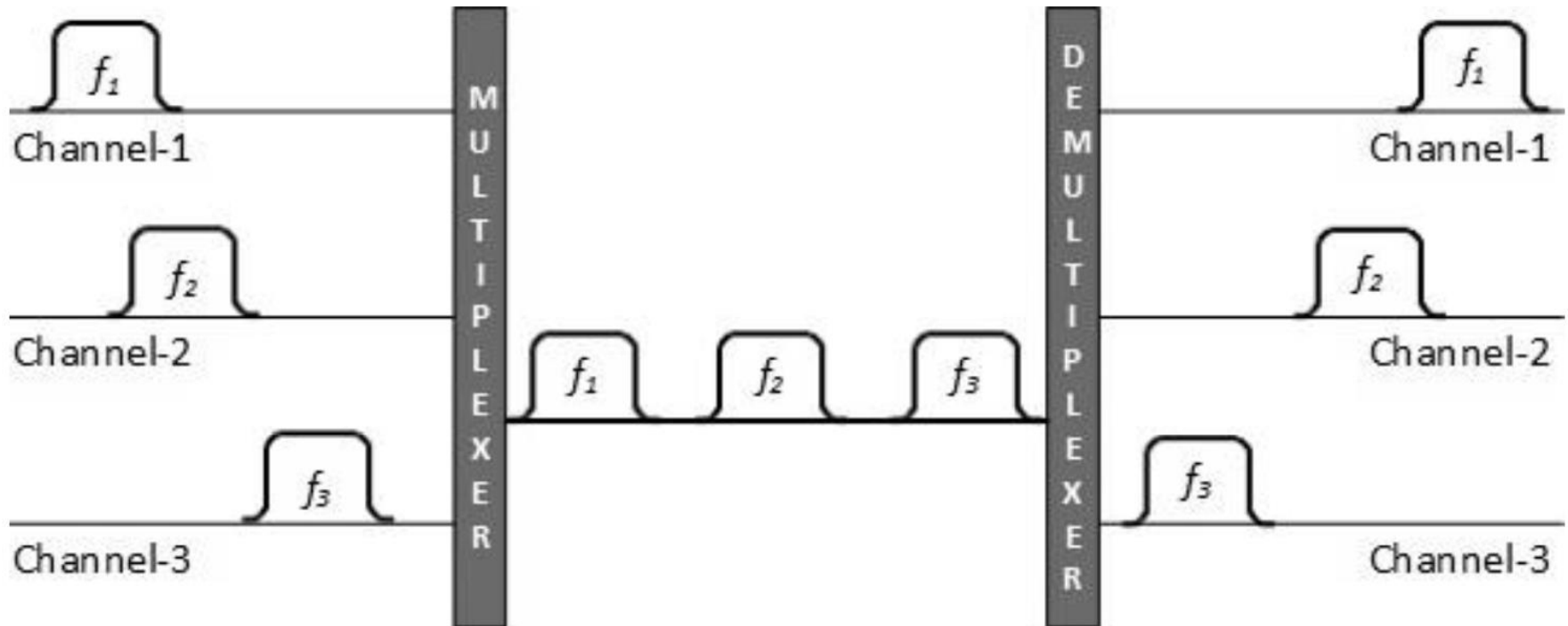
Multiplexing is a technique by which different analog and digital streams of transmission can be simultaneously processed over a shared link. Multiplexing divides the high capacity medium into low capacity logical medium which is then shared by different streams.

Communication is possible over the air (radio frequency), using a physical media (cable), and light (optical fiber). All mediums are capable of multiplexing.

When multiple senders try to send over a single medium, a device called Multiplexer divides the physical channel and allocates one to each. On the other end of communication, a De-multiplexer receives data from a single medium, identifies each, and sends to different receivers.

Frequency Division Multiplexing

When the carrier is frequency, FDM is used. FDM is an analog technology. FDM divides the spectrum or carrier bandwidth in logical channels and allocates one user to each channel. Each user can use the channel frequency independently and has exclusive access of it. All channels are divided in such a way that they do not overlap with each other. Channels are separated by guard bands. Guard band is a frequency which is not used by either channel.



Time Division Multiplexing

TDM is applied primarily on digital signals but can be applied on analog signals as well. In TDM the shared channel is divided among its user by means of time slot.

Each user can transmit data within the provided time slot only.

Digital signals are divided in frames, equivalent to time slot i.e. frame of an optimal size which can be transmitted in given time slot

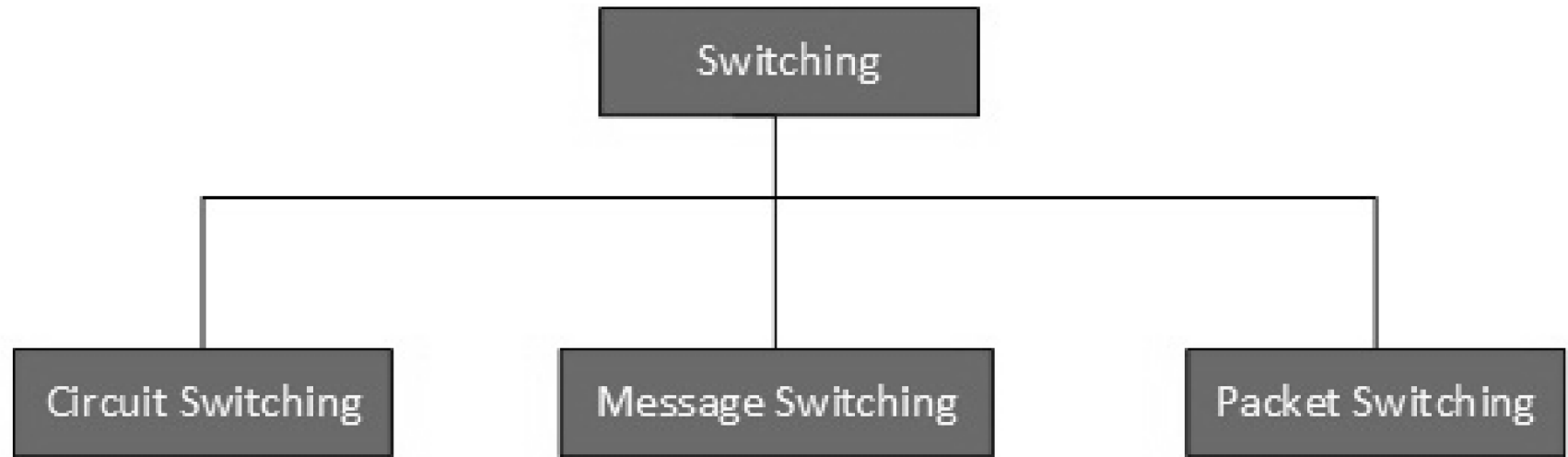
TDM works in synchronized mode. Both ends, i.e. Multiplexer and De-multiplexer are timely synchronized, and both switch to next channel simultaneously.

When channel A transmits its frame at one end, the De-multiplexer provides media to channel A on the other end. As soon as the channel A's time slot expires, this side switches to channel B. On the other end, the De-multiplexer works in a synchronized manner and provides media to channel B. Signals from different channels travel the path in interleaved manner

Switching

Switching is a mechanism by which data/information sent from source towards destination which are not directly connected. Networks have interconnecting devices, which receives data from directly connected sources, stores data, analyze it and then forwards to the next interconnecting device closest to the destination

Switching can be categorized as:



Circuit Switching?

Circuit switching is a communication method where a dedicated communication path, or circuit, is established between two devices before data transmission begins. The circuit remains dedicated to the communication for the duration of the session, and no other devices can use it while the session is in progress. Circuit switching is commonly used in voice communication and some types of data communication.

A circuit-switched network relies on a physical connection between two nodes, which requires the link to be set up before the nodes can communicate.

Circuit switching is another type of switched network communication. They are connection-based networks where data is transferred directly across an already chosen and dedicated channel without interruption until the complete information has been sent to the destination and the communication link ends.

Circuit switching at the physical layer routes the complete message via the dedicated channel built when setting up the connection. Because data is delivered and received in streams, this kind of switching isn't optimal for data transfer since the line would be inactive between transmission waves.

Circuit switching is another type of switched network communication. They are connection-based networks where data is transferred directly across an already chosen and dedicated channel without interruption until the complete information has been sent to the destination and the communication link ends. Circuit switching was created in 1878 and is the technology on which Alexander Bell built his telephone. Circuit switching has been the dominant technology that telecommunications companies have used for over 100 years. This was before the discovery of packet switching networks

There are three phases during the circuit switching process:

- **Establishment of the circuit:** Before sending data across, the network ensures a patent, free and available path between the nodes in the network, joining the sender and receiving device. If available, a circuit is now created and established.
- **Transfer of data:** data of any type, voice, data, analog or digital, is then sent across.
- **Disconnection:** After the data transfer, the circuit ends so that the bandwidth becomes available for other use.

Advantages of Circuit Switching:

- Guaranteed bandwidth: Circuit switching provides a dedicated path for communication, ensuring that bandwidth is guaranteed for the duration of the call.
- Low latency: Circuit switching provides low latency because the path is predetermined, and there is no need to establish a connection for each packet.
- Predictable performance: Circuit switching provides predictable performance because the bandwidth is reserved, and there is no competition for resources.
- Suitable for real-time communication: Circuit switching is suitable for real-time communication, such as voice and video, because it provides low latency and predictable performance.

Disadvantages of Circuit Switching:

- Inefficient use of bandwidth: Circuit switching is inefficient because the bandwidth is reserved for the entire duration of the call, even when no data is being transmitted.
- Limited scalability: Circuit switching is limited in its scalability because the number of circuits that can be established is finite, which can limit the number of simultaneous calls that can be made.
- High cost: Circuit switching is expensive because it requires dedicated resources, such as hardware and bandwidth, for the duration of the call.

A packet-switched network is a digital network that manages data transfer in the form of small and optimized packets, an improvement from older network types.

A packet-switched network is defined as a digital network that manages data transfer in the form of small and optimized packets comprising a payload and a header, an improvement from older, analog-era network types.

Communication technology has evolved over the years – from loud-sounding gongs to the discovery of the telephone and radio waves. In the telecom industry, data is transferred from one location or device to another through packet switching.

Packet switching is a means of organizing and sharing data into bits called packets which are then transmitted across a digital network. These packets consist of two parts; the header and the payload. The header on each packet contains information about the data packet. This information includes the packet's destination address, the source of the extracted payload, and other details. The networking hardware utilizes this information to direct the packets.

Packet switching is the primary basis for transferring data across computer networks. With packet switching, routing and the movement of data packets are done so that a channel is occupied only when a packet is being moved on it. When the data packet has been sent to its designated address, the channel is immediately made available and can transmit further incoming data packets.

Packet switching works by allocating resources as needed. Resources in this instance refer to open channels for transferring bits of data. This allocation is done using statistical multiplexing and dynamic bandwidth dispersing methods. Every time the data packets reach any hardware component, they are received, buffered, assembled, and sent again until they arrive at the next location. This process is repeated throughout the distance covered by the traveling data packet.

The pros of packet switching include:

- In packet switching, data packets can find the destination address without traveling on one dedicated channel.
- Packet switching allows the sender to resend missing or dropped packets. Thus there is reduced packet loss.
- Transmission of Voice over Internet Protocol (VOIP) audio and data is less expensive because dedicated channels are unnecessary.

There are also a few cons to packet switching:

- It does not fully support communication methods that are constantly used, such as high-volume voice calls.
- There is no security protocol for the data packets sent across the network.
- Packet loss is still possible, especially when traffic flow is very high. These packets can not be resent again.

The nature of layers

The layers used by circuit-switched networks and packet-switched networks are different. The circuit network is implemented at the physical layer, whereas the packet-switched network is implemented at the network layer.

The physical layer is the bottom-most layer per the Open Systems Interconnection (OSI) model of computer networking. The circuit fixed network's physical layer means that there must be physical connections between the devices. A good example is the traditional telephone landlines used quite commonly in the past.

The network layer used by packet switching is layer 3 of the OSI. It allows data to move into and through other networks and is responsible for packaging data with the correct labels containing the destination addresses. The network layer also helps direct each data packet to a selected bandwidth

The end terminal

In circuit switching, the logical end terminal is the telephone and modem, while in a packet-switched network, the end terminal is typically a computer.

The technologies in use

Circuit switching and packet switching use different types of technologies. Circuit switching uses time and space division switching, while packet switching uses the datagram and virtual circuit approaches. In addition, the protocols for delivery using circuit switching technologies are more straightforward than packet switching technologies with more complex protocols.

DATA TRANSMISSION

Data or information can be stored in two ways, analog and digital. For a computer to use the data, it must be in discrete digital form. Similar to data, signals can also be in analog and digital form. To transmit data digitally, it needs to be first converted to digital form.

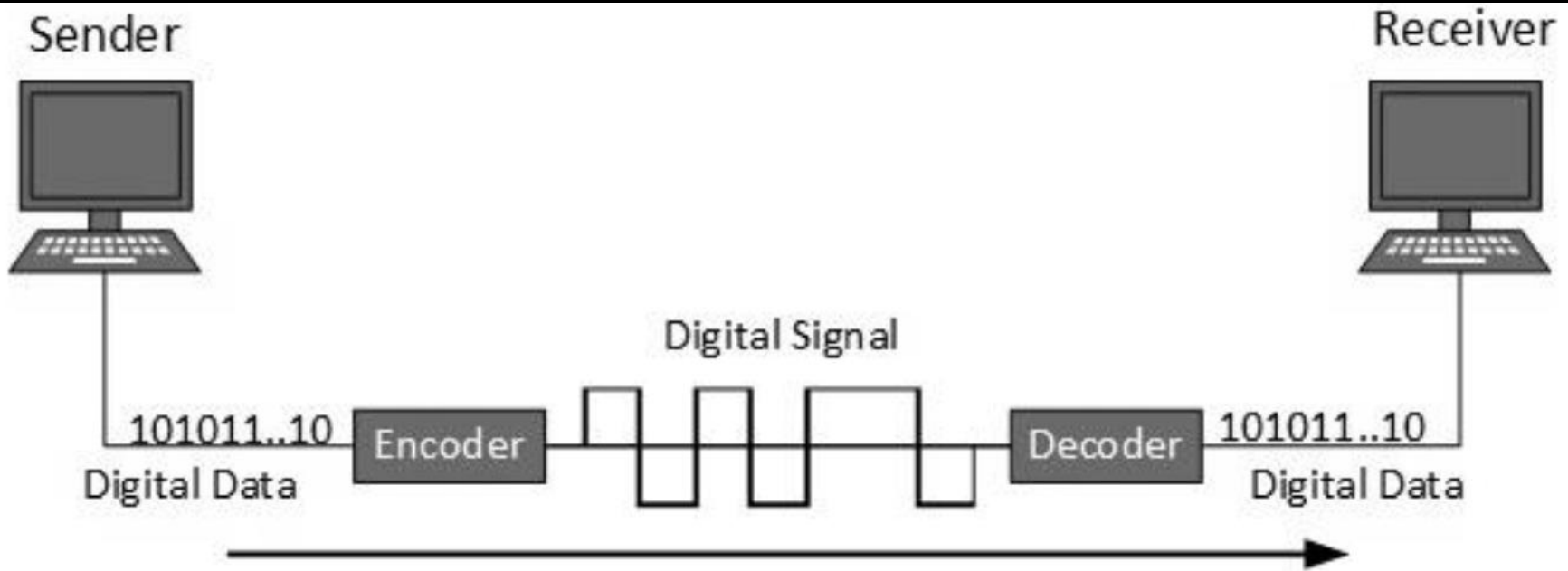
Digital-to-Digital Conversion

This section explains how to convert digital data into digital signals. It can be done in two ways, line coding and block coding. For all communications, line coding is necessary whereas block coding is optional.

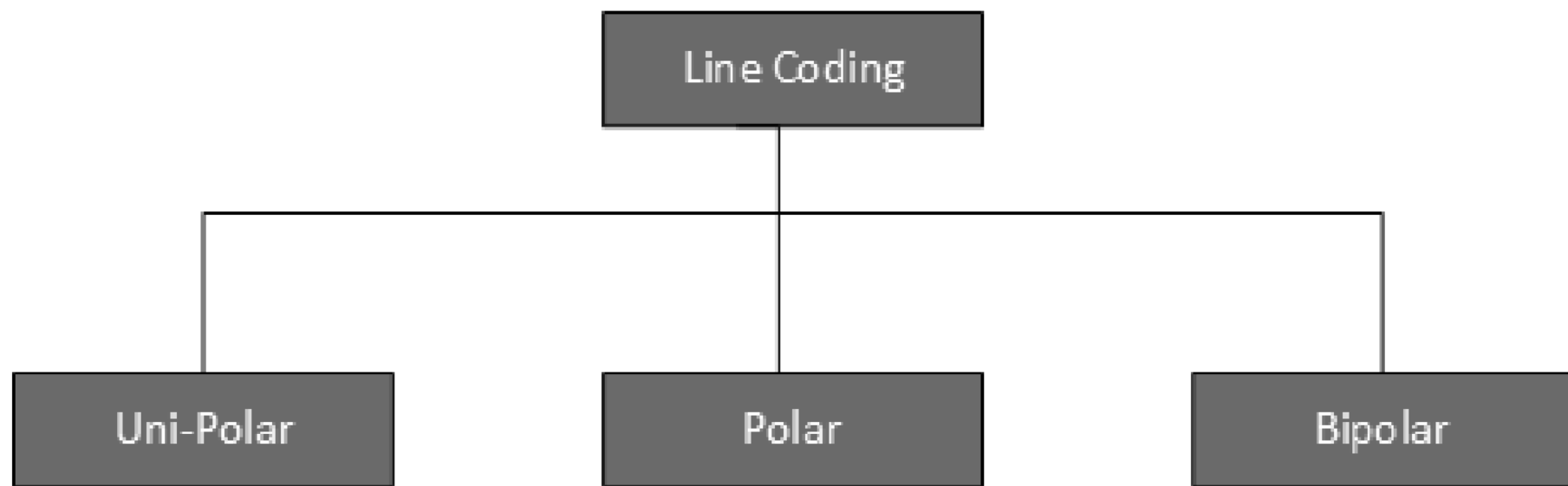
Line Coding

The process for converting digital data into digital signal is said to be Line Coding.

Digital data is found in binary format. It is represented (stored) internally as series of 1s and 0s



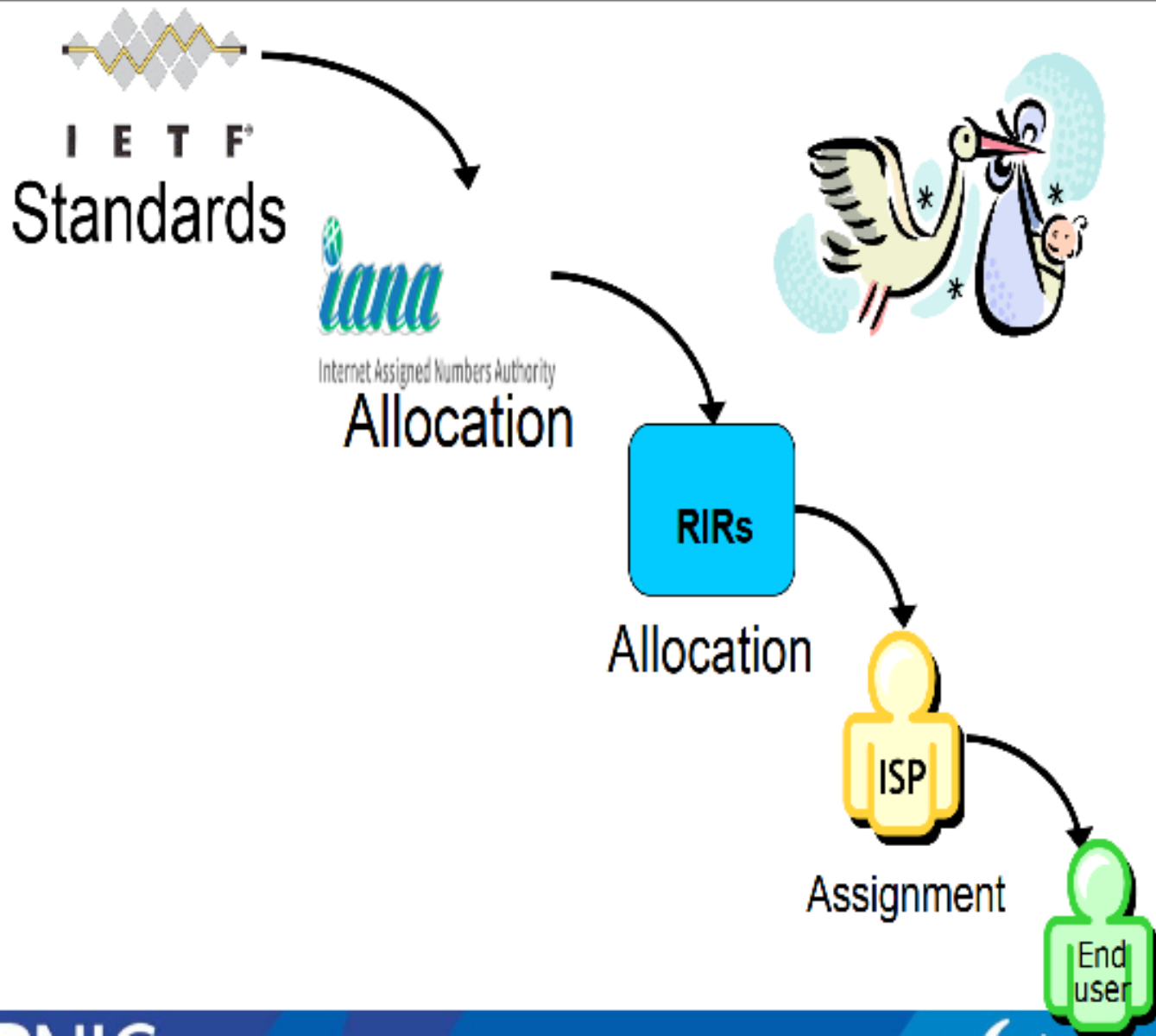
Digital signal is denoted by discrete signal, which represents digital data. There are three types of line coding schemes available:



Internet Addresses and Subnets

The IANA reserves specific IP address blocks for commercial organizations, government departments, and ISPs. When a user connects to the internet, their ISP assigns them an address from within one of the blocks assigned to it. If they only go online from one computer, then they can use the address assigned to it by their ISP.

However, most homes now use routers that share a network connection with multiple devices. So if a router is used to share the connection, then the ISP assigns the IP address to the router, and then a subnet is created for all computers that connect to it.



IP addresses that fall within a subnet have a network and a node. The subnet is identified by the network. The node, also known as the host, connects to the network and needs its own address. Computers separate the network and node via a subnet mask, which filters the appropriate IP address designation. When a large network is set up, the subnet mask that best fits the number of nodes or subnets required is determined.

When it comes to IP addresses within a subnet, the first address is reserved for the subnet, and the final one indicates the broadcast address for the subnet's systems.

IP Address Classes

Some IP addresses are reserved by the Internet Assigned Numbers Authority (IANA). These are typically reserved for networks that carry a specific purpose on the Transmission Control Protocol/Internet Protocol (TCP/IP), which is used to interconnect devices.

Four of these IP address classes include:

1.0.0.0.0: This IP address in IPv4 is also known as the default network. It is the non-routeable meta address that designates an invalid, non-applicable, or unknown network target.

2.127.0.0.1: This IP address is known as the loopback address, which a computer uses to identify itself regardless of whether it has been assigned an IP address.

3.169.254.0.1 to 169.254.254.254: A range of addresses that are automatically assigned if a computer is unsuccessful in an attempt to receive an address from the DHCP.

4.255.255.255.255: An address dedicated to messages that need to be sent to every computer on a network or broadcasted across a network.

IP ADDRESS CLASSES

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 (2^7-2)
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

The router on a TCP/IP network can be configured to ensure it recognizes subnets, then route the traffic onto the appropriate network. IP addresses are reserved for the following subnets:

- 1.Class A:** IP addresses between 10.0.0.0 and 10.255.255.255
- 2.Class B:** IP addresses between 172.16.0.0 and 172.31.255.255
- 3.Class C:** IP addresses between 192.186.0.0 and 192.168.255.255
- 4.Class D or multicast:** IP addresses between 224.0.0.0 and 239.255.255.255
- 5.Class E, which are reserved for experimental usage:** IP addresses between 240.0.0.0 and 254.255.255.254

IP addresses listed under Class A, Class B, and Class C are most commonly used in the creation of subnets. Addresses within the multicast or Class D have specific usage rules outlined in the Internet Engineering Task Force (IETF) guidelines, while the release of Class E addresses for public use was the cause of plenty of debate before the IPv6 standard was introduced.

Types of IP Address

IP Address is of two types:

- 1. IPv4:** Internet Protocol version 4. It consists of 4 numbers separated by the dots. Each number can be from 0-255 in decimal numbers. But computers do not understand decimal numbers, they instead change them to binary numbers which are only 0 and 1. Therefore, in binary, this (0-255) range can be written as (00000000 – 11111111). Since each number N can be represented by a group of 8-digit binary digits. So, a whole IPv4 binary address can be represented by 32-bits of binary digits. In IPv4, a unique sequence of bits is assigned to a computer, so a total of (2^{32}) devices approximately = 4,294,967,296 can be assigned with IPv4.

Classes of IPv4 Address:

There are around 4.3 billion IPv4 addresses and managing all those addresses without any scheme is next to impossible. Let's understand it with a simple example. If you have to find a word from a language dictionary, how long will it take? Usually, you will take less than 5 minutes to find that word. You are able to do this because words in the dictionary are organized in alphabetical order. If you have to find out the same word from a dictionary that doesn't use any sequence or order to organize the words, it will take an eternity to find the word. If a dictionary with one billion words without order can be so disastrous, then you can imagine the pain behind finding an address from 4.3 billion addresses. For easier management and assignment IP addresses are organized in numeric order and divided into the following 5 classes

IPv4 vs. IPv6

IPv4 has not been able to cope with the massive explosion in the quantity and range of devices beyond simply mobile phones, desktop computers, and laptops. The original IP address format was not able to handle the number of IP addresses being created.

To address this problem,

IPv6 was introduced. This new standard operates a hexadecimal format that means billions of unique IP addresses can now be created. As a result, the IPv4 system that could support up to around 4.3 billion unique numbers has been replaced by an alternative that, theoretically, offers unlimited IP addresses.

That is because an IPv6 IP address consists of eight groups that contain four hexadecimal digits, which use 16 distinct symbols of 0 to 9 followed by A to F to represent values of 10 to 15.

How Do I Locate My IP Address?

Windows computer users can look up their IP address by typing "cmd" into the search tab and hitting Enter, then typing "ipconfig" into the pop-up box.

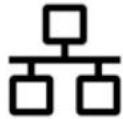
Mac computer users can find their IP address by heading into System Preferences and selecting Network.

To look up an IP address on a mobile phone, users need to head into Settings, then open the Wi-Fi menu and their network menu. The IP address should be listed under the Advanced section, depending on

IP address

Private

LAN



Not recognized over internet

Assigned by LAN admin



Unique only in LAN



PRIVATE IP



LAN

Public

Internet



Recognized over internet

Assigned by Service provider/IANA



Internet Assigned Numbers Authority

Unique Globally

Class	Private Address Range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255

IP addressing



Class	Private Address Range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255



Class	Public IP Ranges
A	1.0.0.0 to 9.255.255.255 11.0.0.0 to 126.255.255.255
B	128.0.0.0 to 171.255.255.255 173.0.0.0 to 191.255.255.255
C	192.0.0.0 to 195.255.255.255 197.0.0.0 to 223.255.255.255
D	224.0.0.0 to 247.255.255.255 Multicast Addresses
E	248.0.0.0 to 255.255.255.254 Experimental Use

The 0.0.0.0 is a Non-routable address is that indicates an invalid, or inapplicable end-user address.

A loopback address is a distinct reserved IP address range that starts from 127.0.0.0 ends at 127.255.255.255 though 127.255.255.255 is the broadcast address for 127.0.0.0/8. The loopback addresses are built into the IP domain system, enabling devices to transmit and receive the data packets. The loopback address 127.0.0.1 is generally known as localhost.

TCP/IP (Transmission Control Protocol/Internet Protocol) is a suite of communication protocols that define the standards for transmitting data over computer networks, including the internet. The TCP/IP protocol is the foundation of the internet and enables devices to communicate with each other using a common language.

The TCP/IP protocol is divided into two layers: the Transport layer and the Internet layer. The Transport layer is responsible for ensuring that data is transmitted reliably from one device to another.

This layer is comprised of two protocols: *the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP)*.

TCP is used for reliable data transmission, while UDP is used for fast transmission of data that can tolerate some packet loss.

The Internet layer is responsible for transmitting data packets between devices. This layer is comprised of two protocols:

the Internet Protocol (IP) and the Address Resolution Protocol (ARP). IP is responsible for routing data packets between devices, while ARP is used to map IP addresses to physical addresses.

IP ADDRESS

A

CLASS

E

1.0.0.0 -127.255.255.255

B

C

D

240.0.0.0-255.255.255.255

128.0.0.0-191.255.255.255

192.0.0.0-223.255.255.255

224.0.0.0-239.255.255.255

192.168.1

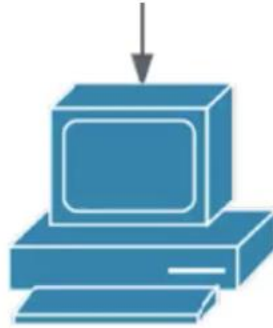
Network

.1

Host

/24

Subnet
Mask



192.168.1.13

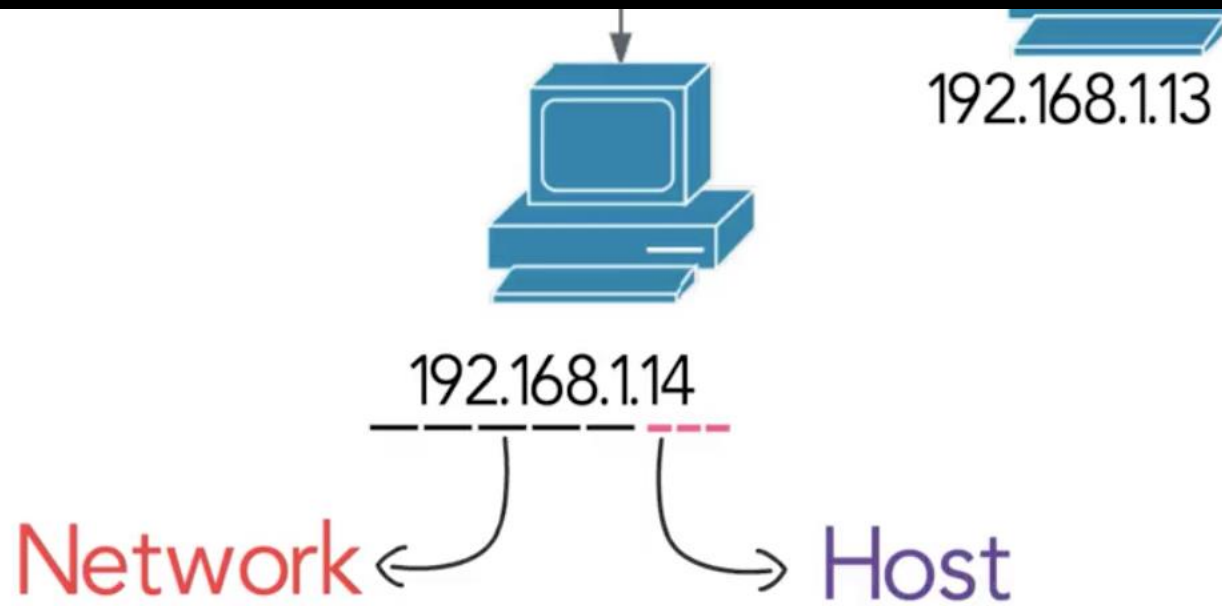
192.168.1.14

Network

Host

Unique Number
Class of network

Assigned to each host.
Uniquely identifies the machine.



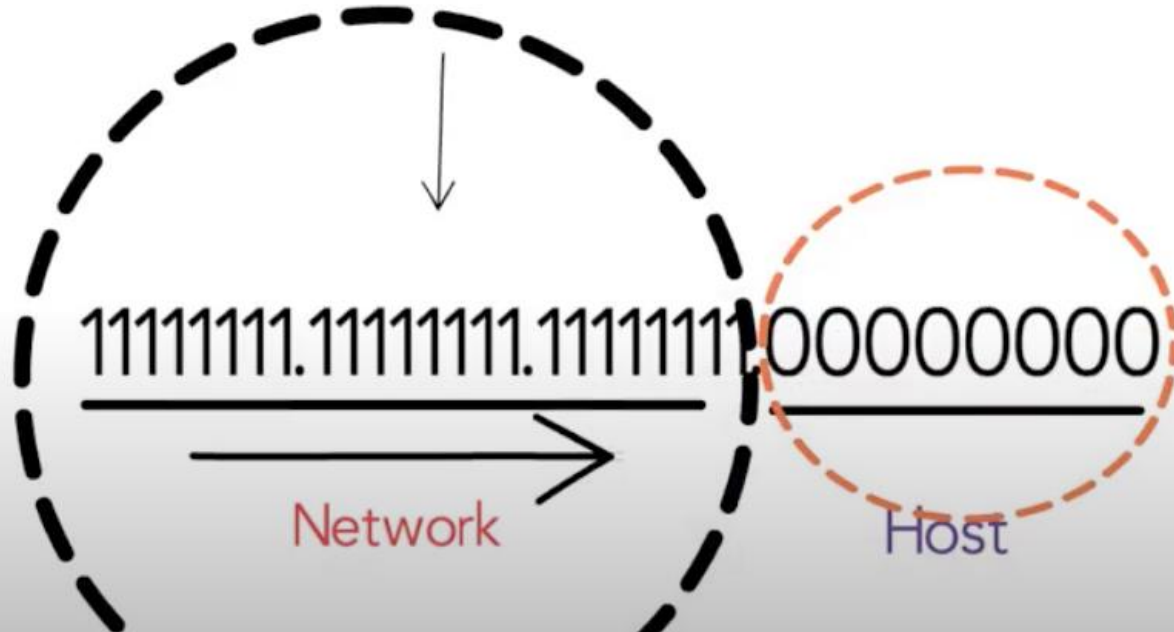
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

■■■■■■■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■



SUBSCRIBE
NETWORKING **H**

255.255.255.0



[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

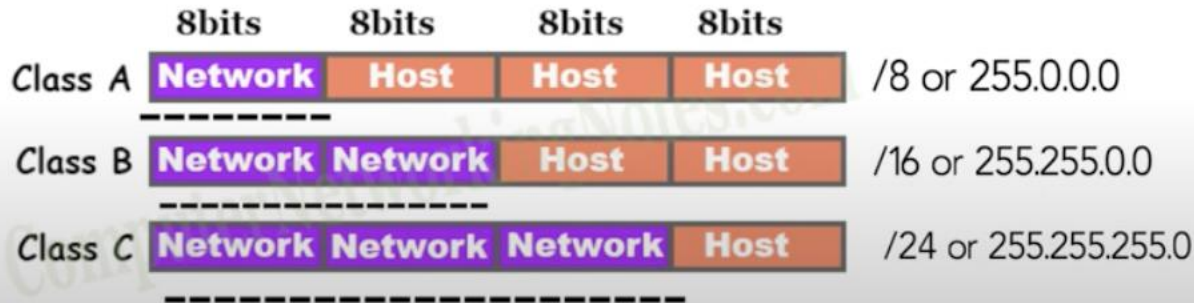
[illegible]

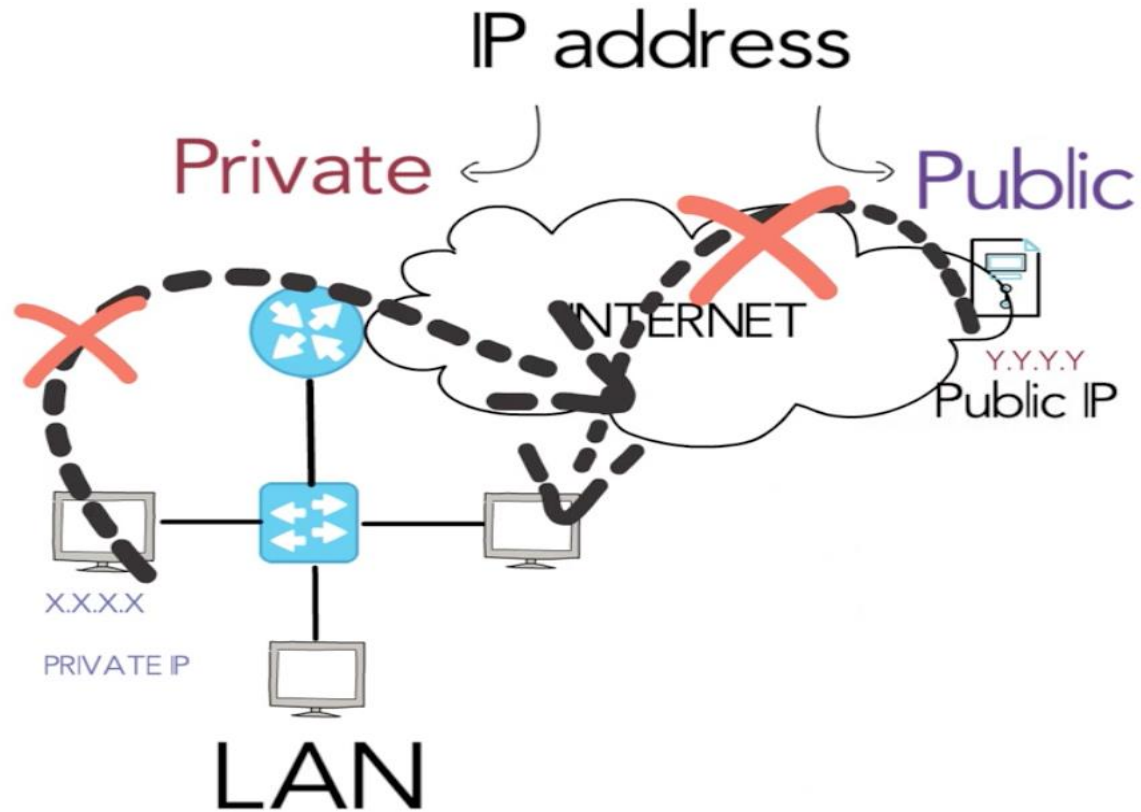
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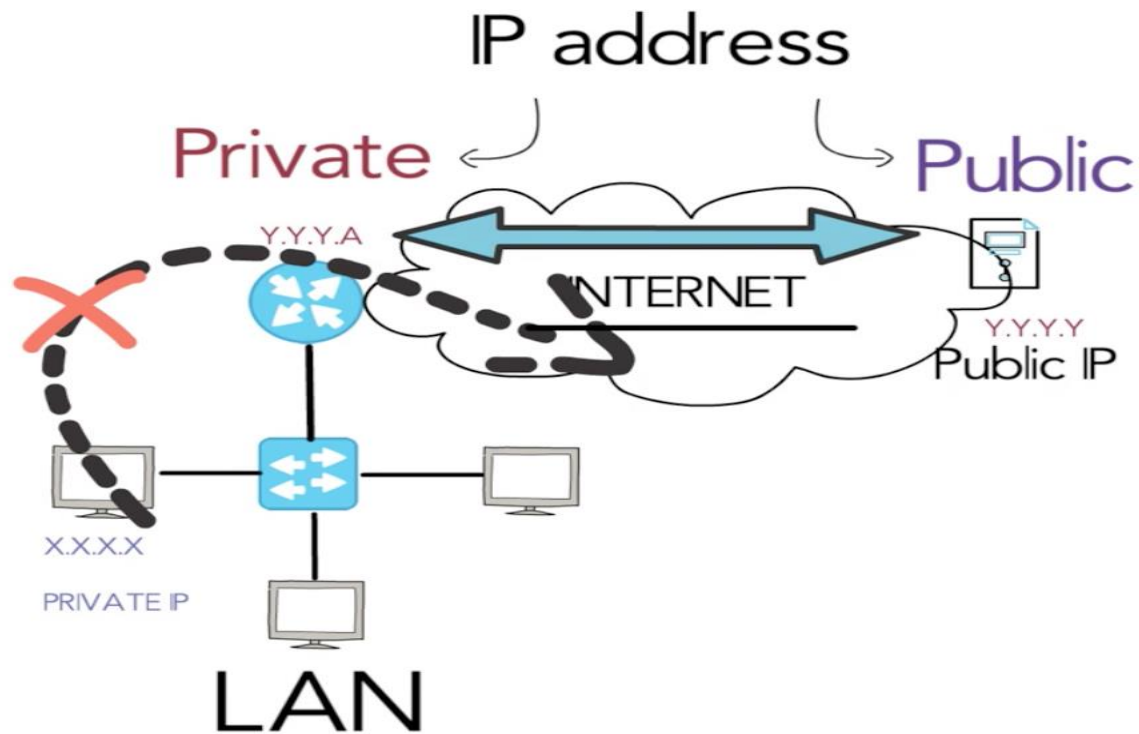
How the classes were created?

192.168.32.170

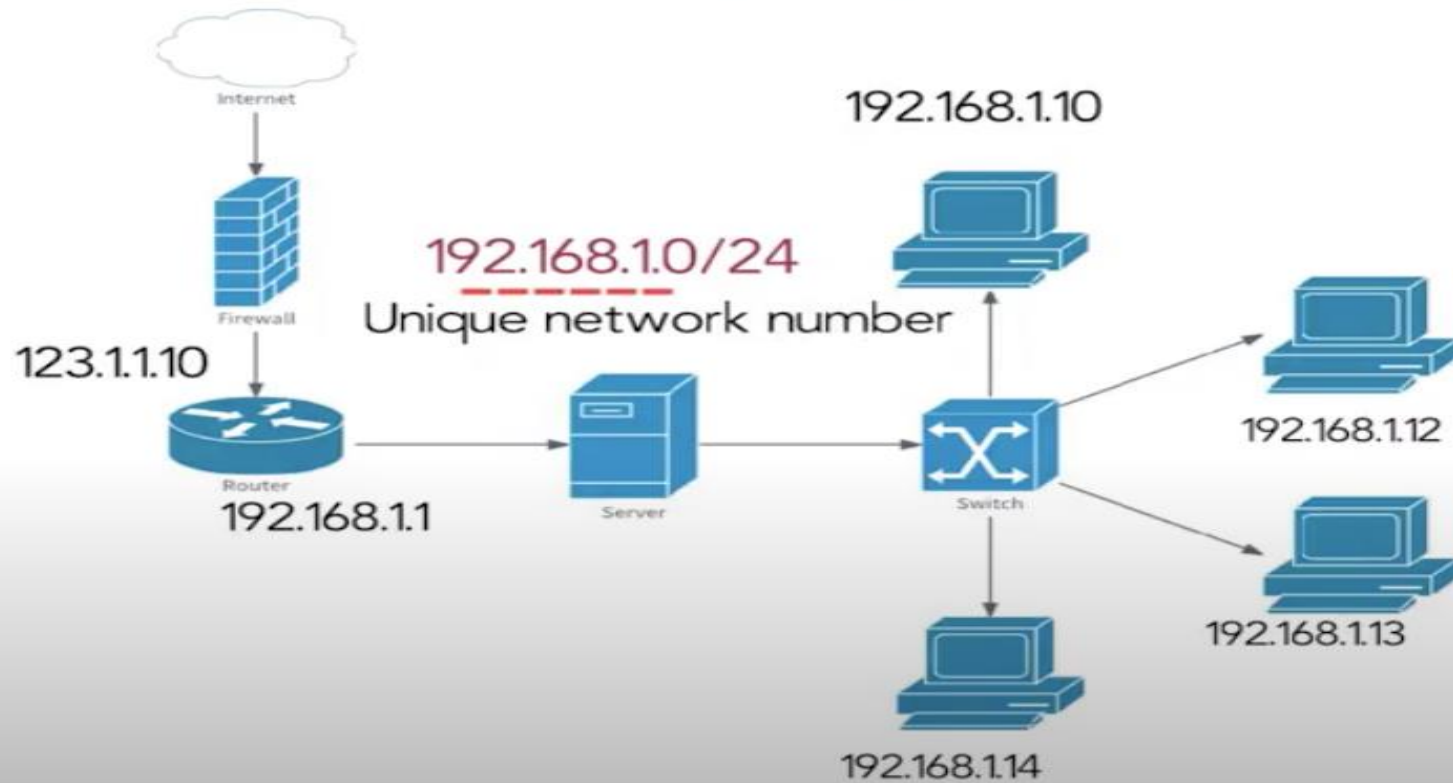
Network ID Host ID







Small office network



What are security threats related to IP addresses?

A variety of security threats are related to IP addresses. Cybercriminals can deceive devices to either reveal your IP address and pretend they are you or stalk it to track activity and take advantage. Online stalking and social engineering are the two leading security threats existing for IP addresses.

Some of the other security threats to an IP address include:

1. Allowing a cybercriminal to use your IP address to track your location
2. Using your IP address to target your network and launch a DDoS attack
3. Using your IP address to download illegal content

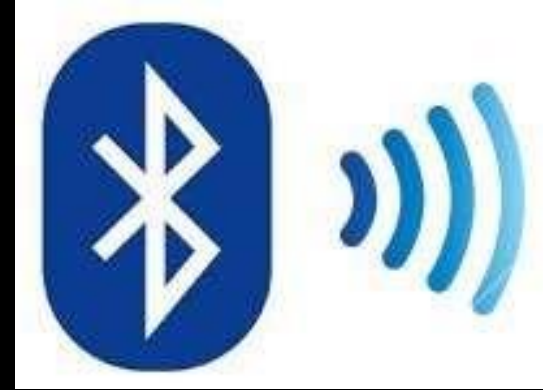
5 ways to protect your IP address

There are multiple ways to protect your IP address from cybercriminals. Some of these options include:

1. Use a VPN
2. Make use of a proxy server
3. Have your ISP make use of dynamic IP addresses
4. Employ a NAT firewall to hide your private IP address
5. Resetting your modem may

Bluetooth

- **Bluetooth** is a short-range wireless technology standard used for exchanging data between fixed and mobile devices over short distances.
- It is using UHF radio waves in the ISM bands, from 2.402 GHz to 2.48 GHz.
- The IEEE standardized Bluetooth as **IEEE 802.15.1**, but no longer maintains the standard.



Wi-Fi

- Wi-Fi Stands for Wireless Fidelity.
- **Wi-Fi**, is a Local Area Wireless technology.
- Wi-Fi networks use radio technologies to transmit and receive data at high speed.
- It is based on the IEEE 802.11 family of standards.
- **Access point:** The access point is a wireless LAN transceiver or “base station” that can connect one or many wireless devices simultaneously to the internet



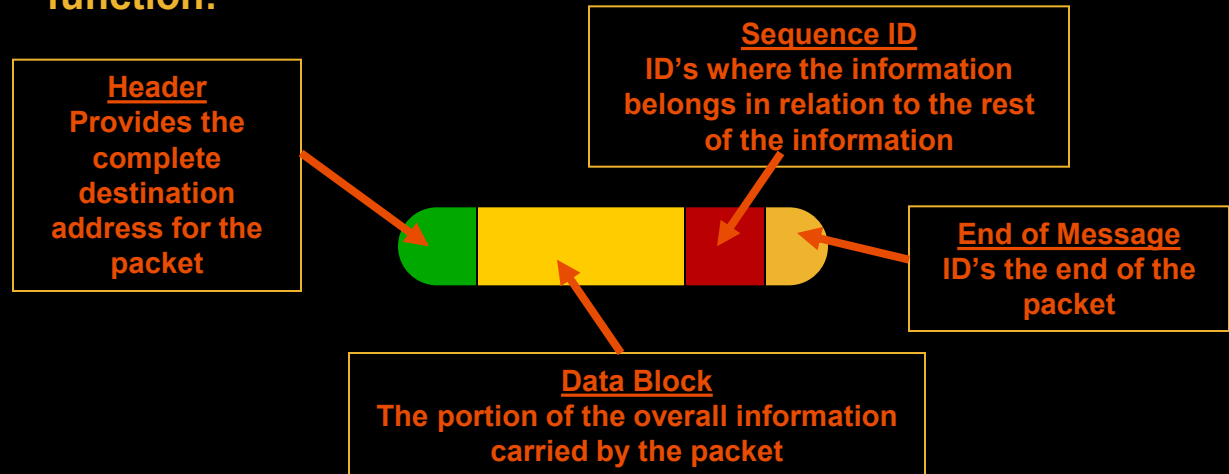
The Internet

The simplest definition of the Internet is that it's a network of computer networks



How Information Travel Through the Internet

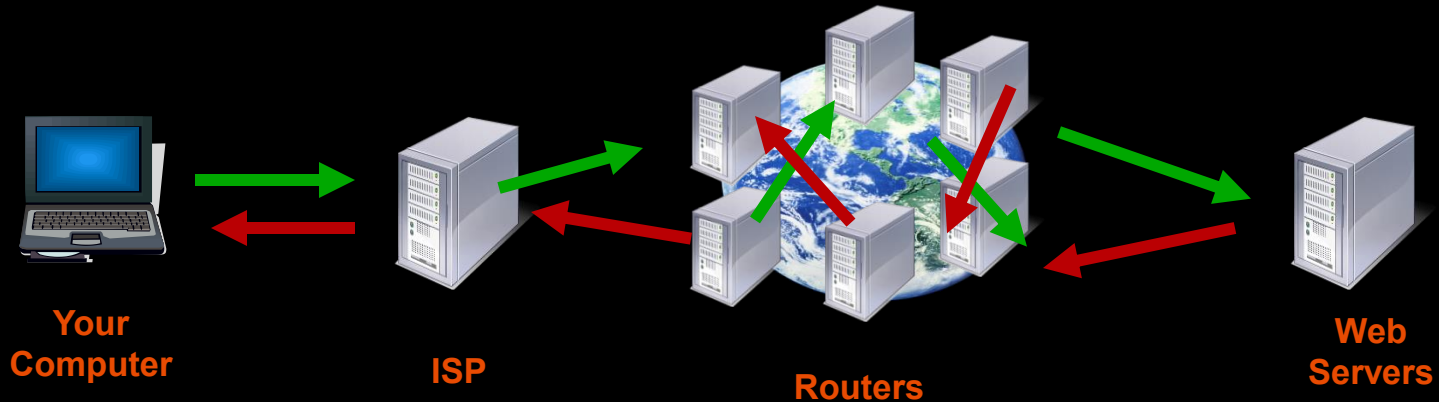
A page on the Internet—whether it's full of words, images or both—doesn't come to you in one shipment. It's translated into digital information, chopped into 1500 byte pieces called **PACKETS**, and sent to you like a puzzle that needs to be reassembled. Each part of the packet has a specific function:



The Internet

How Information Travel Through the Internet

When you connect to a Web site through an ISP and start exchanging information, there isn't a fixed connection between your computer and the Web server computer hosting the Web site. Instead, information is exchanged using the best possible path at that particular time. Special computers called routers determine these paths, avoiding slow links and favoring fast ones.



Cloud Computing

- Cloud computing is **Internet-based computing**, whereby **shared resources, software, and information** are provided to computers and other devices **on demand**, like the electricity grid.
- A Cloud is a type of **parallel and distributed system** consisting of a collection of interconnected and **virtualized computers** that are **dynamically provisioned** and presented as one or more unified computing resources based on **service-level agreements** established through negotiation between the service provider and consumers.

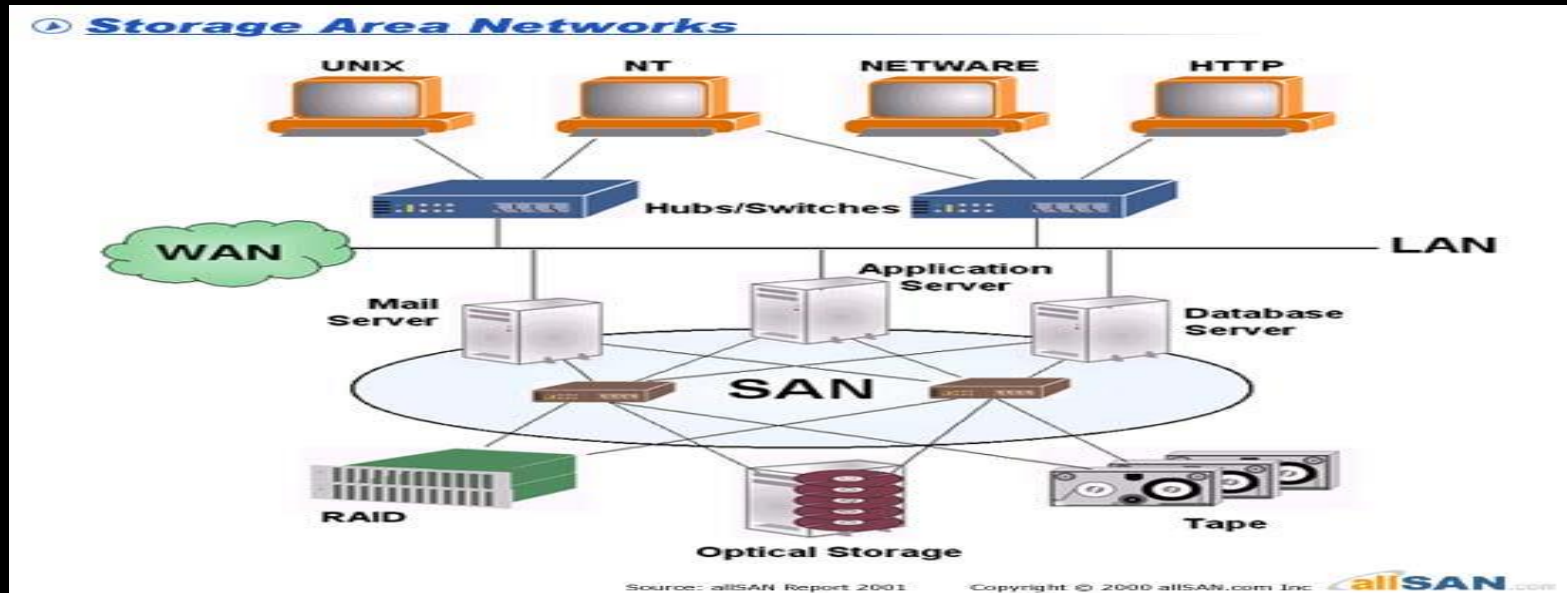






6. Storage Area Network

- SAN may be referred to as a Sub network or special purpose network.
- Its special purpose is to allow users on a larger network to connect various data storage devices with clusters of data servers.
- SANs can be accessed in the same fashion as a drive attached to a server.



Routing is a process in which a layer 3 device (either router or layer 3 switch) finds the best path between the source and destination network. Dynamic routing protocols are used to decrease administrator overhead, i.e., administrators have to configure less but by default, all the routing information is visible to all interested parties as it is not encrypted therefore open to an attack.

We can secure the routing protocols like RIP, EIGRP, and OSPF by putting an authentication to it through creating a key chain and applying it to the interface on which we are advertising the routes. Here, we will not talk about protocol instead we will put authentication on RIP, EIGRP, and OSPF.

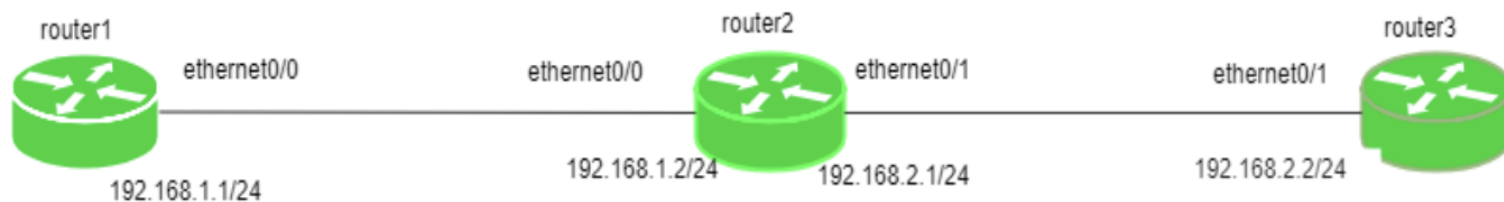
1. Routing Information Protocol (RIP) –

RIP is a distance-vector routing protocol which uses port number 520 and has Administrative Distance 120. It is an application layer protocol and has 3 versions from which only version supports authentication.

Configuration –

There are 3 routers having named
router1 (IP address-192.168.1.1 on ethernet0/0),
router2 (IP address-192.168.1.2 on ethernet0/0 and
192.168.2.1 on ethernet0/1),
router3 (IP address-192.168.2.2 on ethernet0/0).
The router1 will not able to ping router3 as it has no
route to 192.168.2.0 network. For this, we will perform
RIP routing on all the routers and then put
authentication on them.

First configuring router1 for RIP:



```
router1(config)#router rip
router1(config-router)#network 192.168.1.0
router1(config-router)#no auto-summary
router1(config-router)#version 2
```

Now, configuring RIP for router2:

```
router2(config)#router rip
router2(config-router)#network 192.168.1.0
router2(config-router)#network 192.168.2.0
router2(config-router)#no auto-summary
router2(config-router)#version 2
```

Now, configuring RIP for router3:

```
router3(config)#router rip
router3(config-router)#network 192.168.2.0
router3(config-router)#no auto-summary
router3(config-router)#version 2
```

Here, we have enable version 2 of RIP as it supports authentication. Now, able to Ping router3 (192.168.2.2) from router1(192.168.1.1), put Authentication on all routers one by one. First, make a key chain and apply it to the interface:

Here, we have create a key chain named cisco having key l'd 1 and key-string cisco1 and apply it to the interface ethernet0/0 (at which we have advertise RIP). Now creating same key chain on router2:

```
router1(config)#key chain cisco
router1(config-keychain)#key 1
router1(config-keychain-key)#key-string cisco1
router1(config-keychain-key)#exit
router1(config-keychain)#exit
router1(config)#int ethernet0/0
router1(config-if)#ip rip authentication mode md5
router1(config-if)#ip rip authentication key-chain cisco
```

Here, we have create a key chain named cisco having key l'd 1 and key-string cisco1 and apply it to the interface ethernet0/0 (at which we have advertise RIP).

Now creating same key chain on router2:

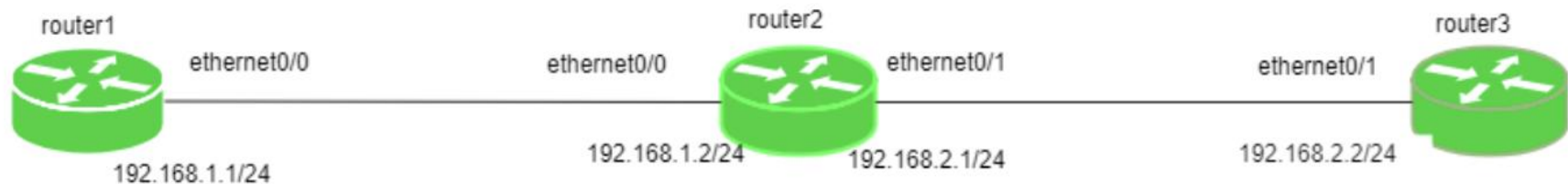
```
router2(config)#key chain cisco
router2(config-keychain)#key 1
router2(config-keychain-key)#key-string cisco1
router2(config-keychain-key)#exit
router2(config-keychain)#exit
router2(config)#int ethernet0/0
router2(config-if)#ip rip authentication mode md5
router2(config-if)#ip rip authentication key-chain cisco
```

Note – The configuration of key chain, its l'd and key-string should be the same on both the routers.

2. Enhanced Interior Gateway Routing Protocol –

EIGRP is an advanced distance routing protocol which uses protocol number 88 and has Administrative Distance 90. It is a network layer protocol which supports clear text and md5 authentication

Configuration –



Administrator: Command Prompt

C:\Windows\system32>nslookup www.google.com

Server: UnKnown

Address: 192.168.43.1

Non-authoritative answer:

Name: www.google.com

Addresses: 2404:6800:4003:c00::6a

74.125.200.106

74.125.200.104

74.125.200.99

74.125.200.103

74.125.200.147

74.125.200.105

C:\Windows\system32>

```
router2(config)#key chain cisco
router2(config-keychain)#key 1
router2(config-keychain-key)#key-string cisco1
router2(config-keychain-key)#exit
router2(config-keychain)#exit
router2(config)#int ethernet0/0
router2(config-if)#ip rip authentication mode md5
router2(config-if)#ip rip authentication key-chain cisco
```

Taking same topology, there are 3 routers having named router1 (IP address-192.168.1.1 on ethernet0/0), router2 (ip address-192.168.1.2 on ethernet0/0 and 192.168.2.1 on ethernet0/1), router3 (ip address-192.168.2.2 on ethernet0/0). Here also, the router1 will not able to ping router3 as it has no route to 192.168.2.0 network. For this we will perform EIGRP routing on all the routers and then put authentication on them.

Configuring EIGRP on router 1:

```
router1(config)#router eigrp 100  
router1(config-router)#network 192.168.1.0  
router1(config-router)#no auto-summary
```

Here, Autonomous System Number 100 is used.

Configuring EIGRP on router2:

```
router2(config)#router EIGRP 100
router2(config-router)#network 192.168.1.0
router2(config-router)#network 192.168.2.0
router2(config-router)#no auto-summary
```

Configuring EIGRP on router3:

```
router3(config)#router eigrp 100  
router3(config-router)#network 192.168.2.0  
router3(config-router)#no auto-summary
```

Now, putting authentication on routers using eigrp.

```
router1(config)#key chain cisco
router1(config-keychain)#key 1
router1(config-keychain-key)#key-string cisco1
router1(config-keychain-key)#exit
router1(config-keychain)#exit
router1(config)#int ethernet0/0
router1(config-if)#ip authentication mode eigrp 100 md5
router1(config-if)#ip authentication key-chain eigrp 100 cisco
```

Configuring same on router2:

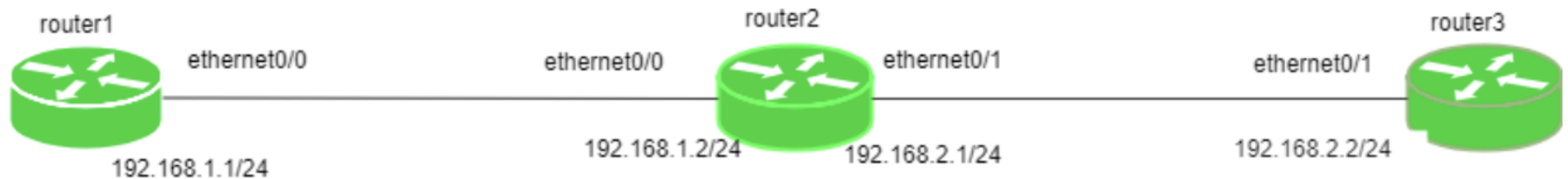
```
router2(config)#key chain cisco
router2(config-keychain)#key 1
router2(config-keychain-key)#key-string cisco1
router2(config-keychain-key)#exit
router2(config-keychain)#exit
router2(config)#int ethernet0/0
router2(config-if)#ip authentication mode eigrp 100 md5
router2(config-if)#ip authentication key-chain eigrp 100 cisco
```

Note – The configuration of key chain, its l'd and key-string should be same on both the routers

3. Open Shortest Path First (OSPF) –

OSPF is a link-state routing protocol which uses protocol number 89 and Administrative Distance 110. It is a network layer protocol which supports clear text and md5 authentication.

Configuration –



Taking same topology, there are 3 routers having named router1 (ip address-192.168.1.1 on ethernet0/0), router2 (ip address-192.168.1.2 on ethernet0/0 and 192.168.2.1 on ethernet0/1), router3 (ip address-192.168.2.2 on ethernet0/0). For this we will perform OSPF routing on all the routers and then put authentication on them.

Configuring OSPF on router1:

```
router1(config)#router ospf 1  
router1(config-router)#network 192.168.1.0 0.0.0.255 area 0
```

Configuring OSPF on router2:

```
router2(config)#router ospf 1  
router2(config-router)#network 192.168.1.0 0.0.0.255 area 0  
router2(config-router)#network 192.168.2.0 0.0.0.255 area 0
```

Configuring ospf on router3:

```
router3(config)#router OSPF 1
router2(config-router)#network 192.168.2.0 0.0.0.255 area 0
```

configuring Authentication on router1:

```
router1(config)#key chain cisco
router1(config-keychain)#key 1
router1(config-keychain-key)#key-string cisco1
router1(config-keychain-key)#exit
router1(config-keychain)#exit
router1(config)#int ethernet0/0
router1(config-if)#ip ospf Authentication message-digest
router1(config-if)#ip ospf authentication-key cisco1
```

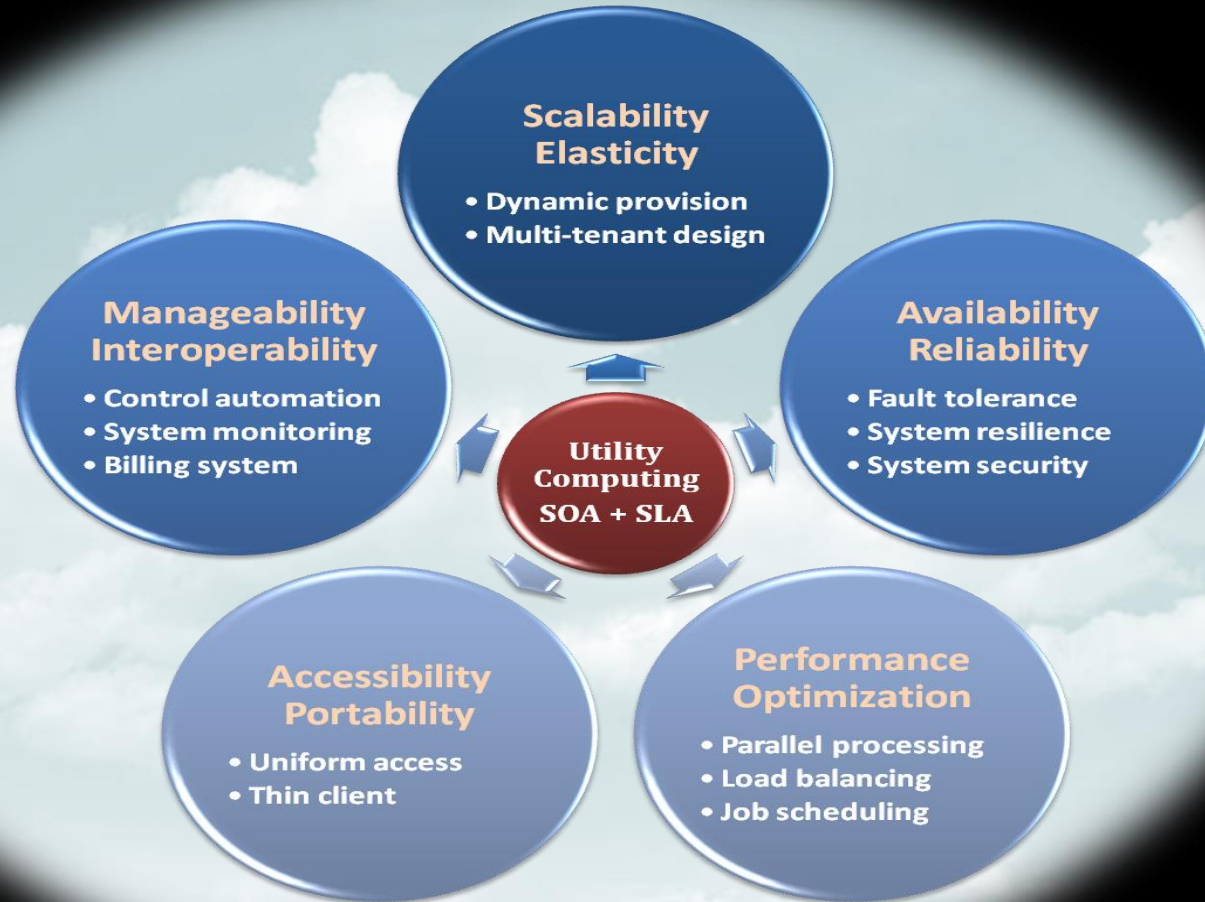
Configuring Authentication on router2:

```
router2(config)#key chain cisco
router2(config-keychain)#key 1
router2(config-keychain-key)#key-string cisco1
router2(config-keychain-key)#exit
router2(config-keychain)#exit
router2(config)#int ethernet0/0
router2(config-if)#ip ospf authentication message-digest
router2(config-if)#ip ospf authentication-key cisco1
```

Note –

- The configuration of key chain, its l'd and key-string should be same on both the routers.
- Md5 is susceptible to brute-force attack therefore it is advised to use passwords containing numbers or special characters and should be long.

Properties & Characteristics



laaS

Infrastructure as a Service – laaS

- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components .

Examples :

Amazon EC2

Eucalyputs

OpenNebula

... etc

PaaS

Platform as a Service – PaaS

- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

Examples :

Microsoft Windows Azure

Google App Engine

Hadoop(High Availability Distributed Object Oriented Platform)

... etc

SaaS

Software as a Service – SaaS

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email).
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

Examples :

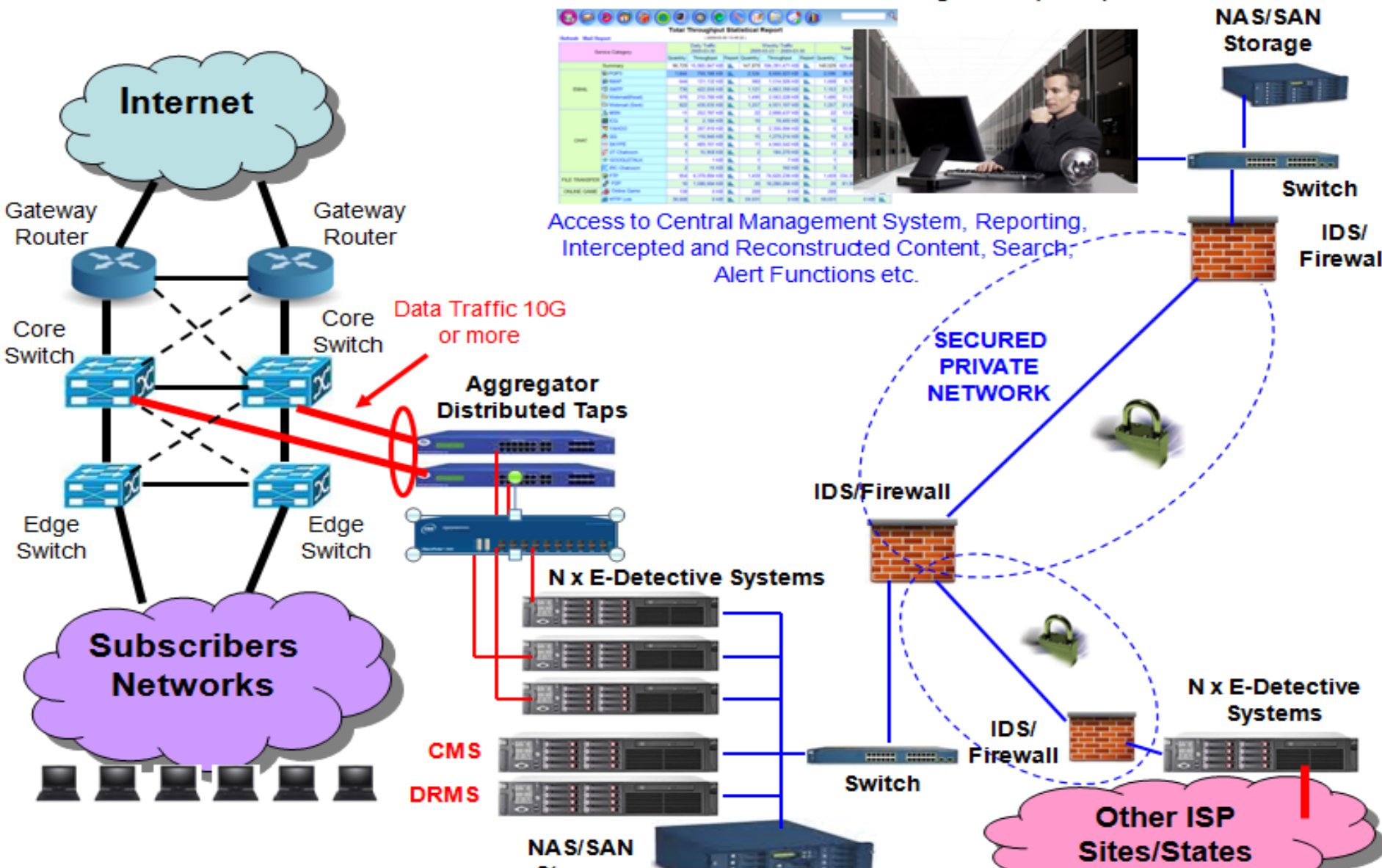
Google Apps (e.g., Gmail, Google Docs, Google sites, ...etc)

Cloud-Deployment Model

There are four primary cloud deployment models :

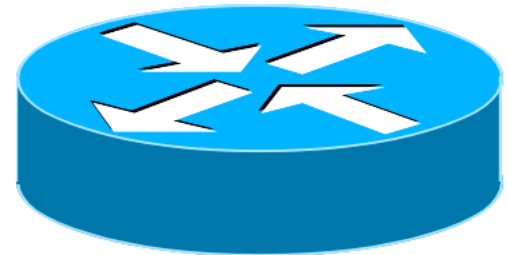
1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud

	Public Cloud	Private Cloud
<i>Infrastructure</i>	<i>Homogeneous</i>	<i>Heterogeneous</i>
<i>Policy Model</i>	<i>Common defined</i>	<i>Customized & Tailored</i>
<i>Resource Model</i>	<i>Shared & Multi-tenant</i>	<i>Dedicated</i>
<i>Cost Model</i>	<i>Operational expenditure</i>	<i>Capital expenditure</i>
<i>Economy Model</i>	<i>Large economy of scale</i>	<i>End-to-end control</i>



A day in a life of a router

- find path
- forward packet, forward packet, forward packet, forward packet...
- find alternate path
- forward packet, forward packet, forward packet, forward packet...
- repeat until powered off



Step 3: Connect Wireless Router to Family PC.

Choose the correct cable to connect Wireless Router Ethernet 1 to Family PC.

If you attached the correct cable, the link lights on the cable turn green.

Part 4: Verify Connections

Step 1: Test the connection from Family PC to netacad.pka.

- Open the Family PC command prompt and ping netacad.pka.
- Open the Web Browser and the web address <http://netacad.pka>.

Step 2: Ping the Switch from Home PC.

Open the Home PC command prompt and ping the Switch IP address of to verify the connection.

Step 3: Open Router0 from Configuration Terminal.

- Open the Terminal of Configuration Terminal and accept the default settings.
- Press Enter to view the Router0 command prompt.
- Type `show ip interface brief` to view interface statuses.

Part 5: Examine the Physical Topology

Step 1: Examine the Cloud.

- Click the Physical Workspace tab or press Shift+P and Shift+L to toggle between the logical and physical workspaces.
- Click the Home City icon.
- Click the Cloud icon.

How many wires are connected to the switch in the blue rack?

- Click Back to return to Home City.

Routing Protocols

- Routers use “routing protocols” to exchange routing information with each other
 - **IGP** is used to refer to the process running on routers inside an ISP's network
 - **EGP** is used to refer to the process running between routers bordering directly connected ISP networks

- At the bottom left, click the orange lightning icon to open the available Connections.
- Choose the correct cable to connect Router0 F0/0 to Cloud Eth6. Cloud is a type of switch, so use a Copper Straight-Through connection. If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect the cloud to Cable Modem.

Choose the correct cable to connect Cloud Coax7 to Modem Port0.

If you attached the correct cable, the link lights on the cable turn green.

Part 2: Connect Router0

Step 1: Connect Router0 to Router1.

Choose the correct cable to connect Router0 Ser0/0/0 to Router1 Ser0/0. Use one of the available Serial cables.

If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect Router0 to netacad.pka.

Choose the correct cable to connect Router0 F0/1 to netacad.pka F0. Routers and computers traditionally use the same wires to transmit (1 and 2) and receive (3 and 6). The correct cable to choose consists of these crossed wires. Although many NICs can now autosense which pair is used to transmit and receive, Router0 and netacad.pka do not have autosensing NICs.

If you attached the correct cable, the link lights on the cable turn green.

Step 3: Connect Router0 to the Configuration Terminal.

Choose the correct cable to connect Router0 Console to Configuration Terminal RS232. This cable does not provide network access to Configuration Terminal, but allows you to configure Router0 through its terminal.

If you attached the correct cable, the link lights on the cable turn black.

Part 3: Connect Remaining Devices

Step 1: Connect Router1 to Switch.

Choose the correct cable to connect Router1 F1/0 to Switch F0/1.

If you attached the correct cable, the link lights on the cable turn green. Allow a few seconds for the light to transition from amber to green.

Step 2: Connect Cable Modem to Wireless Router.

Choose the correct cable to connect Cable Modem Port1 to Wireless Router Internet port.

If you attached the correct cable, the link lights on the cable will turn green.

Step 3: Connect Wireless Router to Family PC.

Choose the correct cable to connect Wireless Router Ethernet 1 to Family PC.

If you attached the correct cable, the link lights on the cable turn green.

Part 4: Verify Connections

Step 1: Test the connection from Family PC to netacad.pka.

- Open the Family PC command prompt and ping netacad.pka.
- Open the Web Browser and the web address <http://netacad.pka>.

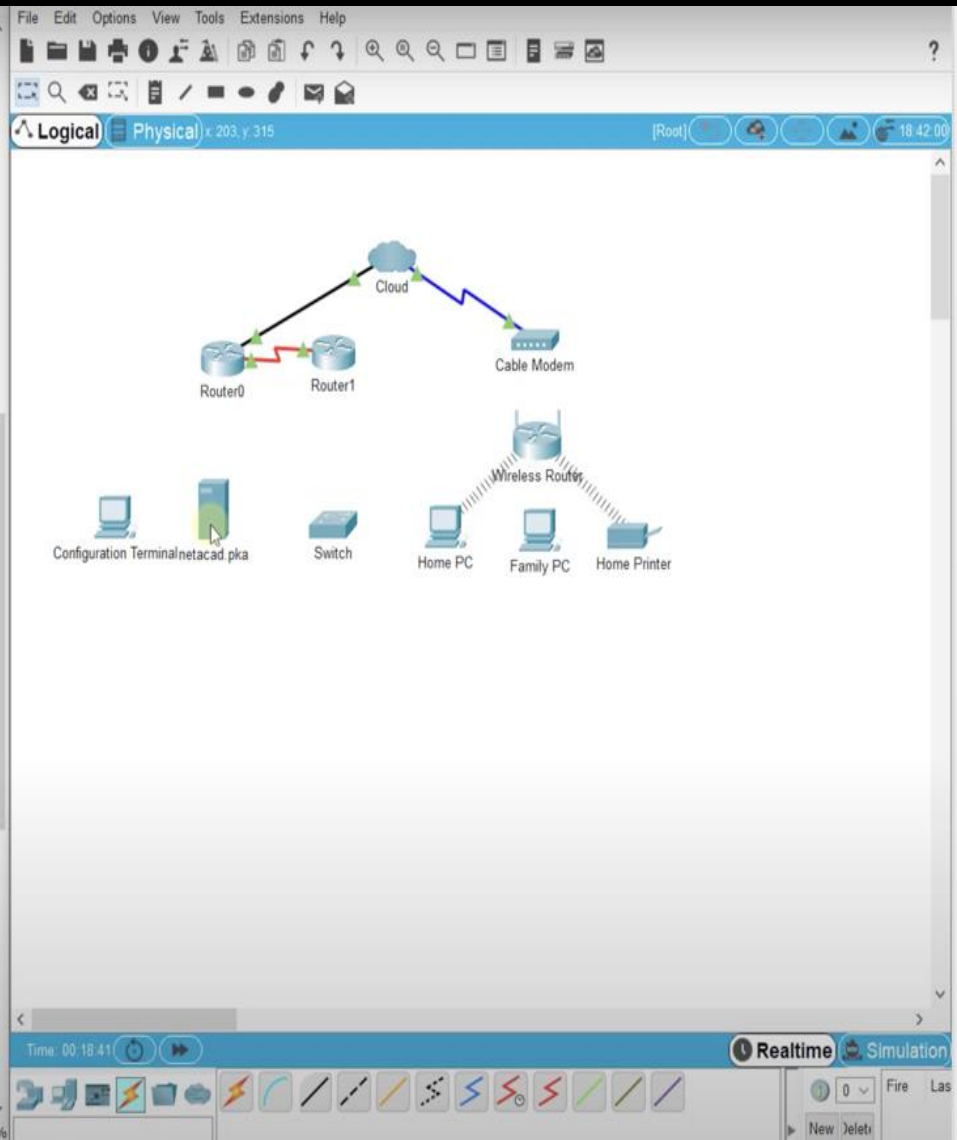
Step 2: Ping the Switch from Home PC.

Open the Home PC command prompt and ping the Switch IP address of to verify the connection.

Step 3: Open Router0 from Configuration Terminal.

Time Elapsed: 00:18:41

Completion: 37%



Step 3: Connect Router0 to the Configuration Terminal.

Choose the correct cable to connect Router0 Console to Configuration Terminal RS232. This cable does not provide network access to Configuration Terminal, but allows you to configure Router0 through its terminal.

If you attached the correct cable, the link lights on the cable turn black.

Part 3: Connect Remaining Devices

Step 1: Connect Router1 to Switch.

Choose the correct cable to connect Router1 F1/0 to Switch F0/1.

If you attached the correct cable, the link lights on the cable turn green. Allow a few seconds for the light to transition from amber to green.

Step 2: Connect Cable Modem to Wireless Router.

Choose the correct cable to connect Cable Modem Port1 to Wireless Router Internet port.

If you attached the correct cable, the link lights on the cable will turn green.

Step 3: Connect Wireless Router to Family PC.

Choose the correct cable to connect Wireless Router Ethernet 1 to Family PC.

If you attached the correct cable, the link lights on the cable turn green.

Part 4: Verify Connections

Step 1: Test the connection from Family PC to netacad.pka.

- Open the Family PC command prompt and ping netacad.pka.
- Open the Web Browser and the web address <http://netacad.pka>.

Step 2: Ping the Switch from Home PC.

Open the Home PC command prompt and ping the Switch IP address of to verify the connection.

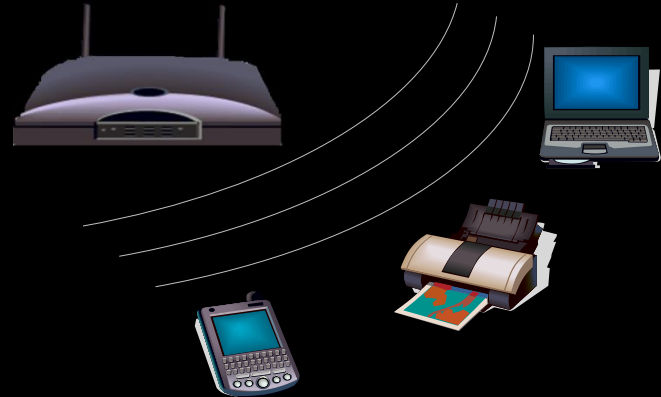
Wireless Networks

Wireless network is a type of computer network that uses wireless data connections for connecting network nodes.

Example

Bluetooth

Wi-Fi



Wi-Fi is a wireless networking technology, by which we can access networks or connect with other computers or mobile using a wireless medium. In Wi-Fi, data are transferred over radio frequencies in a circular range.

Wi-Fi, a brand name given by the Wi-Fi Alliance (formerly Wireless Ethernet Compatibility Alliance), is a generic term that refers to the communication standard for the wireless network which works as a Local Area Network to operate without using the cable and any types of wiring. It is known as **WLAN**. The communication standard is **IEEE 802.11**. Wi-Fi works using Physical Data Link Layer.

- Using Wi-Fi or WLAN we can construct simple wireless connections from one point to another, known as Point to point networks. This can be useful to connect two locations that are difficult to reach by wire, such as two buildings of corporate business.

- One more important application is **VoWi-Fi**, which is known as **voice-over Wi-Fi**. Some years ago telecom companies are introduced VoLTE (Voice over Long-Term Evolution). Nowadays they are introduced to VoWi-Fi, by which we can call anyone by using our home Wi-Fi network, only one thing is that the mobile needs to connect with the Wi-Fi. Then the voice is transferred using the Wi-Fi network instead of using the mobile SIM network, so the call quality is very good. Many mobile phones are already getting the support of VoWi-Fi.

- Wi-Fi in offices: In an office, all the computers are interconnected using Wi-Fi. For Wi-Fi, there are no wiring complexities. Also, the speed of the network is good. For Wi-Fi, a project can be presented to all the members at a time in the form of an excel sheet, ppt, etc. For Wi-Fi, there is no network loss as in cable due to cable break.
- Also using W-Fi a whole city can provide network connectivity by deploying routers at a specific area to access the internet. Already schools, colleges, and universities are providing networks using Wi-Fi because of its flexibility.
- Wi-Fi is used as a *positioning system* also, by which we can detect the positions of Wi-Fi hotspots to identify a device location.

How does Wi-Fi work?

Wi-Fi is a wireless technology for networking, so it uses Electromagnetic waves to transmit networks. We know that there are many divisions of Electromagnetic waves according to their frequency such as X-ray, Gamma-ray, radio wave, microwave, etc, in Wi-Fi, the radio frequency is used. For transmitting Wi-Fi signal there is three medium,

- **Base station network or an Ethernet(802.3) connection:** It is the main host network from where the network connection is provided to the router.
- **Access point or router:** it is a bridge between a wired network and a wireless network. It accepts a wired Ethernet connection and converts the wired connection to a wireless connection and spreads the connection as a radio wave.
- **Accessing devices:** It is our mobile, computer, etc from where we use the Wi-Fi and surfing internet.

PING

A ping is a basic Internet command that allows a user to test and verify whether a given destination IP address exists and can accept requests in computer network administration. Ping is also used for diagnosis to confirm that the computer the user tries to reach is operational. Ping can be used with any operating system (OS) that supports networking, including the majority of embedded network administration software

Ping (Packet Internet Groper) is a method for determining communication latency between two networks or ping is a method of determining the time it takes for data to travel between two devices or across a network. As communication latency decreases, communication effectiveness improves. A low ping time is critical in situations where the timely delivery of data is more important than the quantity and quality of the desired information.

How Does Ping Work?

Ping sends an Internet Control Message Protocol (ICMP) Echo Request to a network interface and then waits for a response. When the ping command is executed, a ping signal is delivered to the provided address. When the target host receives the echo request, it answers with an echo reply packet. This method has two distinct purposes: calculating round-trip time (RTT) or latency and ensuring that the target host is available. RTT is a measure of the time it takes to receive a response. Measured in milliseconds (ms), the process begins when a browser submits a request to a server and concludes when the server responds

How To Get The Ping Value Of Any Site Corresponding To Your Server?

- The ping value represents the strength of a connection between two computers or a network. You can check the ping of any website that corresponds to your computer using a command prompt for Windows or a terminal for Mac.
- Simply type the “ping<space>website name” into the command prompt or terminal to have your system send some data packets to that specific website and then acknowledge you with the value of ping that is occurring within your system and that specific website.
- As you can see in the image below. I entered “> ping youtube.com”, then my system sent and received four packets of data from YouTube to determine the minimum, maximum, and average ping values, which are 20ms, 22ms, and 21ms, respectively.

C:\ Command Prompt

Microsoft Windows [Version 10.0.19041.264]
(c) 2020 Microsoft Corporation. All rights reserved.

C:\Users\H.P> ping youtube.com

Pinging youtube.com [216.58.200.174] with 32 bytes of data:

Reply from 216.58.200.174: bytes=32 time=20ms TTL=120

Reply from 216.58.200.174: bytes=32 time=21ms TTL=120

Reply from 216.58.200.174: bytes=32 time=22ms TTL=120

Reply from 216.58.200.174: bytes=32 time=22ms TTL=120

Ping statistics for 216.58.200.174:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 20ms, Maximum = 22ms, Average = 21ms

C:\Users\H.P>_

- So, if an online game streamer has two network options, one with 10ms of ping and 10mbps internet speed, and the other with 100ms of ping and 500mbps internet speed, the gamer will obviously choose the first because he or she wants to interact with the audience in real-time. However, if a person wants to watch YouTube videos and download them, he or she will obviously select the second option in order to speed up the download process.

How to use ping in troubleshooting?

Echo requests and echo responses are the standard for troubleshooting ICMP messages. Ping is built into almost every operating system with network support for troubleshooting. However, the specific implementation differs slightly between businesses. Ping is the foundation of typical network troubleshooting. Typing 216.58.200.174, for example, will ping an IP address. If the ping is successful, it indicates that the system is operational and that the two machines can communicate with one another. If the ping is successful but the response time is excessively long, this suggests network congestion, routing, or speed problems. Even unsuccessful pings provide useful troubleshooting information. When it comes to network speed tests, ping is standard procedure.

What is Ping Spoofing?

In network security, ping spoofing occurs when attackers provide incorrect data to a server. When this occurs, the fake data duplicates the real data packets, and is sent to the server. The receiver receives fake data and responds to the data packet using a third-party user rather than the original sender. On the receiver side, the server may receive unwanted data from a different address. This helps hide the real third-party user's address. This is known as ping spoofing.

A base station serves as a central connection point for a wireless device to communicate. It further connects the device to other networks or devices, usually through dedicated high bandwidth wire or fiber optic connections.

They are cell towers or cellular antennas. These types of objects are an inevitability since they serve the purpose of providing signal transfer for data and voice between mobile mobiles. The idea of base stations is anchored in their function to provide coverage, capacity, and connectivity, hence allowing for extending the working capabilities of mobile phones and other radio gear.

In essence, a base station provides communication an interactive point within an area of jurisdiction that is commonly referred to as a cell. The name cell in cellular networks emerges from the fact that a coverage area underwent division into small hexagonal cells, which were supplied by one base station. Due to this design, frequency reuse becomes more efficient, and the interference is reduced as well which amplifies performance on a broader scale throughout the spectrum of the network.

Difference between UMTS and GSM

1. Universal Mobile Telecommunications System (UMTS) :

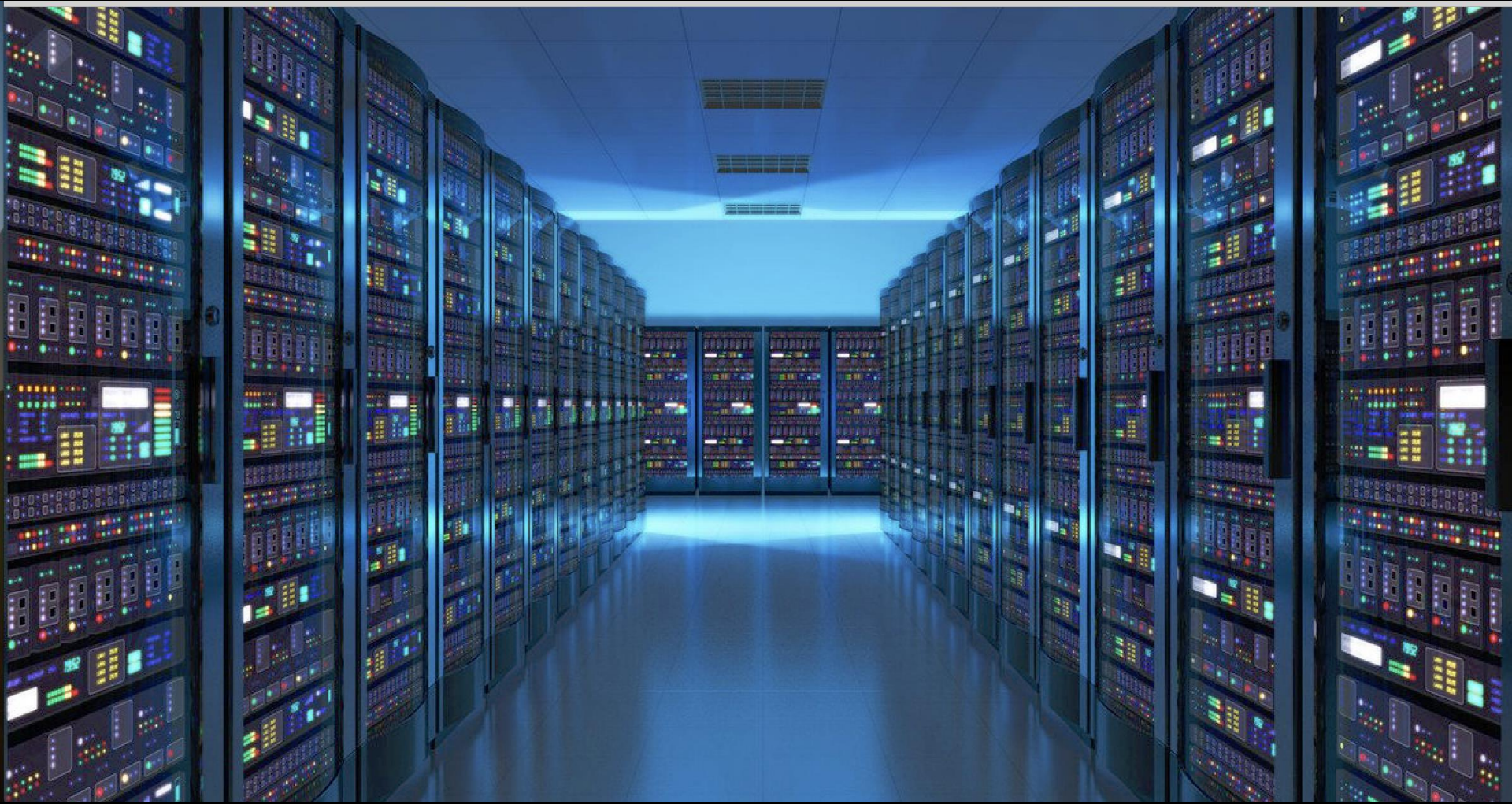
UMTS stands for Universal Mobile Telecommunications System, developed based on 3GPP standards. Its network composed of three main parts UE (User Equipment), Radio Access Network (RAN) and Core Network. There are various technologies which fall under UMTS based on different releases from 3GPP community. It is also referred as 3G.

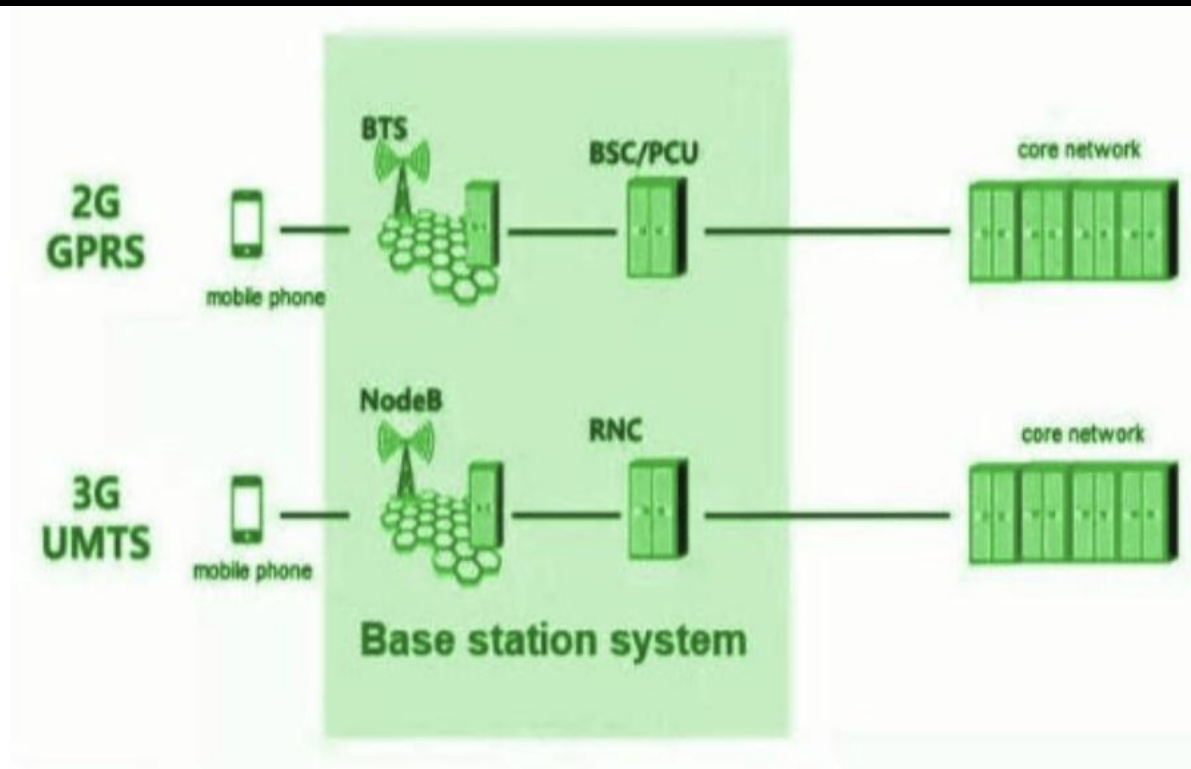
2. Global System for Mobile communication (GSM) :

GSM stands for Global System for Mobile communication. It is majorly used mobile communication system in the world. It uses time division multiple access i.e TDMA to send down the digitized and reduced data through a channel with two different streams of client data, each in its own particular time slot.

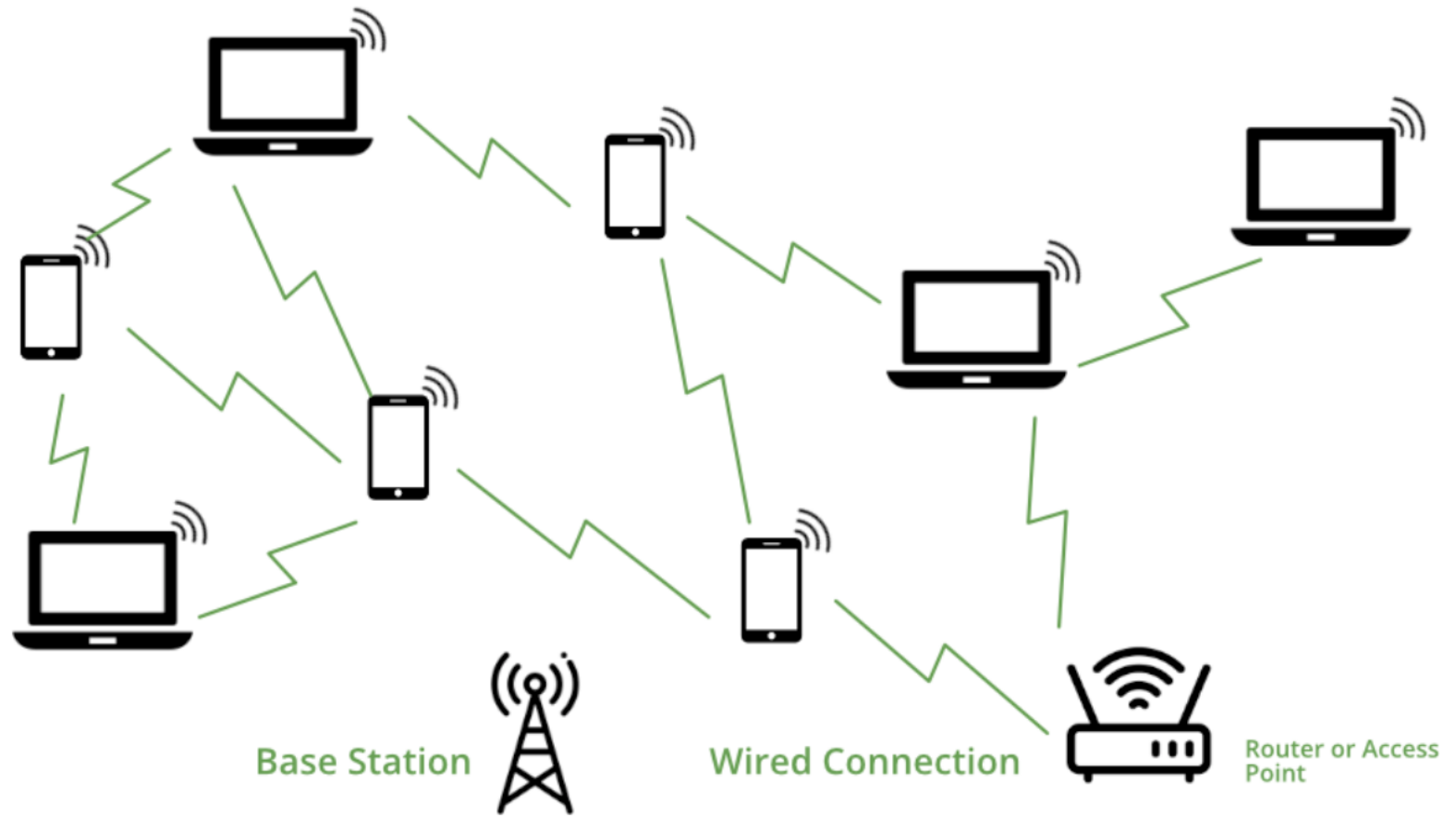
UMTS	GSM
UMTS is already 3G.	GSM is 2G and 2.5G.
It is mainly CDMA-based.	It is typically based on TDMA .
There channels are divided into logical, transport and physical based on where they are used along with their functions.	There channels are divided into traffic channel and control channel based on functionality of use in the system.
Its frame duration is about 10ms,	Its frame duration is about 4.615ms,
It is newer.	It is a rather old technology.
It has various bands from Band-I to Band-VI, Each band specifies frequency and UARFCN.	It has various bands, the major among them are 850MHz, 900MHz, 1800MHz and 1900MHz.
It has various documents specified by ETSI and 3GPP, 3GPP™TS 45-series.	It has various releases such as R99(WCDMA), R5 (HSDPA), R6(HSUPA) etc.
The modulation efficiency is 1.0 b/s/Hz .	The modulation efficiency is 1.35 b/s/Hz.

A base station represents an access point for a wireless device to communicate within its coverage area. It usually connects the device to other networks or devices through a dedicated high bandwidth wire or fiber optic connection. Base stations typically have a transceiver, capable of sending and receiving wireless signals; Otherwise if they only send the trailer it will be considered a transmitter or broadcast point only. The base station will have one or more RF antennas installed to transmit and receive RF signals from other devices.





Base Station System



Working of Wi-Fi

Some more terminologies

- **SSID (Service Set Identifier)**: It is a 32 character name that identifies the Wi-Fi network and differentiates one Wi-Fi from another Wi-Fi. All the devices are attempting to connect a particular SSID. Simply, SSID is the name of the wireless network.

- **WPA-PSK (Wi-Fi Protected Access- Pre-Shared Key)**: It is a program developed by the Wi-Fi Alliance Authority to secure wireless networks with the use of Pre-Shared Key(PSK) authentication. WPA has 3 types, such as WPA. WPA2, WPA3. It is a way of encrypting the Wi-Fi signal to protect from unwanted users.

- Wi-Fi uses **Ad-Hoc** networks to transmit. It is a point-to-point network without any interface.